MATHEMATICS

BINOMIAL THEOREM

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BINOMIAL THEOREM

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BASED ON IMPORTANT TERMS

1. If $\frac{T_2}{T_3}$ in the expansion of $(a+b)^n$ and $\frac{T_3}{T_4}$ in the expansion of $(a+b)^{n+3}$ are equal, then n is equal to :

यदि $(a+b)^n$ के प्रसार में $\frac{T_2}{T_3}$ तथा $(a+b)^{n+3}$ के प्रसार में $\frac{T_3}{T_4}$ बराबर है तब n का मान होगा:

(A) 3 (B) 4 (C) 5 (D) 6

Ans. C

Sol. $\frac{T_2}{T_3}(a+b)^n = \sum_{r=0}^{n} {}^nC_r a^{n-r} b^r$

$$\Rightarrow \frac{T_2}{T_3} = \frac{{}^nC_1 a^{n-1} b^1}{{}^nC_2 a^{n-2} b^2} = \frac{\underline{|n}\,\underline{|2}\,|n-2}{\underline{|n-1}\,\underline{|n}} \cdot \frac{a}{b} = \frac{2a}{(n-1)b}$$

$$(a+b)^{n+3} = \sum_{r=0}^{n+3} {}^{n+3}C_r a^{n+3-r} b^r$$

$$\Rightarrow \frac{T_3}{T_4} = \frac{{}^{n+3}C_2 a^{n+3-2} b^2}{{}^{n+3}C_3 a^{n+3-3} b^3} = \frac{\underline{|n+3}\,\underline{|n}\,\underline{|3}}{\underline{|n+1}\,\underline{|2}\,\underline{|n+3}} \cdot \frac{a^{n+1} \cdot b^2}{a^n b^3}$$

$$= \frac{3}{n+1} \cdot \frac{a}{b}$$

Given $\frac{T_2}{T_3} = \frac{T_3}{T_4}$

$$\Rightarrow \frac{2a'}{(n-1)b'} = \frac{3a'}{(n+1)b'}$$

$\Rightarrow 2n + 2 = 3n - 3$

$\Rightarrow n = 5$

2. If number of terms in the expansion of $(x - 2y + 3z)^n$ are 45, then n is equal to :

(A) 7 (B) 8 (C) 9 (D) None of these

यदि $(x - 2y + 3z)^n$ के प्रसार में कुल पदों की संख्या 45 है तब n का मान होगा :

(A) 7 (B) 8 (C) 9 (D) इनमें से कोई नहीं

Ans. B

Sol. No. of terms in the expansion of $(x-2y+3z)^n$ are 45

$$(x-2y+3z)^n = \sum_{r_1+r_2+r_3=n} \frac{\lfloor n}{\lfloor r_1 \lfloor r_2 \lfloor r_3} \cdot x^{r_1} \cdot (-2)^{r_2} \cdot y^{r_2} \cdot 3^{r_3} \cdot z^{r_3}$$

$$\Rightarrow n + {}^{3-1}C_{3-1} = 45 \Rightarrow {}^{n+2}C_2 = 45 \Rightarrow n^2 + 3n + 2 = 90$$

$$\Rightarrow n = -11 \text{ or } n = 8$$

3. The greatest coefficient in the expansion of $(1 + x)^{2n+2}$ is :

$(1 + x)^{2n+2}$ के प्रसार में महत्तम पद का गुणांक होगा :

(A) $\frac{(2n)!}{(n!)}$ (B) $\frac{(2n+2)!}{[(n+1)!]^2}$ (C) $\frac{(2n+2)!}{n!(n+1)!}$ (D) $\frac{(2n)!}{n!(n+1)!}$

Ans. B

Sol. $(1+x)^{2n+2} = \sum_{r=0}^{2n+2} {}^{2n+2}C_r x^r$

${}^{2n+2}C_r$ is maximum if $r = n + 1$

$\Rightarrow$ Greatest coefficient is ${}^{2n+2}C_{n+1} = \frac{\lfloor 2n+2}{(\lfloor n+1)^2}$

4. The expression $\frac{1}{\sqrt{4x+1}}\left[\left[\frac{1+\sqrt{4x+1}}{2}\right]^7 - \left[\frac{1-\sqrt{4x+1}}{2}\right]^7\right]$ is a polynomial in x of degree :

व्यंजक $\frac{1}{\sqrt{4x+1}}\left[\left[\frac{1+\sqrt{4x+1}}{2}\right]^7 - \left[\frac{1-\sqrt{4x+1}}{2}\right]^7\right]$, x में किस घात का बहुपदीय व्यंजक है :

(A) 7 (B) 5 (C) 4 (D) 3

Ans. D

Sol. Given expression can be rewritten as

$$\frac{2}{2^7\sqrt{4x+1}}\left[{}^7C_1(\sqrt{4x+1})+{}^7C_3(\sqrt{4x+1})^3+\ldots+{}^7C_7(\sqrt{4x+1})^7\right]$$

$$\frac{1}{2^6}\left[{}^7C_1+{}^7C_3(\sqrt{4x+1})^2+\ldots..+{}^7C_7(\sqrt{4x+1})^6\right]$$

$\therefore$ Last term becomes $(4x+1)^3$

Hence degree is 3

5. If the coefficient of $(r+2)^{th}$ and $(3r)^{th}$ term in the exp. of $(1+x)^{2n}$ are equal then :

(A) $n = 2r+1$ (B) $n = 2r-1$ (C) $n = 2r$ (D) None of these

$(1+x)^{2n}$ के प्रसार में $(r+2)$वें तथा $(3r)$ वें पद के गुणांक बराबर हो तो :

(A) $n = 2r+1$ (B) $n = 2r-1$ (C) $n = 2r$;d ds buesa ls dksbZ ugha

Ans. C

Sol. $(r+2)^{th} = (3r)^{th}$

${}^{2n}C_{r+2} = {}^{2n}C_{3r}$

$r+2 = 3r$ or $r+2+2r = 2n$

$r = 1$ or $n = 2r+1$

6. The coefficient of the middle term in the binomial expansion in powers of x of $(1+\alpha x)^4$ and of $(1-\alpha x)^6$ is the same if α equals :

$(1+\alpha x)^4$ तथा $(1-\alpha x)^6$ के द्विपद प्रसार, जो कि x की घातों के रूप में है, के मध्य पदों के गुणांक समान होंगे यदि α का मान है :

(A) $-\frac{5}{3}$ (B) $\frac{10}{3}$ (C) $-\frac{3}{10}$ (D) $\frac{3}{5}$

Ans. C

Sol. Coefficient of middle term in $(1+\alpha x)^4 = {}^4C_2\,\alpha^2$

coefficient of middle term in $(1-\alpha x)^6 = {}^6C_3\,(-\alpha)^3$

${}^4C_2\,\alpha^2 = -{}^6C_3\,\alpha^3$

$-\frac{6}{20} = \alpha$

$a = \frac{-3}{10}$

7. If the coefficients of rth, (r + 1)th and (r + 2)th terms in the binomial expansion of $(1+y)^m$ are in A.P., then m and r satisfy the equation :

एक द्विपद $(1+y)^m$ के प्रसार में यदि r वें, (r + 1) वें तथा (r + 2) वें पदों के गुणांक समान्तर श्रेणी में हैं, तो m तथा r निम्न समीकरण dks lUrq"V djrs gSa

(A) $m^2 - m(4r - 1) + 4r^2 - 2 = 0$ (B) $m^2 - m(4r + 1) + 4r^2 + 2 = 0$

(C) $m^2 - m(4r + 1) + 4r^2 - 2 = 0$ (D) $m^2 - m(4r - 1) + 4r^2 + 2 = 0$

Ans. C

Sol. Coefficient of rth, $(r + 1)^{th}$ and $(r + 2)^{th}$ terms in expansion of $(1 + y)^n$ are ${}^mC_{r-1}, {}^mC_r, {}^mC_{r+1}$

Then $2.\ {}^mC_r = {}^mC_{r-1} + {}^mC_{r+1}$

$\Rightarrow m^2 - m(4r + 1) + 4r^2 - 2 = 0$

Short Method: Let r = 1, hence ${}^mC_0$, ${}^mC_1$ and ${}^mC_2$ are in A.P.

$\Rightarrow 2.{}^mC_1 = {}^mC_0 + {}^mC_2 \Rightarrow 2m = 1 + \frac{m(m-1)}{2}$

$\Rightarrow 4m = 2 + m^2 - m \Rightarrow m^2 - 5m + 2 = 0$

Which is given by (C)

8. If the coefficient of x^7 in $\left[ax^2 + \left(\frac{1}{bx}\right)\right]^{11}$ equals the coefficient of x^{-7} in $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$, then a and b satisfy the relation :

यदि $\left[ax^2 + \left(\frac{1}{bx}\right)\right]^{11}$ के प्रसार में x^7 का गुणांक $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$ के प्रसार में x^{-7} के गुणांक के बराबर है, तो a और b में सम्बन्ध है :

(A) $a - b = 1$ (B) $a + b = 1$ (C) $\frac{a}{b} = 1$ (D) $ab = 1$

Ans. D

Sol. In the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$, the general term is

$$T_{r+1} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{bx}\right)^r = {}^{11}C_r a^{11-r} \frac{1}{b^r} x^{22-3r}$$

For x^7, we must have $22 - 3r = 7 \Rightarrow r = 5$, and the coefficient of $x^7 = {}^{11}C_5 . a^{11-5} \frac{1}{b^5} = {}^{11}C_5 \frac{a^6}{b^5}$.

Similary, in the expansion of $\left(ax^2 - \frac{1}{bx}\right)^{11}$, the general term is $T_{r+1} = {}^{11}C_r (-1)^r \frac{a^{11-r}}{b^r} x^{11-3r}$

For x^{-7} we must have $11 - 3r = -7 \Rightarrow r = 6$, and the coefficient of x^{-7} is ${}^{11}C_6 \frac{a^5}{b^6} = {}^{11}C_5 \frac{a^5}{b^6}$

9. Let x be the 7th term from the beginning and y be the 7th term from the end in the expansion of $\left(3^{1/3}+\frac{1}{4^{1/3}}\right)^n$. If y = 12x then the value of n is :

माना $\left(3^{1/3}+\frac{1}{4^{1/3}}\right)^n$ के प्रसार में शुरू से 7वाँ पद x तथा अंत से 7वाँ पद y है। यदि y = 12x है, तो n बराबर होगा:

(A) 9 (B) 8 (C) 10 (D) 11

Ans. A

Sol. $\left(3^{\frac{1}{3}}+\frac{1}{4^{\frac{1}{3}}}\right)^n=\sum_{r=0}^{n}{}^nC_r\,3^{\frac{n-r}{3}}\cdot4^{\frac{-r}{3}}$

$x={}^nC_6\,3^{\frac{n-6}{3}}\cdot4^{\frac{-6}{3}}$ & $y={}^nC_{n-6}\,3^{\frac{n-(n-6)}{3}}\cdot4^{-\frac{(n-6)}{3}}$

Given y = 12x

$\Rightarrow {}^nC_6\,3^2\cdot4^{-\frac{(n-6)}{3}}={}^nC_6\cdot3^{\frac{n-6}{3}}\cdot4^{-2}\cdot4\cdot3$

$\Rightarrow 4^{-\frac{(n-6)}{3}+2}=3^{\frac{n-6}{3}-2}\cdot4\cdot3$

$\Rightarrow 4^{\frac{-n+6+6}{3}}=3^{\frac{n-12}{3}}\cdot4\cdot3$

$\Rightarrow 4^{\frac{-n+12-3}{3}}=3^{\frac{n-12+3}{3}}$

This is possible if n = 9

10. In the binomial expansion of $(a - b)^n$, $n \geq 5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals :

$(a - b)^n$, $n \geq 5$ के द्विपद प्रसार में, 5वें तथा 6वें पदों का योग 0 है, तब $\frac{a}{b}$ बराबर है:

(A) $\frac{5}{n-4}$ (B) $\frac{6}{n-5}$ (C) $\frac{n-5}{6}$ (D) $\frac{n-4}{5}$

Ans. D

Sol. $(a-b)^n \Rightarrow$ general term $T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r(-1)^r$

${}^nC_4 \cdot a^{n-4} \cdot b^4 - {}^nC_5 \cdot a^{n-5} \cdot b^5 = 0$

$${}^nC_4\left(\frac{a}{b}\right) = {}^nC_5 \Rightarrow \frac{n!}{4!(n-4)!}\left(\frac{a}{b}\right) = \frac{n!}{5!(n-5)!}$$

$$\left(\frac{a}{b}\right) = \frac{(n-4)}{5}$$

BASED ON FINDING COEFFICIENTS

11. If $(1 + ax)^n = 1 + 8x + 24x^2 + \ldots\ldots\ldots$then the value of a and n is :

यदि $(1 + ax)^n = 1 + 8x + 24x^2 + \ldots$ तब a तथा n का मान होगा:

(A) 2, 4 (B) 2, 3 (C) 3, 6 (D) 1, 2

Ans. A

Sol. $(1 + ax)^n = 1 + 8x + 24x^2 + \ldots\ldots$

$\Rightarrow 1 + {}^nC_1 (ax) + {}^nC_2(ax)^2 + \ldots. = 1 + 8x + 24x^2 + \ldots..$

$$\Rightarrow na = 8 \;\&\; \frac{\lfloor n}{\lfloor n-2 \rfloor 2} a^2 = 24$$

$\Rightarrow na = 8 \;\&\; n(n-1)a^2 = 48$

$\Rightarrow na = 8 \;\&\; na\,(na - a) = 48$

$8\,(8 - a) = 48$

$\Rightarrow 8 - a = 6$

$\Rightarrow a = 2$

$\Rightarrow n = 4$

12. The coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^n$ is :

$(1 + x + x^2 + x^3)^n$ के प्रसार में x^4 को गुणांक होगा :

(A) ${}^nC_4$ (B) ${}^nC_4 + {}^nC_2$ (C) ${}^nC_4 + {}^nC_2 + {}^nC_4 \cdot {}^nC_2$ (D) ${}^nC_4 + {}^nC_2 + {}^nC_1 \cdot {}^nC_2$

Ans. D

Sol. $(1 + x + x^2 + x^3)^n = ((1 + x^2)\,(1 + x))^n$

$$=\left(1+x^2\right)^n\left(1+x\right)^n=\left[\sum_{r=0}^{n}{}^nC_r x^{2r}\right]\left[\sum_{r=0}^{n}{}^nC_r x^r\right]$$

Coefficient of x^4

$= {}^nC_0\ {}^nC_4 + {}^nC_1\ {}^nC_2 + {}^nC_2 \cdot {}^nC_0$

$= {}^nC_4 + {}^nC_2 + {}^nC_1 \cdot {}^nC_2$

13. The co-efficient of x^9 in the polynomial given by $(x+1)(x+2)\ldots\ldots(x+10) + (x+2)(x+3)\ldots\ldots(x+11) + \ldots\ldots\ldots + (x+11)(x+12)\ldots\ldots(x+20)$ is :

व्यंजक $(x+1)(x+2)\ldots\ldots(x+10) + (x+2)(x+3)\ldots\ldots(x+11) + \ldots\ldots\ldots + (x+11)(x+12)$

ण्ण्ण्ण्ण्ण्; ग 20द्ध में x^9 का गुणांक है:

(A) 5511 (B) 5151 (C) 1515 (D) 1155

Ans. D

Sol. Coefficient of x^9 in $(x+1)(x+2)\ldots\ldots(x+10) + (x+2)\ldots\ldots(x+11) + \ldots\ldots\ldots + (x+11)(x+12)\ldots\ldots(x+20)$

Let $P(x) = (x+1)(x+2)\ldots\ldots(x+10)$

$\Rightarrow x = -1, \ldots\ldots, -10$ are roots of $P(x)$

Coefficient of $x^9 = -(-1-2\ldots\ldots-10) = 1+\ldots\ldots+10$

$$=\frac{10\times11}{2}=55$$

Coefficient of x^9

$= 55 + 65 + 75 + 85 + 95 + 105 + 115 + 125 + 135 + 145 + 155 \rightarrow$ A.P. with $d = 10$

$$=\frac{11}{2}\left[110+10\times10\right]=\frac{210}{2}\times11$$

$= 105 \times 11 = 1155$

14. The coefficient of x^{39} in the expansion of $\left(x^4-\frac{1}{x^3}\right)^{15}$ is :

$\left(x^4-\frac{1}{x^3}\right)^{15}$ के प्रसार में x^{39} का गुणांक है:

(A) – 455 (B) – 105 (C) + 455 (D) + 105

Ans. A

Sol. $\left(x^4-\frac{1}{x^3}\right)^{15}=\sum_{r=0}^{15}{}^{15}C_r\left(x^4\right)^{15-r}\cdot\left(\frac{-1}{x^3}\right)^r$

$=\sum_{r=0}^{15}{}^{15}C_r(-1)^r x^{60-7r}$

For x^{39}; $60-7r=39$

$\Rightarrow 7r=60-39$

$\Rightarrow r=3$

Coefficient of x^{39} is $(-1)^3\cdot{}^{15}C_3$

$=-455$

15. The coefficient of x^n in expansion of $(1+x)(1-x)^n$ is :

$(1+x)(1-x)^n$ के प्रसार में x^n का गुणांक होगा :

(A) $(n-1)$ (B) $(-1)^n(1-n)$ (C) $(-1)^{n-1}(n-1)^2$ (D) $(-1)^{n-1}n$

Ans. B

Sol. Coefficient of x^n in expansion of $(1+x)(1-x)^n$ i.e., coefficient of x^n in expansion of $(1-x)^n$ + coefficient of x^{n-1} in expansion of $(1-x)^n$]

Now, $(-1)^n\,{}^nC_n+(-1)^{n-1}\,{}^nC_{n-1}$

$(-1)^n[{}^nC_n-{}^nC_{n-1}]=(-1)^n[1-n]$

16. For natural numbers m, n if $(1-y)^m(1+y)^n=1+a_1y+a_2y^2+....$, and $a_1=a_2=10$, then (m, n) is :

प्राकृत संख्याओं m, n के लिए यदि $(1-y)^m(1+y)^n=1+a_1y+a_2y^2+....$, तथा $a_1=a_2=10$, तो (m, n) है :

(A) (35, 20) (B) (45, 35) (C) (35, 45) (D) (20, 45)

Ans. C

Sol. $(1-y)^m(1+y)^n=1+a_1y+a_2y^2+.....$

$(1-my+{}^mC_2y^2.....)(1+ny+{}^nC_2y^2.....)=1+a_1y+a_2y^2+.....$

$a_1=n-m=10$ ..... (1)

$a_2={}^mC_2+{}^nC_2-mn=10$ ..... (2)

solving (1) & (2), we get $(m, n)\equiv(35, 45)$

17. The coefficient of x^7 in the expansion of $(1-x-x^2+x^3)^6$ is : **[AIEEE 2011]**

$(1-x-x^2+x^3)^6$ के प्रसार में x^7 का गुणांक है :

(A) – 144 (B) 132 (C) 144 (D) –132

Ans. A

Sol. $(1-x-x^2+x^3)^6 \Rightarrow [(1-x)-x^2(1-x)]$

$\Rightarrow (1-x)^6(1-x^2)^6$

$\Rightarrow (1-{}^6C_1\,x+{}^6C_2x^2-{}^6C_3x^3+{}^6C_4\,x^4-{}^6C_5+{}^6C_6\,x^6)\times(1-{}^6C_1x^2+{}^6C_2x^4-{}^6C_3x^6+....)$

$\Rightarrow$ cofficient of x^7 is $\rightarrow$

${}^6C_1.\,{}^6C_3-{}^6C_3.\,{}^6C_2+{}^6C_5\,.\,{}^6C_1$

$\Rightarrow \frac{6.5.4}{3.2}-\frac{6.5.4}{3.2.1}+6.6=120-300+36$

$\Rightarrow -144$

18. If the coffcicnet of x^3 and x^4 in the expansion of $(1+ax+bx^2)\ (1-2x)^{18}$ in powers of x are both zero then (a, b) is equal to : **[JEE(Main) 2014]**

यदि $(1+ax+bx^2)\ (1-2x)^{18}$ के x की घातों में प्रसार में x^3 तथा x^4 दोनों के गुणांक शून्य हैं, तो (a, b) बराबर है :

(A) $\left(14,\frac{272}{3}\right)$ (B) $\left(16,\frac{272}{3}\right)$ (C) $\left(16,\frac{251}{3}\right)$ (D) $\left(14,\frac{251}{3}\right)$

Ans. B

Sol. $(1+ax+bx^2)(1-2x)^{18}$

coeff of $x^3={}^{18}C_3(-2)^3+a.(-2)^2\,.\,{}^{18}C_2+b.(-2).\,{}^{18}C_1=0$

coeff of $x^4\equiv{}^{18}C_4(-2)^4+a(-2)^3\,.\,{}^{18}C_3+b(-2)^2\,.\,{}^{18}C_2=0$

$\Rightarrow$ $51a-3b=544$ and $32a-3b=240$

Subtracting we get $a=16$

$\Rightarrow$ $b=\frac{272}{3}$

REMAINDER BASED PROBLEMS

19. The number 7^{1995} when divided by 100 leaves the remainder :

(A) 43 (B) 45 (C) 50 (D) None of these

जब संख्या 7^{1995} को 100 से विभाजित किया जाता है तब शेषफल होगा

(A) 43 (B) 45 (C) 50 ;क द्धइनमें से कोई नहीं

Ans. A

Sol. $7^{1995} = 7[(7)^2]^{997} = 7(50-1)^{997}$

using expansion

$7\{^{997}C_0.50^{997} + {}^{997}C_1.50^{996} + ...+^{997}C_{996}.50-1\}$

$= 100K + 7(997 \times 50 - 1)$

$= 100K + 7(900 \times 50 + 7 \times 50 - 1)$

$= 100K_1 + 49 \times 50 - 7 = 100K_1 + 2450 - 7$

$= 100K_1 + 2400 + 43$

$\therefore$ Remainder is 43

20. The last digit of (3^P+2), where $P = 3^{4n+2}$ is :

(3^P+2) का अन्तिम अंक क्या होगा जबकि $P = 3^{4n+2}$:

(A) 1 (B) 9 (C) 4 (D) 5

Ans. D

Sol. $(3^P + 2)$; where $P = 3^{4n+2}$

$P = 3^{4n}\cdot 9 = 81^n\cdot 9$

$P = 9\cdot(80+1)^n$

$$= 9\sum_{r=0}^{n} {}^nC_r (80)^{n-r}$$

$P = 10\lambda + 9\cdot{}^nC_n$

$\Rightarrow$ Remainder = 9 last digit of $\left(3^{3^{4n+2}} + 2\right)$ = last digit of $3^9 + 2$

$= 9^4\cdot 3 + 2$

$= (10-1)^4\cdot 3 + 2$

$= 3 + 2 + 10\lambda'$

\ Remainder = 5

21. The remainder left out when $8^{2n}-(62)^{2n+1}$ is divided by 9 is : **[AIEEE-2009]**

$8^{2n}-(62)^{2n+1}$ को 9 से भाग देने पर शेषफल बचता है :

(A) 0 (B) 2 (C) 7 (D) 8

Ans. B

Sol. $8^{2n} - (62)^{2n+1}$

$= (9-1)^{2n} - (63-1)^{2n+1}$

$= \sum_{r=0}^{2n} {}^{2n}C_r 9^{2n-r}\cdot(-1)^r - \sum_{r=0}^{2n+1} {}^{2n+1}C_0 (63)^{2n+1-r}(-1)^r$

$= 9\lambda + 1 + 1$

$= 9\lambda + 2$

$\therefore$ Remainder $= 2$

22. If $X = \{4^n - 3n - 1 : n \in N\}$ and $Y = \{9(n-1) : n \in N\}$, where N is the set of natural numbers, then

$X \cup Y$ is equal to : **[JEE(Main) 2014]**

यदि $X = \{4^n - 3n - 1 : n \in N\}$ तथा $Y = \{9(n-1) : n \in N\}$, हैं, जहाँ N प्राकृत संख्याओं का समुच्चय है, तो

$X \cup Y$ बराबर है :

(A) X (B) Y (C) N (D) $Y - X$

Ans. B

Sol. $4^n - 3n - 1$

$= (1+3)^n - 3n - 1$

$= {}^nC_2 3^2 + {}^nC_3 3^3 + ...$

$= 9[{}^nC_2 + 3 \times {}^nC_3 + ...]$

Set X contains numbers which are multiples of 9 but not all multiples of 9

$\therefore X \cup Y = Y$

NUMERICALLY GREATEST TERM

23. Only 4th term is numerically greatest term in $\left(2+\frac{3x}{8}\right)^{10}$. Then the range of x is :

$\left(2+\frac{3x}{8}\right)^{10}$ के प्रसार में केवल 4th पद महत्तम पद हो तो x का परिसर है :

(A) $\left(\frac{-64}{21},-2\right)\cup\left(1,\frac{64}{25}\right)$ (B) $\left(\frac{-64}{21},-2\right)\cup\left(2,\frac{64}{21}\right)$

(C) $\left(-2,\frac{-64}{21}\right)\cup\left(2,\frac{64}{21}\right)$ (D) $\left(\frac{-64}{21},-2\right)\cup\left(1,\frac{64}{21}\right)$

Ans. B

Sol. $\left(2+\frac{3x}{8}\right)^{10}$

T_4 is only NGT

$\Rightarrow [P] = 3 \;\&\; P \notin Z$

$\Rightarrow 3 < P < 4$

$\Rightarrow 3 < \frac{11}{1+\left|\frac{16}{3x}\right|} < 4$

$\Rightarrow 3\left(1+\left|\frac{16}{3x}\right|\right)-11<0$ & $11<4\left(1+\left|\frac{16}{3x}\right|\right)$

$\Rightarrow \frac{16}{|x|}<8$ $\quad 7<\frac{64}{3|x|}$

$\Rightarrow |x|>2\{\because |x|>0\}$ $\quad \Rightarrow |x|<\frac{64}{21}$

$\Rightarrow x\in(-\infty,-2)\cup(2,\infty)$ $\quad x\in\left(-\frac{64}{21},0\right)\cup\left(0,\frac{64}{21}\right)$

$\Rightarrow x\in\left(-\frac{64}{21},-2\right)\cup\left(2,\frac{64}{21}\right)$

INTEGRAL PART AND FRACTIONAL PART BASED CONCEPT

24. If n is a positive integer, then $\left(\sqrt{3}+1\right)^{2n}-\left(\sqrt{3}-1\right)^{2n}$ is : **[AIEEE-2012]**

(A) A rational number other than positive integers

(B) An irrational number

(C) An odd positive integer

(D) An even positive integer

यदि n एक धनपूर्णांक है, तो $\left(\sqrt{3}+1\right)^{2n}-\left(\sqrt{3}-1\right)^{2n}$: **[AIEEE-2012]**

(A) धनपूर्णांकों को छोड़कर एक परिमेय संख्या हैं (B) एक अपरिमेय संख्या है

(C) एक विषम धनपूर्णांक है (D) एक सम धनपूर्णांक है।

Ans. B

Sol. $\left(\sqrt{3}+1\right)^{2n}-\left(\sqrt{3}-1\right)^{2n}$

$= 2[T_2 + T_2 + T_6 + ... + T_{2n}]$

$= 2[{}^{2n}C_1(\sqrt{3})^{2n-1} + {}^{2n}C_3(\sqrt{3})^{2n-3} + ...]$

= An Irrational Number

SUMMATION OF SERIES

25. $\sum_{r=1}^{n} r.\ {}^{2n}C_r$ is equal to :

$\sum_{r=1}^{n} r.\ {}^{2n}C_r$ का मान है :

(A) $n.2^{n-1}$ (B) 2^{2n-1} (C) $2^{n-1}+1$ (D) $n.2^{2n-1}$

Ans. D

Sol. $\sum_{r=1}^{n} r\cdot{}^{2n}C_r = \sum_{r=1}^{n} r\cdot\frac{2n}{r}\cdot{}^{2n-1}C_{r-1}$

$= 2n\sum_{r=1}^{n} {}^{2n-1}C_{r-1}$

$= 2n\cdot\left({}^{2n-1}C_0 + {}^{2n-1}C_1 + ... + {}^{2n-1}C_{n-1}\right)$

$= 2\cdot n$

$= 2n\cdot\frac{2^{2n-1}}{2} = n\cdot 2^{2n-1}$

26. If $C_0, C_1, C_2, ...C_n$ denote the coefficients in the expansion of $(1 + x)^n$, then the value of $\sum_{r=0}^{n}(r+1)C_r$ is :

यदि $(1 + x)^n$ के प्रसार में $C_0, C_1, C_2, ...C_n$ द्विपद गुणांक है तब $\sum_{r=0}^{n}(r+1)C_r$ है :

(A) $n.2^n$ (B) $(n+1)2^{n-1}$ (C) $(n+2)2^{n-1}$ (D) $(n+2)2^{n-2}$

Ans. C

Sol. $\sum_{r=0}^{n}(r+1)C_r = \sum_{r=0}^{n} r\cdot C_r + \sum_{r=0}^{n} C_r$

$= n\sum_{r=1}^{n} {}^{n-1}C_{r-1} + 2^n$

$= n\cdot 2^{n-1} + 2^n$

$= 2^{n-1}(n+2)$

27. If $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \ldots\ldots\ldots + C_n x^n$, then $C_0 + 5C_1 + 9C_2 + \ldots\ldots\ldots + (4n+1)C_n$ is equal to :

यदि $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \ldots\ldots\ldots + C_n x^n$, तब $C_0 + 5C_1 + 9C_2 + \ldots\ldots\ldots + (4n+1)C_n$ बराबर होगा :

(A) $n.2^n$ (B) $(n+1)2^n$ (C) $(2n+1)2^n$ (D) $(4n+1)2^n$

Ans. C

Sol. $C_0 + {}^5C_1 + {}^9C_2 + \ldots\ldots + (4n+1)\,C_n$

$= \sum_{r=0}^{n}(4r+1)^n\, C_r$

$= 4\sum_{r=0}^{n} r\cdot {}^nC_r + \sum_{r=0}^{n} {}^nC_r$

$= 4\sum_{r=1}^{n} r\cdot\frac{n}{r}\,{}^{n-1}C_{r-1} + \sum_{r=0}^{n} {}^nC_r$

$= 4n\cdot 2^{n-1} + 2^n$

$= (4n+2)2^{n-1}$

$= (2n+1)\,2^n$

28. If $(1 - x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \ldots\ldots\ldots + a_{2n} x^{2n}$, then $a_0 + a_2 + a_4 + \ldots\ldots + a_{2n}$ equals :

यदि $(1 - x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \ldots\ldots\ldots + a_{2n} x^{2n}$, तब $a_0 + a_2 + a_4 + \ldots\ldots + a_{2n}$ बराबर होगा:

(A) $\frac{3^n+1}{2}$ (B) $\frac{3^n-1}{2}$ (C) $\frac{1-3^n}{2}$ (D) $3^n + \frac{1}{2}$

Ans. A

Sol. $(1 - x + x^2)^n = a_0 + a_1x + ... + a_{2n}x^{2n}$

Put $x = 1$

$1 = a_0 + a_1 + ... + a_{2n}$ ...(i)

Put $x = -1$

$3^n = a_0 - a_1 + ... + a_{2n}$ ...(ii)

(i) + (ii)

$2[a_0 + a_2 + ... + a_{2n}] = 3^n + 1$

$$\Rightarrow a_0 + a_2 + ... + a_{2n} = \frac{3^n + 1}{2}$$

29. If $(1+ x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, then $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + ... + \frac{nC_n}{C_{n-1}} =$

यदि $(1+ x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, तो $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + ... + \frac{nC_n}{C_{n-1}} =$

(A) $\frac{n}{2}$ (B) $n(n+1)$ (C) $\frac{n(n+1)}{12}$ (D) $\frac{n(n+1)}{2}$

Ans. D

Sol. $$\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + ... + \frac{n \cdot C_n}{C_{n-1}}$$

$$= \sum_{r=1}^{n} r \cdot \frac{C_r}{C_{r-1}}$$

$$\left\{ \because \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} \right\}$$

$$= \sum_{r=1}^{n} r \cdot \left(\frac{n-r+1}{r} \right) = \sum_{r=1}^{n} (n-r+1)$$

$$= (n+1)\sum_{r=1}^{n} 1 - \sum_{r=1}^{n} r$$

$$= n(n+1) - \frac{n(n+1)}{2}$$

$= \frac{n(n+1)}{2}$

30. If $S_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$ and $t_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r}$, then $\frac{t_n}{s_n}$ is equal to :

यदि $S_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$ एवं $t_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r}$, तो $\frac{t_n}{s_n}$ बराबर है :

(A) $\frac{1}{2}n$ (B) $\frac{1}{2}n-1$ (C) n–1 (D) $\frac{2n-1}{2}$

Ans. A

Sol. $s_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$

$t_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r}$

$t_n = \sum_{r=0}^{n} \frac{n-r}{{}^nC_r}$ $\therefore$ $2t_n = n\sum_{r=0}^{n} \frac{1}{{}^nC_r} = ns_n$ $\Rightarrow$ $\frac{t_n}{s_n} = \frac{n}{2}$

31. The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots\ldots - \ldots.. + {}^{20}C_{10}$ is:
श्रेणी, ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots\ldots - \ldots.. + {}^{20}C_{10}$ का योग है :

(A) $-{}^{20}C_{10}$ (B) $\frac{1}{2}{}^{20}C_{10}$ (C) 0 (D) ${}^{20}C_{10}$

Ans. B

Sol. $s = {}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 \ldots\ldots + {}^{20}C_{10}$
$s = {}^{20}C_{10} - {}^{20}C_{11} + \ldots\ldots\ldots + {}^{20}C_{20}$
$2s = 0 + {}^{20}C_{10}$

$s = \frac{1}{2}({}^{20}C_{10})$

32. The sum of coefficients of integral powers of x in the binomal expansion of $\left(1-2\sqrt{x}\right)^{50}$ is :

$\left(1-2\sqrt{x}\right)^{50}$ के द्विपद प्रसार में x को पूर्णांकीय घातों के गुणांको का योग है : **[JEE(Main) 2015]**

(A) $\frac{1}{2}(3^{50}+1)$ (B) $\frac{1}{2}(3^{50})$ (C) $\frac{1}{2}(3^{50}-1)$ (D) $\frac{1}{2}(2^{50}+1)$

Ans. A

Sol. $(1-2\sqrt{x})^{50} = 1 - {}^{50}C_1(2\sqrt{x}) + {}^{50}C_2(x^2)(\sqrt{x})^2 + ...$

Put $\sqrt{x} = 1$ & $\sqrt{x} = -1$ & then add

Ans, $\frac{3^{50}+1}{2}$

33. The value of $({}^{21}C_1 - {}^{10}C_1) + ({}^{21}C_2 - {}^{10}C_2) + ({}^{21}C_3 - {}^{10}C_3) + ({}^{21}C_4 - {}^{10}C_4) + ... + ({}^{21}C_{10} - {}^{10}C_{10})$ is :

; ${}^{21}C_1 - {}^{10}C_1) + ({}^{21}C_2 - {}^{10}C_2) + ({}^{21}C_3 - {}^{10}C_3) + ({}^{21}C_4 - {}^{10}C_4) + ... + ({}^{21}C_{10} - {}^{10}C_{10})$ का मान है –

[JEE(Main) 2017]

(A) $2^{20} - 2^9$ (B) $2^{20} - 2^{10}$ (C) $2^{21} - 2^{11}$ (D) $2^{21} - 2^{10}$

Ans. B

Sol. $\left({}^{21}C_0 + {}^{21}C_1 + {}^{21}C_2 + ... + {}^{21}C_{10}\right) - \left({}^{10}C_0 + {}^{10}C_1 + ... + {}^{10}C_{10}\right) - \left({}^{21}C_0 - {}^{10}C_0\right)$

$\frac{2^{21}}{2} - 2^{10} = 2^{20} - 2^{10}$

34. The sum of the co-efficients of all odd degree terms in the expansion of $\left(x+\sqrt{x^3-1}\right)^5 + \left(x-\sqrt{x^3-1}\right)^5$,

$(x > 1)$ is : **[JEE(Main) 2018]**

$\left(x+\sqrt{x^3-1}\right)^5 + \left(x-\sqrt{x^3-1}\right)^5$, $(x > 1)$ के प्रसार में सभी विषम घातों वाले पदों के गुणांको का योग है –

(A) 1 (B) 2 (C) –1 (D) 0

Ans. B

Sol. $f(x) = \left(x+\sqrt{x^3-1}\right)^5 + \left(x-\sqrt{x^3-1}\right)^5$

$= 2(x^5 + 10x^3(x^3-1) + 5x(x^3-1)^2)$

Sum of coefficients of odd degree $= \frac{f(1)-f(-1)}{2} = 2$

35. For r = 0, 1, ... 10, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r\left(B_{10}\ B_r - C_{10}\ A_r\right)$ is equal to : **[JEE 2010]**

$(1+x)^{10}$, $(1+x)^{20}$ और $(1+x)^{30}$ के प्रसार में माना x^r के गुणांक क्रमशः A_r, B_r और C_r है जहां r = 0, 1, ... 10, तब $\sum_{r=1}^{10} A_r \left(B_{10}\ B_r - C_{10}\ A_r\right)$ बराबरः

(A) $B_{10} - C_{10}$ (B) $A_{10}\left(B_{10}^2 - C_{10}\,A_{10}\right)$

(C) 0 (D) $C_{10} - B_{10}$

Ans. D

Sol. $A_r = {}^{10}C_r$, $B_r = {}^{20}C_r$, $C_r = {}^{30}C_r$

$$\sum_{r=1}^{10}\left({}^{20}C_{10}\ {}^{10}C_r\ {}^{20}C_r - {}^{30}C_{10}\left({}^{10}C_r\right)^2\right)$$

$$= {}^{20}C_{10}\left({}^{10}C_1\ {}^{20}C_1 + {}^{10}C_2\ {}^{20}C_2 + ... + {}^{10}C_{10}\ {}^{20}C_{10}\right)$$

$$-{}^{30}C_{10}\left({}^{10}C_1^2 + {}^{10}C_2^2 + ... + {}^{10}C_{10}^2\right)$$

$$= {}^{20}C_{10}\left({}^{30}C_{10} - 1\right) - {}^{30}C_{10}\left({}^{20}C_{10} - 1\right)$$

$$= {}^{30}C_{10} - {}^{20}C_{10} = C_{10} - B_{10}$$

BINOMIAL THEOREM FOR ANY INDEX

36. Number of terms in the expansion of $(1-4x)^{-30}$ are :

(A) 29 (B) 30 (C) 31 (D) infinitely many

$(1-4x)^{-30}$ के प्रसार में पदों की संख्या होगी :

(A) 29 (B) 30 (C) 31 ;क द्धअनन्त

Ans. D

Sol. No. of terms in the expansion of $(1-4x)^{-30}$ are infinite.

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON IMPORTANT TERMS

1. (a)If the coefficients of $(2r+4)^{th}$, $(r-2)^{th}$ terms in the expansion of $(1+x)^{18}$ are equal, find r.

(b) If the coefficients of the rth, $(r+1)^{th}$ & $(r+2)^{th}$ terms in the expansion of $(1+x)^{14}$ are in AP, find r.

(c) If the coefficients of 2nd, 3rd & 4th terms in the expansion of $(1+x)^{2n}$ are in AP, show that $2n^2 - 9n + 7 = 0$.

(a) यदि $(1+x)^{18}$ के प्रसार में $(2r+4)^{th}$, $(r-2)^{th}$ पदों के गुणांक समान है तो r का मान ज्ञात कीजिए।

(b) यदि $(1+x)^{14}$ के प्रसार में r^{th}, $(r+1)^{th}$ तथा $(r+2)^{th}$ पद समान्तर श्रेणी में है तो r का मान ज्ञात कीजिए।

(c) $(1+x)^{2n}$ के प्रसार में 2nd, 3rd तथा 4th पद समान्तर श्रेणी में है तो सिद्ध कीजिए $2n^2 - 9n + 7 = 0$.

Ans. (a) $r = 6$ (b) $r = 5$ or 9

Sol. (a) Coefficient of T_{2r+4} term = Coefficient of T_{r-2} in $(1+x)^{18}$

$\Rightarrow {}^{18}C_{2r+3} = {}^{18}C_{r-3}$

$\Rightarrow 2r + 3 = r - 3$ or $2r + r + 3 - 3 = 18$

$\Rightarrow r = 6$ or $r = 6$

$\Rightarrow r = 6$

(b) $T_r, T_{r+1}, T_{r+2} \rightarrow$ A.P. $(1+x)^{14}$

$\Rightarrow {}^{14}C_{r-1}, {}^{14}C_r, {}^{14}C_{r+1} \rightarrow$ A.P.

$2 \cdot {}^{14}C_r = {}^{14}C_{r-1} + {}^{14}C_{r+1}$

$\Rightarrow 4 \cdot {}^{14}C_r = {}^{14}C_r + {}^{14}C_{r-1} + {}^{14}C_{r+1} + {}^{14}C_r$

$= {}^{15}C_r + {}^{15}C_{r+1}$

$4 \cdot {}^{14}C_r = {}^{16}C_{r+1}$

$$\Rightarrow 4 \cdot \frac{\lfloor 14}{\lfloor 14-r \rfloor r} = \frac{\lfloor 16}{\lfloor 15-r \rfloor r+1} \Rightarrow 4 = \frac{16 \times 15}{(15-r)(r+1)}$$

$\Rightarrow (r+1)(r-15) = -60 \Rightarrow r^2 - 14r + 45 = 0 \Rightarrow r = 9 \text{ or } 5$

(c) $T_2, T_3, T_4 \rightarrow$ A.P. $(1+x)^{2n}$

$\Rightarrow 2 \cdot {}^{2n}C_2 = {}^{2n}C_1 + {}^{2n}C_3$

$\Rightarrow 4 \cdot {}^{2n}C_2 = {}^{2n}C_2 + {}^{2n}C_1 + {}^{2n}C_3 + {}^{2n}C_2$

$\Rightarrow \frac{4 \cdot \lfloor 2n}{\lfloor 2n-2 \rfloor 2} = {}^{2n+1}C_2 + {}^{2n+1}C_3 = {}^{2n+2}C_3$

$\Rightarrow \frac{4 \cdot \lfloor 2n}{\lfloor 2n-2 \rfloor 2} = \frac{(2n+2)(2n+1)\lfloor 2n}{(2n-1)\lfloor 2n-2 \rfloor 3}$

$\Rightarrow 12(2n-1) = 4n^2 + 6n + 2$

$\Rightarrow 4n^2 + 6n + 2 - 24n + 12 = 0$

$\Rightarrow 4n^2 - 18n + 14 = 0$

$\Rightarrow 2n^2 - 9n + 7 = 0$

2. (a) Find the term independent of x in the expansion of (i) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$ (ii) $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$

(b) Find the value of x for which the fourth term in the expansion, $\left(5^{\frac{2}{5}\log_5\sqrt{4^x+44}} + \frac{1}{5^{\log_5\sqrt[3]{2^{x-1}+7}}}\right)^8$ is 336.

(a) निम्न के प्रसार में x रहित पद ज्ञात कीजिए। **(i)** $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$ **(ii)** $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$

(b) x का मान ज्ञात कीजिए जिसके लिए $\left(5^{\frac{2}{5}\log_5\sqrt{4^x+44}} + \frac{1}{5^{\log_5\sqrt[3]{2^{x-1}+7}}}\right)^8$ के प्रसार में 4th पद 336 है।

Ans. (a) (i) $\frac{5}{12}$ (ii) $T_6 = 7$, (b) $x = 0$ or 1

Sol. (a) (i) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$

$$= \sum_{r=0}^{10} {}^{10}C_r \left(\frac{x}{3}\right)^{\frac{10-r}{2}} \left(\frac{\sqrt{3}}{2x^2}\right)^r$$

Term independent of x if

$$5 - \frac{r}{2} - 2r = 0 \Rightarrow 10 - 5r = 2$$

$$\Rightarrow r = 2$$

$$\text{Term} = {}^{10}C_2 \cdot \left(\frac{1}{3}\right)^4 \cdot \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{45}{27} \cdot \frac{1}{4} = \frac{5}{12}$$

(ii) $$\left[\frac{1}{2}x^{\frac{1}{3}} + x^{-\frac{1}{5}}\right]^8 = \sum_{r=0}^{8} {}^8C_r \left(\frac{1}{2}\right)^{8-r} (x)^{\frac{8-r}{3}} \cdot x^{-\frac{r}{5}}$$

Term independent of x if $\frac{8-r}{3} - \frac{r}{5} = 0$

$$\Rightarrow 40 - 5r - 3r = 0$$

$$\Rightarrow r = 5$$

$$\text{Term } T_6 = {}^8C_5 \left(\frac{1}{2}\right)^3 = \frac{\lfloor 8}{\lfloor 5 \lfloor 3} \cdot \frac{1}{8} = 7$$

(b) $$\left(5^{\frac{2}{5}\log_5 \sqrt{4^x+44}} + \frac{1}{5^{\log_5 \sqrt[3]{2^{x-1}+7}}}\right)^8$$

$$= \left(\left(4^x + 44\right)^{\frac{2}{5}\cdot 2} + \frac{1}{\left(2^{x-1}+7\right)^{\frac{1}{3}}}\right)^8$$

Given $T_4 = 336$

$$\Rightarrow {}^{8}C_3\left(4^x+44\right)^{\frac{8-3}{5}}\cdot\left(\frac{1}{\left(2^{x-1}+7\right)^{\frac{1}{3}}}\right)^3=336$$

$$\Rightarrow \frac{4^x+44}{2^{x-1}+7}=6$$

Let $2^x = t$

$\Rightarrow t^2 + 44 = 3t + 42$

$\Rightarrow t^2 - 3t + 2 = 0$

$\Rightarrow t = 1$ or $t = 2$

$\Rightarrow x = 0$ or $x = 1$

BASED ON FINDING COEFFICIENTS

3. Prove that the ratio of the coefficient of x^{10} in $(1-x^2)^{10}$ & the term independent of x in $\left(x-\frac{2}{x}\right)^{10}$ is 1 : 32.

सिद्ध कीजिए $(1-x^2)^{10}$ के प्रसार में x^{10} का गुणांक तथा $\left(x-\frac{2}{x}\right)^{10}$ के प्रसार में x रहित पद के गुणांक का अनुपात 1 : 32 है।

Ans.

Sol. $\left(1-x^2\right)^{10}=\sum_{r=0}^{10}{}^{10}C_r\left(-x^2\right)^r$

Coefficient of x^{10} is ${}^{10}C_5(-1)^5$

$$\left(x-\frac{2}{x}\right)^{10}=\sum_{r=0}^{10}{}^{10}C_r x^{10-r}\cdot(-2)^r\cdot x^{-r}$$

Term independent of x if $r = 5$

$\Rightarrow$ Coefficient of term independent of $x = {}^{10}C_5(-2)^5$

$$\Rightarrow \frac{{}^{10}C_5(-1)^5}{{}^{10}C_5(-2)^5}=\frac{1}{32}$$

4. Let a and b be the coefficient of x^3 in $(1+x+2x^2+3x^3)^4$ and $(1+x+2x^2+3x^3+4x^4)^4$ respectively. Find the value of $(a-b)$.

यदि $(1 + x + 2x^2 + 3x^3)^4$ तथा $(1 + x + 2x^2 + 3x^3 + 4x^4)^4$ के प्रसार में x^3 के गुणांक क्रमशः a तथा b है तो (a – b) का मान होगा ।

Ans. 0

Sol. Coefficient of x^3 in $(1 + x + 2x^2 + 3x^3)^4$

$$= \frac{\lfloor 4}{\lfloor P_1 \lfloor P_2 \lfloor P_3 \lfloor P_4} \cdot 1^{P_1} \cdot x^{P_2} \cdot 2^{P_3} \cdot x^{2P_3} \cdot 3^{P_4} \cdot x^{3P_4}$$

$$= \frac{\lfloor 4}{\lfloor P_1 \lfloor P_2 \lfloor P_3 \lfloor P_4} \cdot 1^{P_1} \cdot 2^{P_3} \cdot 3^{P_4} \cdot x^{P_2+2P_3+3P_4}$$

$P_1 + P_2 + P_3 + P_4 = 4$

$P_2 + 2P_3 + 3P_4 = 3$

$\Rightarrow a = 40$

$$(1 + x + 2x^2 + 3x^3 + 4x^4)^4 = \frac{\lfloor 4 \cdot 2^{P_3} \cdot 3^{P_4} \cdot 4^{P_5}}{\lfloor P_1 \lfloor P_2 \lfloor P_3 \lfloor P_4 \lfloor P_5} x^{P_2+2P_3+3P_4+4P_5}$$

For x^3

$P_1 + P_2 + P_3 + P_4 + P_5 = 4$

$P_2 + 2P_3 + 3P_4 + 4P_5 = 3$

$\Rightarrow b = 40$

$\Rightarrow b - a = 0$

5. Find the coefficient of x^r in the expression of

$(x + 3)^{n-1} + (x + 3)^{n-2}(x + 2) + (x + 3)^{n-3}(x + 2)^2 + ... + (x + 2)^{n-1}$.

$(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + ... + (x+2)^{n-1}$ में x^r का गुणांक ज्ञात कीजिए।

Ans. ${}^nC_r(3^{n-r} - 2^{n-r})$

Sol. $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + ... + (x+2)^{n-1}(x+3)^{n-1}$

G.P. with $r = \dfrac{(x+2)}{x+3}$

$$= (x+3)^{n-1}\dfrac{\left[\left[\dfrac{(x+2)}{x+3}\right]^n - 1\right]}{\dfrac{x+2}{x+3} - 1}$$

$$= (x+3)^{n-1}\cdot\dfrac{(x+3)}{-1}\cdot\left[\left(\dfrac{x+2}{x+3}\right)^n - 1\right]$$

$$= -(x+3)^n\left[\left(\dfrac{x+2}{x+3}\right)^n - 1\right]$$

$= (x+3)^n - (x+2)^n$

Coefficient of $x^r = {}^nC_r(3^{n-r} - 2^{n-r})$

6. Find the term independent of x in the expansion of $(1 + x + 2x^3)\left(\dfrac{3x^2}{2} - \dfrac{1}{3x}\right)^9$.

$(1 + x + 2x^3)\left(\dfrac{3x^2}{2} - \dfrac{1}{3x}\right)^9$ के प्रसार में x रहित पद ज्ञात कीजिए।

Ans. $\dfrac{17}{54}$

Sol. Co-efficient of x^0 in $(1 + x + 2x^3)\left(\dfrac{3}{2}x^2 - \dfrac{1}{3x}\right)^9$

$$= \text{Co-efficient of } x^0 \text{ in } \left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9 + \text{Co-efficient of } x^{-1} \text{ in } \left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$$
$$+ 2 \text{ Co-efficient of } x^{-3} \text{ in } \left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$$

$$= {}^9C_{r_1}\left(\frac{3}{2}x^2\right)^{9-r_1}\left(-\frac{1}{3x}\right)^{r_1} + {}^9C_{r_2}\left(\frac{3}{2}x^2\right)^{9-r_2}\left(-\frac{1}{3x}\right)^{r_2} + 2\ {}^9C_{r_3}\left(\frac{3}{2}x^2\right)^{9-r_3}\left(-\frac{1}{3x}\right)^{r_3}$$

$= {}^{9}C_6\left(\frac{3}{2}\right)^3\left(-\frac{1}{3}\right)^6 + 2\ {}^{9}C_7\left(\frac{3}{2}\right)^2\left(-\frac{1}{3}\right)^7 = \frac{17}{54}$

7. Let $(1+x^2)^2 . (1+x)^n = \sum_{K=0}^{n+4} a_K . x^K$. If a_1, a_2 & a_3 are in AP, find n.

माना $(1+x^2)^2 . (1+x)^n = \sum_{K=0}^{n+4} a_K . x^K$. यदि a_1, a_2 तथा a_3 समान्तर श्रेणी में है तो n का मान ज्ञात कीजिए।

Ans. n = 2 or 3 or 4

Sol. $\left(1+x^2\right)^2(1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$

$\Rightarrow \left(1+2x^2+x^4\right)(1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$

$\Rightarrow \left(1+2x^2+x^4\right)\sum_{r=0}^{n} {}^{n}C_r x^r = a_0 + a_1 x + ...$

$a_1 = {}^{n}C_1 = n$

$a_2 = {}^{n}C_2 + 2\cdot{}^{n}C_0 = \frac{n(n-1)}{2} + 2 = \frac{n^2-n+4}{2}$

$a_3 = {}^{n}C_3 + 2\cdot{}^{n}C_1 = \frac{n(n-1)(n-2)}{6} + 2\cdot n$

$\Rightarrow a_1 = n, a_2 = \frac{n^2-n+4}{2}, a_3 = \frac{n(n-1)(n-2)}{6} + 2n$

Given $2a_2 = a_1 + a_3$

$\Rightarrow n^2 - n + 4 = n + \frac{n(n-1)(n-2)}{6} + 2n$

$\Rightarrow n^2 - 2n + 4 = \frac{\left(n^2-n\right)(n-2)}{6} + 2n$

$\Rightarrow n^2 - 2n + 4 - 2n = \frac{n(n-1)(n-2)}{6}$

$\Rightarrow n(n-2) + (-2)(n-2) = \frac{n(n-1)(n-2)}{6}$

$\Rightarrow 6(n-2)^2 - (n-2)\,n\,(n-1) = 0$

$\Rightarrow (n-2)[6n - 12 - n^2 + n] = 0$

$(n-2)[7n - 12 - n^2] = 0$

$n = 2, 3, 4$

8. Find the sum of the roots (real or complex) of the equation $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0.$

समीकरण $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$ के सभी मूलों (वास्तविक या काल्पनिक) का योगफल ज्ञात कीजिए।

Ans. 500

Sol. $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$

$$\because \left(\frac{1}{2} - x\right)^{2001} = \sum_{r=0}^{2001} {}^{2001}C_r \left(\frac{1}{2}\right)^{2001-r} \cdot (-x)^r$$

$$= {}^{2001}C_0 \left(\frac{1}{2}\right)^{2001} + {}^{2001}C_1 \left(\frac{1}{2}\right)^{2000}(-x) + {}^{2001}C_2 \left(\frac{1}{2}\right)^{2002} \cdot (-x)^2 + \ldots + {}^{2001}C_{2000} \left(\frac{1}{2}\right)^{1} (-x)^{2000}$$

$$+ {}^{2001}C_{2001} (-x)^{2001}$$

$$\Rightarrow \sum_{r}^{2000} {}^{2001}C_r \left(\frac{1}{2}\right)^{2001-r} (-x)^r = 0$$

Sum of the roots of this equation

$$= -\frac{\text{Coefficient of } x^{1999}}{\text{coefficient of } x^{2000}}$$

$$= +\frac{\left(\frac{1}{2}\right)^2 \cdot {}^{2001}C_{1999}}{\left(\frac{1}{2}\right)^1 \cdot {}^{2001}C_{2000}}$$

$$= \frac{1}{4} \times \frac{2}{1} \times \frac{|2001|1|2000}{|2|1999|2001}$$

$$= \frac{1}{4} \times 2000 = 500$$

NUMERICALLY GREATEST TERM

9. Find numerically greatest term in the expansion of

(i) $(2 + 3x)^9$ when $x = \frac{3}{2}$ (ii) $(3 - 5x)^{15}$ when $x = \frac{1}{5}$

निम्न के प्रसार में संख्यात्मक महत्तम पद होगा

(i) $(2 + 3x)^9$ जब $x = \frac{3}{2}$ (ii) $(3 - 5x)^{15}$ जब $x = \frac{1}{5}$

Ans. (i) $T_7 = \frac{7.3^{13}}{2}$ (ii) 455×3^{12}

Sol. (i) $(2 + 3x)^9$; $x = \frac{3}{2}$

$$p = \frac{n+1}{1+\left|\frac{2}{3x}\right|} = \frac{10}{1+\left|\frac{2}{3\times\frac{3}{2}}\right|} = \frac{10}{1+\frac{4}{9}} = \frac{90}{13}$$

$p \notin Z$

$\Rightarrow$ NGT is $T_{[P]+1} = T_7 = {}^9C_6(2)^3 \cdot 3^6$

$$= \frac{9\times8\times7}{6} \times 8 \times 3^6 \times \left(\frac{3}{2}\right)^6$$

$$= \frac{3^{13}\cdot 7}{2}$$

(ii) $(3 - 5x)^{15}$ when $x = \frac{1}{5}$

$$P = \frac{15+1}{1+\left|\frac{3}{-5x}\right|} = \frac{16}{1+3} = 4 \in Z$$

$\Rightarrow T_4$ & $T_{4+1} = T_5$ are NGT

$$T_5 = {}^{15}C_4 (3)^{11} \cdot (-5x)^4 \Big|_{x=\frac{1}{5}}$$

$$= \frac{15\times14\times13\times12}{4\times3\times2}\times3^{11}$$

$$= 3^{12}\times455 = T_5$$

$$T_4 = {}^{15}C_3 3^{12}\cdot(-5x)^4$$

$$\Rightarrow T_4 = \frac{15\times14\times13}{3\times2}\times-3^{12}$$

$$\Rightarrow T_4 = -455\times3^{12}$$

$$\Rightarrow |T_4| = |T_5|$$

10. Find the index n of the binomial $\left(\frac{x}{5}+\frac{2}{5}\right)^n$ if the 9th term of the expansion has numerically the greatest coefficient $(n \in N)$.

n का मान ज्ञात कीजिए जिसके लिए $\left(\frac{x}{5}+\frac{2}{5}\right)^n$ के प्रसार में 9th पद संख्यात्मक महत्तम है $(n \in N)$.

Ans. $n = 12$

Sol. For T_9 to be the numerically greatest term, $r = \left[\frac{n+1}{1+\left|\frac{x}{a}\right|}\right] = \left[\frac{n+1}{1+\left|\frac{1}{2}\right|}\right] = 8$

$$\Rightarrow \quad 8 < \frac{2(n+1)}{3} < 9 \Rightarrow 11 < n < 12.5 \Rightarrow n = 12$$

11. For which positive values of x is the fourth term in the expansion of $(5 + 3x)^{10}$ is the greatest.

x के किस मान के लिए $(5 + 3x)^{10}$ के प्रसार में 4th पद संख्यात्मक महत्तम है।

Ans. $\frac{5}{8} < x < \frac{20}{21}$

Sol. 4th term is greatest in the expansion of $(5 + 3x)^{10}$

$$\Rightarrow [P] = 3$$

$$\Rightarrow 3 < P < 4$$

$$\Rightarrow 3 < \frac{10+1}{1+\left|\frac{5}{3x}\right|} < 4$$

$\Rightarrow 3+3\left|\frac{5}{3x}\right|<11$ & $11<4+4\left|\frac{5}{3x}\right|$

$\frac{15}{3|x|}<8$ & $\frac{7}{20}<\frac{1}{3|x|}$

$\Rightarrow 8|x|>5$ & $|x|<\frac{20}{21}$

$\Rightarrow |x|>\frac{5}{8}$ & $|x|<\frac{20}{21}$

$\Rightarrow x\in\left(-\infty,-\frac{5}{8}\right)\cup\left(\frac{5}{8},\infty\right)$ & $x\in\left(-\frac{20}{21},\frac{20}{21}\right)$

$\Rightarrow x\in\left(-\frac{20}{21},-\frac{5}{8}\right)\cup\left(\frac{5}{8},\frac{20}{21}\right)$

Given $x>0$

$\Rightarrow x\in\left(\frac{5}{8},\frac{20}{21}\right)$

MULTINOMIAL THEOREM

12. Find the coefficient of

(a) $x^2y^3z^4$ in the expansion of $(ax-by+cz)^9$.

(b) $a^2b^3c^4d$ in the expansion of $(a-b-c+d)^{10}$.

निम्न के गुणांक ज्ञात कीजिए।

(a) $(ax-by+cz)^9$ के प्रसार में $x^2y^3z^4$ का गुणांक

(b) $(a-b-c+d)^{10}$ के प्रसार में $a^2b^3c^4$ का गुणांक

Ans. (a) $-1260\cdot a^2b^3c^4$; (b) -12600

Sol. (a) $(ax-by+cz)^9=\sum_{P_1+P_2+P_3}\frac{\lfloor 9}{\lfloor P_1\lfloor P_2\lfloor P_3}a^{P_1}\cdot(-b)^{P_2}(+c)^{P_3}x^{P_1}y^{P_2}z^{P_3}$

For $x^2y^3z^4$

$P_1=2, P_2=3, P_3=4$

$\Rightarrow$ Coefficient $= -\frac{\underline{|9}}{\underline{|2}\underline{|3}\underline{|4}} a^2b^3c^4$

$= -\frac{9\times8\times7\times6\times5}{2\times6} a^2b^3c^4$

$= -63 \times 20\ a^2b^3c^4$

$= -1260\ a^2b^3c^4$

(b) $(a - b - c + d)^{10}$

$= \sum_{P_1+P_2+P_3+P_4=10} \frac{\underline{|10}\ aa^{P_1}(-b)^{P_2}(-c)^{P_3}d^{P_4}}{\underline{|P_1}\underline{|P_2}\underline{|P_3}\underline{|P_4}}$

For $a^2b^3c^4d$

$P_1 = 2, P_2 = 3, P_3 = 4, P_4 = 1$

Coefficient of $a^2b^3c^4d$

$= \frac{\underline{|10}}{\underline{|2}\underline{|3}\underline{|4}\underline{|1}} \cdot (-1)^3 \cdot (-1)^4$

$= -\frac{10\times9\times8\times7\times6\times5}{2\times6} = -20\times63\times10 = -12600$

13. Find the coefficient of

(a) x^4 in the expansion of $(1 + x + x^2 + x^3)^{11}$ (b) x^4 in the expansion of $(2 - x + 3x^2)^6$.

निम्न के गुणांक ज्ञात कीजिए।

(a) $(1 + x + x^2 + x^3)^{11}$ के प्रसार में x^4 का गुणांक

(b) $(2 - x + 3x^2)^6$ के प्रसार में x^4 का गुणांक

Ans. (a) 990 (b) 3660

Sol. $\left(1+x+x^2+x^3\right)^{11} = \sum_{P_1+P_2+P_3+P4} \frac{\underline{|11}\cdot x}{\underline{|P_1}\underline{|P_2}\underline{|P_3}\underline{|P_4}}$

For x^4

$P_4 + 2P_3 + 3P_4 = 4$

(b) $\left(2-x+3x^2\right)^6 = \sum_{r_1+r_2+r_3=6} \frac{\lfloor 6 2^{r_1}\cdot(-1)^{r_2}(3)^{r_3}}{\lfloor r_1 \lfloor r_2 \lfloor r_3} x^{r_2+2r_3}$

For x^4

$r_1 + r_2 + r_3 = 6$ & $r_2 + 2r_3 = 4$

INTEGRAL PART AND FRACTIONAL PART BASED CONCEPT

14. (a) Show that the integral part in each of the following is odd. $n \in N$

(A) $\left(5+2\sqrt{6}\right)^n$ (B) $\left(8+3\sqrt{7}\right)^n$

(b) Show that the integral part in each of the following is even. $n \in N$

(A) $\left(3\sqrt{3}+5\right)^{2n+1}$ (B) $\left(5\sqrt{5}+11\right)^{2n+1}$

(a) सिद्ध कीजिए प्रत्येक का पूर्णांक भाग विषम है जहां $n \in N$

(A) $\left(5+2\sqrt{6}\right)^n$ (B) $\left(8+3\sqrt{7}\right)^n$

(b) सिद्ध कीजिए प्रत्येक का पूर्णांक भाग सम है जहां $n \in N$

(A) $\left(3\sqrt{3}+5\right)^{2n+1}$ (B) $\left(5\sqrt{5}+11\right)^{2n+1}$

Ans.

Sol. (a) (A) $\left(5+2\sqrt{6}\right)^n = I + f; 0 < f < 1$

$\left(5-2\sqrt{6}\right)^n = 0 + f'; 0 < f' < 1$

$\Rightarrow I + f + f' = \left(5+2\sqrt{6}\right)^n + \left(5-2\sqrt{6}\right)^n$

$= 2\left[{}^nC_0 5^n + {}^nC_2 5^{n-2}\left(2\sqrt{6}\right)^2 + ...\right]$

$\Rightarrow I + f + f' =$ even Integer

$\because\ 0 < f < 1$ & $0 < f' < 1$

$\Rightarrow 0 < f + f' < 2$

Case-I : If I is even integer then $f + f'$ must be even integer.

but $0 < f + f' < 2 \rightarrow$ There is no even integer

Case-II : If I is odd integer then $f + f'$ must be odd integer

$\Rightarrow f + f' =$ as $0 < f + f' < 2$

$\therefore$ Integral part of $\left(5+2\sqrt{6}\right)^n$ is odd.

(B) $\left(8+3\sqrt{7}\right)^n = I + f; 0 < f < 1$

$\left(8-3\sqrt{7}\right)^n = 0 + f'; 0 < f' < 1$

$\Rightarrow \left(8+3\sqrt{7}\right)^n + \left(8-3\sqrt{7}\right)^n = 2\left[{}^nC_0 8^n + {}^nC_2 8^{n-2}\left(3\sqrt{7}\right)^2 + ...\right] = I + f + f'$

$\Rightarrow I + f + f' =$ even integer

$\because\ 0 < f < 1 \quad \& \quad 0 < f' < 1$

$\Rightarrow 0 + f + f' < 2 \Rightarrow f + f' = 1$

$\Rightarrow$ Integral part of $\left(8+3\sqrt{7}\right)^{n}$ is odd.

(b) (A) $\left(3\sqrt{3}+5\right)^{2n+1} = I + f; 0 < f < 1$

$\left(3\sqrt{3}-5\right)^{2n+1} = f'; 0 < f' < 1$

$\downarrow$

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$\Rightarrow I - f - f' = \left(3\sqrt{3}+5\right)^{2n+1} - \left(3\sqrt{3}-5\right)^{2n+1}$

$= 2\left[{}^{2n+1}C_1\left(3\sqrt{3}\right)^{2n}(5)^1 + ...\right]$

$\Rightarrow I + f - f' =$ even integer

$-1 < -f' < 0$

$-1 < f - f' < 1$

$\Rightarrow f - f' = 0$

$\Rightarrow I$ must be even

(B) $\left(5\sqrt{5}+11\right)^{2n+1} = I + f$

$\left(5\sqrt{5}-11\right)^{2n+1} = f'$

$\Rightarrow I + f - f' = \left(5\sqrt{5}+1\right)^{2n+1} - \left(5\sqrt{5}-1\right)^{2n+1}$

$= 2\left[{}^{2n+1}C_1\left(5\sqrt{5}\right)^{2n}\cdot 1 + ...\right]$

$\because -1 < f - f' < 0$

$\Rightarrow f - f' = 0$

$\Rightarrow I$ must be even

15. If $\left(7+4\sqrt{3}\right)^n$ = p+β where n & p are positive integers and β is a proper fraction show that $(1-\beta)(p+\beta)=1$.

यदि $\left(7+4\sqrt{3}\right)^n = p+\beta$ जहां n & p धनात्मक पूर्णांक है तथा β भिन्नात्मक भाग है तो सिद्ध कीजिए

$(1-\beta)(p+\beta)=1$.

Ans.

Sol. $\left(7+4\sqrt{3}\right)^n = p+\beta;\ n \in N \ \&\ p \in N \ \&\ \beta$ is a proper fraction.

$$\left(7-4\sqrt{3}\right)^n = \beta';\ 0<\beta'<1$$

$$\Rightarrow \beta' = \frac{(49-48)^n}{\left(7+4\sqrt{3}\right)^n} = \frac{1}{p+\beta}$$

$$\Rightarrow p+\beta+\beta' = \left(7+4\sqrt{3}\right)^n + \left(7-4\sqrt{3}\right)^n$$

$$= 2\left[{}^nC_0\,7^n + {}^nC_2\,7^{n-2}\left(4\sqrt{3}\right)^2 + \ldots.\right]$$

$\Rightarrow \beta + \beta' < 2$

$\Rightarrow \beta + \beta' = 1$

$\Rightarrow 1 - \beta = \beta'$

$$\Rightarrow 1-\beta = \frac{1}{p+\beta}$$

$$\Rightarrow (p+\beta)(1-\beta) = 1$$

16. If $\left(6\sqrt{6}+14\right)^{2n+1} = N \ \&\ F$ be the fractional part of N, prove that $NF = 20^{2n+1}$ $(n \in N)$

यदि $\left(6\sqrt{6}+14\right)^{2n+1} = N$ तथा F, N का भिन्नात्मक भाग है तो सिद्ध कीजिए $NF = 20^{2n+1}$ $(n \in N)$

Ans.

Sol. $\left(6\sqrt{6}+14\right)^{2n+1} = I + f = N;\ 0<f<1$

$\left(6\sqrt{6}-14\right)^{2n+1} = f'$

$\Rightarrow I + f - f' = 2\left({}^{2n+1}C_1\left(6\sqrt{6}\right)^{2n} + {}^{2n+1}C_3\left(6\sqrt{6}\right)^{2n-2} + ...\right)$

$\Rightarrow f - f' = 0$

$\Rightarrow f = f'$

$\Rightarrow Nf = Nf' = \left(6\sqrt{6}+14\right)^{2n+1}\cdot\left(6\sqrt{6}-14\right)^{2n+1}$

$= \left(216-196\right)^{2n+1}$

$= 20^{2n+1};\ n \in N$

17. Let I denotes the integral part and F the proper fractional part of $\left(3+\sqrt{5}\right)^n$ where $n \in N$ and if ρ denotes the rational part and σ the irrational part of the same, show that

$\rho = \frac{1}{2}(I+1)$ and $\sigma = \frac{1}{2}(I+2F-1)$.

यदि I तथा F, $\left(3+\sqrt{5}\right)^n$ के प्रसार में क्रमशः पूर्णांक तथा भिन्नात्मक भाग हैं जहां $n \in N$ तथा यदि ρ परिमेय भाग तथा σ अपरिमेय भाग है तो सिद्ध कीजिए

$\rho = \frac{1}{2}(I+1)$ and $\sigma = \frac{1}{2}(I+2F-1)$.

Ans.

Sol. $\left(3+\sqrt{5}\right)^n = I + F$

$\left(3-\sqrt{5}\right)^n = F'$

$\Rightarrow I + F + F' = 2\left[{}^nC_0 3^n + {}^nC_2 3^{n-2}\left(\sqrt{5}\right)^2 + ...\right]$

$\Rightarrow 0 < F + F' < 2$

$\Rightarrow F + F' = 1$

$\Rightarrow F' = 1 - F$

$\because (I+F)=\left(3+\sqrt{5}\right)^n=\left[{}^nC_0 3^n\left(\sqrt{5}\right)^0+{}^nC_1 3^{n-1}\left(\sqrt{5}\right)^1+{}^nC_2 3^{n-2}\left(\sqrt{5}\right)^2+....\right]$

$(F')=\left(3-\sqrt{5}\right)^n=\left[{}^nC_0 3^n\left(\sqrt{5}\right)^0-{}^nC_1 3^{n-1}\left(\sqrt{5}\right)^1+....\right]$

$\Rightarrow 2\rho=I+F+F'\Rightarrow \rho=\dfrac{I+1}{2}$

& $2\sigma=I+F-F'\Rightarrow \sigma=\dfrac{I+2F-1}{2}$

SUMMATION OF SERIES

18. Given $s_n=1+q+q^2+.....+q^n$ & $S_n=1+\dfrac{q+1}{2}+\left(\dfrac{q+1}{2}\right)^2+....+\left(\dfrac{q+1}{2}\right)^n$, $q\neq 1$, prove that ${}^{n+1}C_1+{}^{n+1}C_2.s_1+{}^{n+1}C_3.s_2+....+{}^{n+1}C_{n+1}.s_n=2^n\,.\,S_n$.

दिया गया है $s_n=1+q+q^2+.....+q^n$ और $S_n=1+\dfrac{q+1}{2}+\left(\dfrac{q+1}{2}\right)^2+....+\left(\dfrac{q+1}{2}\right)^n$, $q\neq 1$, तो सिद्ध कीजिए ${}^{n+1}C_1+{}^{n+1}C_2.s_1+{}^{n+1}C_3.s_2+....+{}^{n+1}C_{n+1}.s_n=2^n\,.\,S_n$.

Ans.

Sol. $s_n=\dfrac{q^{n+1}-1}{q-1}$ and $S_n=\dfrac{\left(\dfrac{q+1}{2}\right)^{n+1}-1}{\dfrac{q+1}{2}-1}$

${}^{n+1}C_1+{}^{n+1}C_2 s_1+.....+{}^{n+1}C_{n+1}\,s_n$

$={}^{n+1}C_1+{}^{n+1}C_2\left(\dfrac{q^2-1}{q-1}\right)+.....+{}^{n+1}C_{n+1}\,\dfrac{q^{n+1}-1}{q-1}$

$=\dfrac{1}{q-1}[{}^{n+1}C_1 q+{}^{n+1}C_2 q^2+....+{}^{n+1}C_{n+1}q^{n+1}-{}^{n+1}C_1-{}^{n+1}C_2-....-{}^{n+1}C_{n+1}]$

$=\dfrac{1}{q-1}[(1+q)^{n+1}-1-2^{n+1}+1]=\dfrac{1}{q-1}[(1+q)^{n+1}-2^{n+1}]=\left(\dfrac{\left(\dfrac{q+1}{2}\right)^{n+1}-1}{\dfrac{q-1}{2}}\right).2^n=2^n\,S_n$

19. Find the sum of the series $\sum_{r=0}^{n}(-1)^r.{}^nC_r\left[\dfrac{1}{2^r}+\dfrac{3^r}{2^{2r}}+\dfrac{7^r}{2^{3r}}+\dfrac{15^r}{2^{4r}}+.....\text{up to m terms}\right]$

श्रेणी $\sum_{r=0}^{n}(-1)^r.{}^nC_r\left[\dfrac{1}{2^r}+\dfrac{3^r}{2^{2r}}+\dfrac{7^r}{2^{3r}}+\dfrac{15^r}{2^{4r}}+.....\text{up to m terms}\right]$ का योगफल ज्ञात कीजिए।

Ans. $\dfrac{\left(2^{mn}-1\right)}{\left(2^{n}-1\right)\left(2^{mn}\right)}$

Sol. $\sum_{r=0}^{n}(-1)^{r}.\ {}^{n}C_{r}\left[\dfrac{1}{2^{r}}+\left(\dfrac{3}{4}\right)^{r}+\left(\dfrac{7}{8}\right)^{r}+\ldots\ldots\ldots .m \text{ terms}\right]$

$=\sum_{r=0}^{n}\left[(-1)^{r}\ {}^{n}C_{r}\dfrac{1}{2^{r}}+(-1)^{r}\ {}^{n}C_{r}\left(\dfrac{3}{4}\right)^{r}+\ldots\ldots\ldots .m \text{ terms}\right]$

$=[\ {}^{n}C_{0}-{}^{n}C_{1}\left(\dfrac{1}{2}\right)+{}^{n}C_{2}\left(\dfrac{1}{2}\right)^{2}+\ldots\ldots\ldots\ldots]+$

$[\ {}^{n}C_{0}-{}^{n}C_{1}\left(\dfrac{3}{4}\right)+{}^{n}C_{2}\left(\dfrac{3}{4}\right)^{2}+\ldots\ldots]+\ldots\ldots\text{m terms}$

$=\left(1-\dfrac{1}{2}\right)^{n}+\left(1-\dfrac{3}{4}\right)^{n}+\left(1-\dfrac{7}{8}\right)^{n}+\ \ldots\ldots\text{m terms}$

$=\ \dfrac{1}{2^{n}}\left(\dfrac{1-\dfrac{1}{2^{mn}}}{1-\dfrac{1}{2^{n}}}\right)=\dfrac{1}{2^{n}-1}\left(1-\dfrac{1}{2^{mn}}\right)$

20. Using the theory of permutation or otherwise prove that

${}^{100}C_{10}+5\cdot{}^{100}C_{11}+10\cdot{}^{100}C_{12}+10\cdot{}^{100}C_{13}+5\cdot{}^{100}C_{14}+{}^{100}C_{15}\ ={}^{105}C_{90}$

क्रमचय सिद्धान्त या अन्यथा प्रयोग कर सिद्ध कीजिये।

${}^{100}C_{10}+5\cdot{}^{100}C_{11}+10\cdot{}^{100}C_{12}+10\cdot{}^{100}C_{13}+5\cdot{}^{100}C_{14}+{}^{100}C_{15}\ ={}^{105}C_{90}.$

Ans.

Sol. ${}^{100}C_{10}+5\cdot{}^{100}C_{11}+10\cdot{}^{100}C_{12}+10\cdot{}^{100}C_{13}+5\cdot{}^{100}C_{14}+{}^{100}C_{15}$

$={}^{5}C_{0}\cdot{}^{100}C_{10}+{}^{5}C_{1}\cdot{}^{100}C_{11}+{}^{5}C_{2}\cdot{}^{100}C_{12}+{}^{5}C_{3}\cdot{}^{100}C_{13}+{}^{5}C_{4}\cdot{}^{100}C_{14}+{}^{5}C_{5}\cdot{}^{100}C_{15}$

$|r_{1}-r_{2}|=-10=\text{Constant}$

$(x+1)^{5}={}^{5}C_{0}\,x^{5}+{}^{5}C_{1}\,x^{4}+{}^{5}C_{2}\,x^{3}+{}^{5}C_{3}\,x^{2}+{}^{5}C_{4}\,x^{1}+{}^{5}C_{5}$

$(1+x)^{100}={}^{100}C_{0}+{}^{100}C_{1}\,x+\ldots..+{}^{100}C_{15}\,x^{15}+\ldots.+{}^{100}C_{100}\,x^{100}$

$(1+x)^{105}=\left[\sum_{r=0}^{5}{}^{5}C_{r}x^{r}\right]\left[\sum_{r=0}^{100}{}^{100}C_{r}x^{r}\right]$

Comparing coefficient of x^{15}

$^{100}C_{10} + 5 \cdot {}^{100}C_{11} + 10 \cdot {}^{100}C_{12} + 100 \cdot {}^{100}C_{13} + 5 \cdot {}^{100}C_{14} + {}^{100}C_{15}$

$= {}^{105}C_{90}$

21. Let $f(n) = \sum_{r=0}^{n} \sum_{k=r}^{n} \binom{k}{r}$. Find the total number of divisors of $f(11)$.

माना $f(n) = \sum_{r=0}^{n} \sum_{k=r}^{n} \binom{k}{r}$. तब $f(11)$ के कुल विभाजक ज्ञात कीजिये।

Ans. 24

Sol. $f(n) = \sum_{r=0}^{n} \sum_{k=r}^{n} \binom{k}{r} = \sum_{r=0}^{n} \left[\sum_{k=r}^{n} {}^{k}C_r \right]$

$= \sum_{r=0}^{n} \left[{}^{r}C_r + {}^{r+1}C_r + \ldots + {}^{n}C_r \right]$

$\sum_{r=0}^{n} \left[{}^{r+1}C_{r+1} + {}^{r+1}C_r + {}^{r+2}C_r + \ldots + {}^{n}C_r \right]$

$= \sum_{r=0}^{n} \left[{}^{r+2}C_{r+1} + {}^{r+2}C_r + \ldots + {}^{n-1}C_r + {}^{n}C_r \right]$

$= \sum_{r=0}^{n} {}^{n+1}C_{r+1} = {}^{n+1}C_1 + {}^{n+1}C_2 + \ldots + {}^{n+1}C_{n+1}$

$= 2^{n+1} - 1$

$\Rightarrow f(11) = 2^{12} - 1 = 4095 = 5 \times 9 \times 7 \times 13 = 5 \times 7 \times 13 \times 3^2$

$\Rightarrow$ Total no. of divisor $= 2 \times 2 \times 2 \times 3 = 24$

22. Given that $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$, find the values of :

(i) $a_0 + a_1 + a_2 + \ldots + a_{2n}$; (ii) $a_0 - a_1 + a_2 - a_3 \ldots + a_{2n}$; (iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots + a_{2n}^2$

दिया गया है $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$, तो निम्न के मान ज्ञात कीजिए –

(i) $a_0 + a_1 + a_2 + \ldots + a_{2n}$; (ii) $a_0 - a_1 + a_2 - a_3 \ldots + a_{2n}$; (iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots + a_{2n}^2$

Ans. (i) 3^n (ii) 1, (iii) a_n

Sol. $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$

(i) $a_0 + a_1 + a_2 + ... + a_{2n} = (1 + 1 + 1)^n = 3^n$

(ii) $a_0 - a_1 + a_2 - a_3 + ... + a_{2n} = (1 - 1 + (-1)^2)^n = 1$

(iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2 =$

$\because \left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + ... + a_{2n} x^{2n}$ _____(i)

$x \to \frac{1}{x}$

$\left(1 + \frac{1}{x} + \frac{1}{x^2}\right)^n = a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + ... + \frac{a_{2n}}{x^{2n}}$

$\left[\frac{\left(1 + x + x^2\right)^n}{x^{2n}}\right] = \frac{a_0 x^{2n} + a_1 x^{2n-1} + a_2 x^{2n-2} + ... + a_{2n}}{x^{2n}}$

$x \to -x$

$\left[\frac{\left(1 - x + x^2\right)^n}{x^{2n}}\right] = \frac{a_0 x^{2n} - a_1 x^{2n-1} + a_2 x^{2n-2} + ... + a_{2n}}{x^{2n}}$ _____(ii)

By (i) × (ii); we get

$\Rightarrow \left[\left(1 + x^2\right)^2 - x^2\right]^n = ... + x^{2n}\left(a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2\right)$

Comparing coefficient of x^{2n} in $[1 + x^4 + x^2]^n$

$\because (1 + x + x^2)^n = a_0 + a_1 x + ... + a_{2n} x^{2n}$

$x \to x^2$

$(1 + x^2 + x^4)^n = a_0 + a_1 x^2 + ... + a_n x^{2n} + ... + a_{2n} x^{4n}$

$\Rightarrow a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2 = a_n$

23. Prove that $\sum_{K=0}^{n} {}^nC_K \sin Kx . \cos(n - K)x = 2^{n-1} \sin nx$.

सिद्ध कीजिए $\sum_{K=0}^{n} {}^nC_K \sin Kx . \cos(n - K)x = 2^{n-1} \sin nx$

Ans.

Sol. $S = {}^nC_0.\sin(0x).\cos nx + {}^nC_1.\sin x.\cos(n-1)x$

$\quad\quad\quad

Ans.

Sol. (a) $C_0^2 + C_1^2 + C_2^2 + ... + C_n^2$

$$(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$$

$$(x+1)^n = C_0x^n + C_1x^{n-1} + ... + C_n$$

$$\Rightarrow (1+x)^{2n} = \left[\sum_{r=0}^{n} C_r x^r\right]\left[\sum_{r=0}^{n} C_r x^{n-r}\right]$$

Comparing coefficient of x^n in $(1+x)^{2n}$

$$\Rightarrow C_0^2 + C_1^2 + C_2^2 + ...C_n^2 = {}^{2n}C_n = \frac{\lfloor 2n}{\lfloor n \lfloor n}$$

(b) $C_0C_1 + C_1C_2 + C_2C_3 + ... + C_{n-1}C_n$

Comparing coefficient of x^{n-1} in $(1+x)^{2n}$

$$\Rightarrow C_0C_1 + C_1C_2 + ... + C_{n-1}C_n = {}^{2n}C_{n-1} = \frac{\lfloor 2n}{\lfloor n-1 \lfloor n+1}$$

(c) We know that

$(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_2x^r + ... + C_nx^n$ _____(i)

$$\left(1+\frac{1}{x}\right)^n = C_0 + C_1\frac{1}{x} + C_2\frac{1}{x^2} + ...$$

$$+ C_r\frac{1}{x^r} + ... + C_n\frac{1}{x^n} \quad \text{_____(ii)}$$

Multiply (i) & (ii) we get

$$\frac{(1+x)^{2n}}{x^n} = (C_0 + C_1x + ... C_rx^r + ... + C_nx^n)$$

$$\left(C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \frac{C_n}{x^n}\right) \quad \text{_____(iii)}$$

Now coefficient of $\frac{1}{x^r}$ in RHS

$= C_0C_r + C_1C_{r+1} + ... C_{n-r}C_n$

$\therefore$ Coefficient of $\frac{1}{x^r}$ in LHS

= coefficient of x^{n-r} in $(1+x)^{2n} = {}^{2n}C_{n-r}$

$= \dfrac{2n!}{(n-r)!(n+r)!}$

But (iii) is an identity

$\therefore$ Coefficient of $\dfrac{1}{x^r}$ in RHS

= coefficient of $\dfrac{1}{x^r}$ in LHS

$\Rightarrow \quad {}^nC_0\,{}^nC_r + {}^nC_1\,{}^nC_{r+1} + ... + {}^nC_{n-r}\,{}^nC_n = \dfrac{2n!}{(n-r)!(n+r)!}$

(d) $\sum_{r=0}^{n-2} {}^nC_r\,{}^nC_{r+2} = {}^2C_0\,{}^nC_2 + {}^nC_1\,{}^nC_3 + {}^nC_2\,{}^nC_4 + ... + {}^nC_{n-2}\,{}^nC_n$

Comparing coefficient of x^{n-2} in

$\sum_{r=0}^{n-2} {}^nC_r\,{}^nC_{r+2} = \dfrac{\lfloor 2n}{\lfloor n-2 \rfloor n+2}$

26. Prove the following identities using the theory of permutation where $C_0, C_1, C_2, ..., C_n$ are the combinatorial coefficients in the expansion of $(1 + x)^n$, $n \in N$, then prove the following:

क्रमचय सिद्धान्त का प्रयोग कर सर्वसमिकाओं को सिद्ध कीजिये। यदि $C_0, C_1, C_2, ..., C_n$, $(1 + x)^n$, $n \in N$ के प्रसार में द्विपद गुणांक है। तब निम्न को सिद्ध कीजिए।

(a) $C_1 + 2C_2 + 3C_3 + ...+ n \,.\, C_n = n \,.\, 2^{n-1}$

(b) $C_0 + 2C_1 + 3C_2 + ...+ (n+1)C_n = (n+2)2^{n-1}$

(c) $C_0 + 3C_1 + 5C_2 + ...+ (2n+1)C_n = (n+1)\,2^n$

(d) $(C_0+C_1)(C_1+C_2)(C_2+C_3) ..... (C_{n-1}+C_n) = \dfrac{C_0 \,.\, C_1 \,.\, C_2 .... C_{n-1}\,(n+1)^n}{n!}$

(e) $1 \,.\, C_o^{\,2} + 3 \,.\, C_1^{\,2} + 5 \,.\, C_2^{\,2} + ... + (2n+1)\, C_n^{\,2} = \dfrac{(n+1)\,(2n)!}{n!\,n!}$

Ans.

Sol. $(1+x)^n = {}^nC_0 + {}^nC_1 x + ... + {}^nC_n x^n$

(a) $\sum_{r=1}^{n} r \cdot {}^nC_r = \sum_{r=1}^{n} r \cdot \dfrac{n}{r}\,{}^{n-1}C_{r-1}$

$= n\left[{}^{n-1}C_0 + {}^{n-1}C_1 + ... + {}^{n-1}C_{n-1}\right]$

$= n \cdot 2^{n-1}$

(b) $C_0 + 2C_1 + 3C_2 + ... + (n+1)C_n$

$$= \sum_{r=0}^{n} (r+1)^{n} C_r = \sum_{r=0}^{n} r \cdot {}^{n}C_r + \sum_{r=0}^{n} {}^{n}C_r$$

$= n \cdot 2^{n-1} + 2^n = 2^{n-1}(n+2)$

(c) $C_0 + 3C_1 + 5C_2 + ... + (2n+1)C_n$

$$= \sum_{r=0}^{n} (2r+1) {}^{n}C_r$$

$$= 2\sum_{r=0}^{n} r \cdot {}^{n}C_r + \sum_{r=0}^{n} {}^{n}C_r$$

$= 2^n \cdot 2^{n-1} + 2^n$

$= 2^n (n+1)$

(d) $(C_0 + C_1)(C_1 + C_2).....(C_{n-1} + C_n)$

$= ({}^{n+1}C_1)({}^{n+1}C_2).....({}^{n+1}C_n) = ((n+1).C_0)\left(\frac{n+1}{2}.C_1\right)..... \left(\frac{n+1}{n}.C_{n-1}\right)$

$= \frac{(n+1)^n}{n!} C_0 C_1 ..... C_{n-1}$

(e) $1 \cdot C_0^2 + 3 \cdot C_1^2 + 5 \cdot C_2^2 + ... + (2n+1)C_n^2$

$$= \sum_{r=0}^{n} (2r+1) C_r^2$$

$$= 2\sum_{r=0}^{n} r \cdot C_r^2 + \sum_{r=0}^{n} C_r^2$$

$$= 2\sum_{r=0}^{n} r \cdot \frac{n}{r} {}^{n-1}C_{r-1} \cdot {}^{n}C_r + \sum_{r=0}^{n} \left({}^{n}C_r\right)^2$$

$$= 2n \cdot \sum_{r=1}^{n} {}^{n-1}C_{r-1} \cdot {}^{n}C_r + \sum_{r=0}^{n} \left({}^{n}C_r\right)^2$$

$$= 2n \left[\underbrace{{}^{n-1}C_0}_{\text{Coefficient of } x^{n-1}} \underbrace{{}^{n}C_1}_{\text{Coefficient of } x} + {}^{n-1}C_1 {}^{n}C_2 + ... + {}^{n-1}C_{n-1} \cdot {}^{n}C_n \right] + \sum_{r=0}^{n} \left({}^{n}C_r\right)^2$$

$$= 2n \cdot \text{coefficient of } x^n \text{ in } (1+x)^{2n-1} + \frac{|2n}{|n|n}$$

$$= 2n\cdot\frac{\lfloor 2n-1}{\lfloor n-1\lfloor n}+\frac{\lfloor 2n}{\lfloor n\lfloor n}$$

$$= 2n\left[\frac{\lfloor 2n-1}{\lfloor n-1\lfloor n}+\frac{\lfloor 2n-1}{\lfloor n\lfloor n}\right]=\frac{\lfloor 2n}{\lfloor n\lfloor n-1}\left[1+\frac{1}{n}\right]$$

$$=\frac{\lfloor 2n}{\lfloor n\lfloor n}\cdot(n+1)$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 27 to 29

A path of length n is a sequence of points $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$ with integer coordinates such that for all i between 1 and $n-1$ both inclusive,

either $x_{i+1} = x_i + 1$ and $y_{i+1} = y_i$ (in which case we say the ith step is rightward)

or $x_{i+1} = x_i$ and $y_{i+1} = y_i+1$ (in which case we say that the ith step is upward).

This path is said to start at (x_1, y_1) and end at (x_n, y_n). Let P(a, b), for a and b non-negative integers, denotes the number of paths that start at (0, 0) and end at (a, b).

n लम्बाई का पाथ एक अनुक्रम है जिसके बिन्दु पूर्णांक निर्देशांकों $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$ है। इस प्रकार की 1 और $n-1$ के मध्य, जहां 1 और $n-1$ भी शामिल है, सभी i के लिए

या तो $x_{i+1} = x_i + 1$ और $y_{i+1} = y_i$ (में i^{th} स्टेप दांयी ओर है)

या $x_{i+1} = x_i$ और $y_{i+1} = y_i+1$ (में i^{th} स्टेप ऊपर की ओर है)

इस पथ का प्रारम्भिक बिन्दु (x_1, y_1) और अन्त बिन्दु (x_n, y_n) है। माना P(a, b), जहां a, b अऋणात्मक पूर्णांक है। उन पाथों की संख्या दर्शाते हैं जिसका प्रारम्भिक बिन्दु (0, 0) और अन्त बिन्दु (a, b) है।

27. The value of $\sum_{i=0}^{10} P(i, 10-i)$, is :

$\sum_{i=0}^{10} P(i, 10-i)$ का मान होगा:

(A) 1024 (B) 512 (C) 256 (D) 128

Ans. A

Sol. $$\sum_{r=0}^{10} P(i,10-i)=\sum_{i=0}^{10}\frac{\lfloor 10}{\lfloor i\lfloor 10-i}$$

$$=\sum_{i=0}^{10} {}^{10}C_i = 2^{10} = 1024$$

28. Number of ordered pairs (i, j) where $i \neq j$ for which $P(i, 100-i) = P(j, 100-j)$, is :

$P(i, 100-i) = P(j, 100-j)$ के लिए क्रमित युग्मों (i, j) की संख्या (जहां $i \neq j$) है:

(A) 50 (B) 99 (C) 100 (D) 101

Ans. C

Sol. $i \neq j \quad P(i, 100-i) = P(j, 100-j)$

$\neq {}^{100}C_i = {}^{100}C_j$

$\Rightarrow i = j$ (wrong) or $\underbrace{i+j=100}_{\substack{(0,100)\\(1,99)\\(50,50)\\(49,51)\\(99,1)\\(100,0)}}$ & $i \neq j$

$\Rightarrow$ No. of ordered pair = 101 – 1 = 100

29. The sum $P(43, 4) + \sum_{j=1}^{5} P(49-j, 3)$ is equal to :

$P(43, 4) + \sum_{j=1}^{5} P(49-j, 3)$ का योग बराबर :

(A) P(4, 48) (B) P(3, 49) (C) P(4, 47) (D) P(5, 47)

Ans. A

Sol. $P(43,4) + \sum_{j=1}^{5} P(49-j,3)$

$= \frac{\lfloor 47}{\lfloor 43 \lfloor 4} + \sum_{j=1}^{5} \frac{\lfloor 52-j}{\lfloor 49-j \lfloor 3}$

$= {}^{47}C_4 + \sum_{j=1}^{5} {}^{52-j}C_3$

$= {}^{47}C_4 + {}^{47}C_3 + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3 + {}^{51}C_3$

$= {}^{48}C_4 + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3 + {}^{51}C_3$

$= {}^{52}C_4$

$= \frac{\lfloor 52}{\lfloor 4 \lfloor 48}$

= P(4, 48)

Paragraph for Question Nos. 30 to 32

Let $S_k = 3^k \binom{100}{0}\binom{100}{k} - 3^{k-1}\binom{100}{1}\binom{99}{k-1} + 3^{k-2}\binom{100}{2}\binom{98}{k-2} - \ldots\ldots\ldots + (-1)^k \binom{100}{k}\binom{100-k}{0}$

where $\binom{n}{r}$ represents the coefficient of x^r in the binomial expansion of $(1+x)^n \ \forall\, n \in N$

and $V_k = \left(\frac{1}{2}\right)^k S_k = M(100, k)$.

माना $S_k = 3^k \binom{100}{0}\binom{100}{k} - 3^{k-1}\binom{100}{1}\binom{99}{k-1} + 3^{k-2}\binom{100}{2}\binom{98}{k-2} - \ldots\ldots\ldots + (-1)^k \binom{100}{k}\binom{100-k}{0}$

जहां $\binom{n}{r}$ $(1+x)^n \ \forall\, n \in N$ के प्रसार में x^r का गुणांक है और $V_k = \left(\frac{1}{2}\right)^k S_k = M(100, k)$.

30. The value of $\sum_{k=0}^{100} S_k\, S_{100-k}$ equals :

$\sum_{k=0}^{100} S_k\, S_{100-k}$ का मान है :

(A) $\binom{200}{100}$ (B) $2^{100}\binom{200}{100}$ (C) $2^{200}\binom{200}{100}$ (D) $2^{99}\binom{200}{99}$

Ans. B

Sol. $S_K = 3^K\ {}^{100}C_0\ {}^{100}C_K - 3^{K-1}\ {}^{100}C_1\ {}^{99}C_{K-1} + \ldots. + (-1)^K \cdot {}^{100}C_K\ {}^{100-K}C_0$

$(1+x)^{100} = {}^{100}C_0 + {}^{100}C_1\, x + \ldots. + {}^{100}C_K\, x^K + \ldots. + {}^{100}C_{100}\, x^{100}$

$$S_K = \sum_{r=0}^{K} 3^{K-r}\ {}^{100}C_r\ {}^{100-r}C_{K-r} \cdot (-1)^r$$

$$= \sum_{r=0}^{K} 3^{K-r}\, \frac{\lfloor 100}{\lfloor 100-r \lfloor r}\, \frac{\lfloor 100-r}{\lfloor 100-K \lfloor K-r}\, (-1)^r$$

$$= \sum_{r=0}^{K} 3^{K-r} \cdot (-1)^r\ {}^{100}C_K \cdot {}^{K}C_r$$

$$= {}^{100}C_K\, (3-1)^K = 2^K \cdot {}^{100}C_K$$

$$\Rightarrow V_K = \frac{1}{2^K} S_K = {}^{100}C_K = M(100, K)$$

$$\sum_{K=0}^{100} S_K S_{100-K} = \sum_{K=0}^{100} {}^{100}C_K \cdot {}^{100}C_{100-K} \cdot 2^K \cdot 2^{100-K}$$

$$= 2^{100} \sum_{K=0}^{100} \left({}^{100}C_K\right)^2 \rightarrow \text{Coefficient of } x^{100} \text{ in } (1+x)^{200}$$

$$= 2^{100} \cdot {}^{200}C_{100}$$

$$= 2^{100} \cdot \binom{200}{100}$$

31. The value of the $M(100, 49) + M(100, 50)$ is equal to :

$M(100, 49) + M(100, 50)$ का मान है :

(A) $M(99, 50)$ (B) $M(100, 51)$ (C) $M(101, 49)$ (D) $M(101, 51)$

Ans. D

Sol. $\because M(100, K) = {}^{100}C_K$

$\Rightarrow M(100, 49) + M(100, 50)$

$= {}^{100}C_{49} + {}^{100}C_{50}$

$= {}^{100}C_{50} = M(101, 50) = M(101, 51)$

32. In an examination, there are 100 true-false type questions. The number of ways in which one student can make his choice to attempt or not to attempt the questions is given by :

किसी परीक्षा में 100 सत्य असत्य प्रकार के प्रश्न है। तब कोई विद्यार्थी कुल कितने प्रकार से प्रश्न का उत्तर दे या नहीं दे सकता है :

(A) $\sum_{k=0}^{100} S_k$ (B) $\sum_{k=0}^{100} V_k$ (C) $\sum_{k=1}^{100} S_k$ (D) $\sum_{k=1}^{100} V_k$

Ans. A

Sol. $100 \rightarrow$ question (True/False)

$= {}^{100}C_0 \cdot 2^0 + {}^{100}C_1 \cdot 2 + ... + {}^{100}C_{100} \cdot 2^{100}$

$$= \sum_{K=0}^{100} {}^{100}C_K \cdot 2^K = \sum_{K=0}^{100} S_K$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

33. In the expansion of $\left(x-\frac{\alpha}{x}\right)^n$ and $\left(x+\frac{\beta}{x^2}\right)^n$ in powers of x have one term independent of x, then n is divisible by :

यदि $\left(x-\frac{\alpha}{x}\right)^n$ तथा $\left(x+\frac{\beta}{x^2}\right)^n$ के प्रसार में x की घातों के रूप में केवल एक पद ही x से स्वतन्त्र हो तब n विभाजित होगा:

(A) 2 (B) 3 (C) 4 (D) 6

Ans. ABD

Sol. $\left(x-\frac{\alpha}{x}\right)^n=\sum_{r=0}^{n}{}^nC_r x^{n-2r}\cdot(-\alpha)^r \rightarrow$ Term independent of x if $n=2r$

$\left(x+\frac{\beta}{x^2}\right)^n=\sum_{r=0}^{n}{}^nC_r x^{n-3r}\cdot(\beta)^r \rightarrow$ Term independent of x if $n=3r$

$\Rightarrow$ n is divisible by 2, 3 or 6.

34. In the expansion of $(a+b+c)^{10}$:

(A) total number of terms is 66 (B) coefficient of a^8 bc is 90

(C) coefficient of $a^4 b^5 c^3$ is 0 (D) none of these

; . इ. बद्ध्^{0} के प्रसार में :

(A) कुल पदों की संख्या 66 है। (B) a^8 bc का गुणांक 90 है।

(C) $a^4 b^5 c^3$ का गुणांक 0 है। (D) इनमकें से कोई नहीं

Ans. ABC

Sol. $(a+b+c)^{10}$

Total terms are ${}^{10+3-1}C_2={}^{12}C_2=\frac{12\times11}{2}=66$

$$(a+b+c)^{10}=\sum_{P_1+P_2+P_3=10}\frac{\lfloor 10}{\lfloor P_1\lfloor P_2\lfloor P_3}a^{P_1}b^{P_2}c^{P_3}$$

Coefficient of a^8bc is $\frac{\lfloor 10}{\lfloor 8\lfloor 1\lfloor 1}=90$

Coefficient of $a^4b^5c^3$ is 0 as $P_1+P_2+P_3=12\neq10$

35. In the expansion of $\left(x+\frac{\alpha}{x^2}\right)^n$, $a \neq 0$, if no term is independent of x, then n is :

$\left(x+\frac{a}{x^2}\right)^n$, $a \neq 0$ के प्रसार में कोई भी पद x से स्वतन्त्र नहीं है तब n =

(A) 10 (B) 12 (C) 16 (D) 20

Ans. ACD

Sol. $\left(x+\frac{a}{x^2}\right)^n ; a \neq 0 = \sum_{r=0}^{n} x^{n-3r} \cdot {}^nC_r \cdot a^r$

If no term is independent of x then

$n - 3r \neq 0$

$\Rightarrow n \neq 3r$

$\Rightarrow n \neq 0, 3, 6, 9, 12, 15$ .....

$\Rightarrow n = 10, 16 \& 20$

36. In the expansion of $\left(x+\frac{1}{x}+2\right)^{2009}$

(A) Middle term is ${}^{4018}C_{2009}$
(B) The term independent of x is ${}^{4018}C_{2009}$
(C) Coefficient of x in the expansion is ${}^{4018}C_{2010}$
(D) Coefficient of 1/x in the expansion is ${}^{4018}C_{2008}$

$\left(x+\frac{1}{x}+2\right)^{2009}$ के प्रसार में –

(A) मध्य पद ${}^{4018}C_{2009}$ है। (B) x से स्वतंत्र पद ${}^{4018}C_{2009}$ है।
(C) x का गुणांक ${}^{4018}C_{2010}$ है। (D) 1/x का गुणांक ${}^{4018}C_{2008}$ है।

Ans. ABCD

Sol. $\left(x+\frac{1}{x}+2\right)^{2009}$

$=\left(\frac{x^2+2x+1}{x}\right)^{2009} = \frac{1}{x^{2009}}(x+1)^{4018}$

$=1\left[\sum_{r=0}^{4018} {}^{4018}C_r x^{r-2009}\right]$

Term independent of x if r = 2009

$\Rightarrow$ Coefficient of this is ${}^{4018}C_{2009}$.

Coefficient of x = ${}^{4018}C_{2010}$

Coefficient of $\frac{1}{x}$ = ${}^{4018}C_{2008}$

Coefficient of middle term is ${}^{4018}C_{2009}$

37. Which of the following statement(s) is/are correct?

(A) $1+\frac{2}{2}+\frac{3}{2^2}+\frac{4}{2^3}+.....+\infty=4$

(B) Integral part of $\left(9+4\sqrt{5}\right)^n$, is even, $n \in N$.

(C) $\left({}^nC_0+{}^nC_1+{}^nC_2+.....+{}^nC_n\right)^2=1+{}^{2n}C_1+{}^{2n}C_2+.....+{}^{2n}C_{2n}$

(D) $\frac{1}{(3+2x)^2}$ can be expanded as infinite series in ascending powers of x only if $|x|<\frac{2}{3}$

निम्न में से कौनसा/से कथन सत्य है ?

(A) $1+\frac{2}{2}+\frac{3}{2^2}+\frac{4}{2^3}+.....+\infty=4$

(B) $\left(9+4\sqrt{5}\right)^n$ का पूर्णांक मान सम है, जहाँ $n \in N$.

(C) $\left({}^nC_0+{}^nC_1+{}^nC_2+.....+{}^nC_n\right)^2=1+{}^{2n}C_1+{}^{2n}C_2+.....+{}^{2n}C_{2n}$

(D) यदि $|x|<\frac{2}{3}$ तो $\frac{1}{(3+2x)^2}$ को x के बढ़ते घात में अनन्त पदों तक लिखा जा सकता है

Ans. AC

Sol. A) $\left(1-\frac{1}{2}\right)^{-2}=\left(\frac{1}{2}\right)^{-2}=\frac{1}{2^{-2}}=4 \quad \Rightarrow$ **(A)** is correct

(B) it should be odd $\Rightarrow$ **(B)** is incorrect

(C) L.H.S. = $(2^n)^2$; R.H.S. = $2^{2n} \quad \Rightarrow$ **(C)** is correct

(D) $|x|<\frac{3}{2} \quad \Rightarrow$ **(D)** is incorrect

38. If $\left(9+\sqrt{80}\right)^n = 1+f$ where I, n are integers and $0<f<1$, then

(A) I is an odd integer (B) I is an even integer

(C) $(I+f)(1-f)=1$ (D) $1-f=\left(9-\sqrt{80}\right)^n$

यदि $\left(9+\sqrt{80}\right)^n = I + f$ जहाँ I तथा n पूर्णांक है जहाँ $0 < f < 1$तो

(A) I विषम पूर्णांक (B) I सम पूर्णांक है

(C) $(I + f)(1 - f) = 1$ (D) $1 - f = \left(9-\sqrt{80}\right)^n$

Ans. ACD

Sol. $\left(9+4\sqrt{5}\right)^n = 3 + f$

let $\left(9-4\sqrt{5}\right)^n = f'$

$I + f + f' = \left(9+4\sqrt{5}\right)^n + \left(9-4\sqrt{5}\right)^n$ = even integer

$0 < f + f' < 2 \Rightarrow f + f' = 1$

$\therefore$ I is an odd integer $\Rightarrow$ (A)

39. Let $(1 + x^2)^2 (1 + x)^n = A_0 + A_1 x + A_2 x^2 + ...$ If A_0, A_1, A_2 are in A.P. then the value of n is :

माना $(1 + x^2)^2 (1 + x)^n = A_0 + A_1 x + A_2 x^2 + ...$ यदि A_0, A_1, A_2 समान्तर श्रेणी में हो, तो n बराबर होगा :

(A) 2 (B) 3 (C) 5 (D) 7

Ans. AB

Sol. $(1 + x^2)^2 (1 + x)^n = (1 + 2x^2 + x^4)\,({}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + ...)$

$= {}^nC_0 + {}^nC_1 x + ({}^nC_2 + 2 \cdot {}^nC_0) x^2 + ...$

Hence $A_0 = 1$; $A_1 = {}^nC_1$; $A_2 = {}^nC_2 + 2$ which are in A.P.

40. If ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$ has the value equal to ${}^xC_y$; then the possible value(s) of x + y can be :

यदि ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$ का मान ${}^xC_y$ के बराबर है ; तो x + y के संभव मान हो सकते है :

(A) 112 (B) 114 (C) 196 (D) 198

Ans. BD

Sol. ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$

$= ({}^{100}C_7 + {}^{100}C_6) + 3\,({}^{100}C_7 + {}^{100}C_8) + 3\,({}^{100}C_8 + {}^{100}C_9) + {}^{100}C_9 + {}^{100}C_{10}$

$= {}^{101}C_7 + 3 \cdot {}^{101}C_8 + 3 \cdot {}^{101}C_9 + {}^{101}C_{10}$

$= ({}^{101}C_7 + {}^{101}C_8) + 2\,[{}^{101}C_8 + {}^{101}C_9] + [{}^{101}C_{10} + {}^{101}C_9]$

$= {}^{102}C_8 + 2 \cdot {}^{102}C_9 + {}^{102}C_{10}$

$= {}^{103}C_9 + {}^{103}C_{10}$

$= {}^{104}C_{10} = {}^{104}C_{94}$

$\Rightarrow x + y = 114$ or $x + y = 198$

MATCH THE COLUMN TYPE QUESTION(S)

41. **Match the column**

	Column I	**Column II**
(A)	If the coefficients of x^7 and x^8 in $\left(2+\frac{x}{3}\right)^n$ are equal, then n is	(P) 2
(B)	If the coefficient of x in the expansion of $\left(x-\frac{1}{ax^2}\right)^{10}$ is –15, then the value of a, is	(Q) 01
(C)	The remainder when 2^{2003} in divided by 17 is	(R) 55
(D)	The last two digits of 3^{400}, are	(S) 8

	Column I	**Column II**
(A)	यदि $\left(2+\frac{x}{3}\right)^n$ के प्रसार में x^7 व x^8 के गुणांक बराबर है तब n =	(P) 2
(B)	$\left(x-\frac{1}{ax^2}\right)^{10}$ के प्रसार में x का गुणांक –15 है तब a =	(Q) 01
(C)	2^{2003} को 17 से विभाजित करने पर शेषफल	(R) 55
	;क द्ध3^{400} के अन्तिम दो अंक	(S) 8

Ans. (A) R (B) P (C) S (D) Q

Sol. (A) Coefficient of x^7 & x^8 are equal in $\left(2+\frac{x}{3}\right)^n$

$$\Rightarrow {}^nC_7\,2^{n-7}\cdot\frac{x^7}{3^7} = {}^nC_8\,2^{n-8}\cdot\frac{x^8}{3^8}$$

$$\Rightarrow \frac{\lfloor n}{\lfloor n-7 \rfloor 7}\frac{2^n}{2^7}\cdot\frac{1}{3^7} = \frac{\lfloor n}{\lfloor n-8 \rfloor 8}\frac{2^n}{2^8}\cdot\frac{1}{3^8}$$

$$\Rightarrow \frac{1}{(n-7)} = \frac{1}{6\cdot 8} \qquad \Rightarrow n-7 = 48$$

$$\Rightarrow n = 55$$

$$(B)\ \left(x-\frac{1}{ax^2}\right)^{10} = \sum_{r=0}^{10} {}^{10}C_r\,x^{10-r}\cdot\frac{(-1)^r}{a^r\cdot x^{2r}}$$

$$= \sum_{r=0}^{10} {}^{10}C_r\cdot x^{10-3r}\cdot\left(\frac{-1}{a}\right)^r$$

For x^1 ; $\quad 10-3r = 1$

$3r = 9$

r = 3

$$\Rightarrow {}^{10}C_3\left(\frac{-1}{a}\right)^3 = -15$$

$$\Rightarrow \frac{10\times 9\times 8}{6} = 15\cdot a^3$$

$\Rightarrow 9 \times 8 \times 2 = 18a^3$

$\Rightarrow a^3 = 8$

$\Rightarrow a = 2$

(C) $2^{2003} = 2 \cdot 2^{2002} = 2^3 \cdot (2)^{2000} = 8 \cdot (16)^{500}$

$= 8 \cdot [17 - 1]^{500}$

$$= 8\sum_{r=0}^{500} {}^{500}C_r 17^{500-r}\cdot(-1)^r$$

$= 17\lambda + {}^{8500}C_{500}$

$\therefore$ Remainder = 8

(D) $3^{400} = (81)^{100} = (80 + 1)^{100}$

$$= \sum_{r=0}^{100} {}^{100}C_r (80)^r$$

$= {}^{100}C_0 + {}^{100}C_1 \cdot 80 + 100\lambda$

$= {}^{100}C_0 + 100\lambda' \; ; \lambda \in Z$

$= 100\,\lambda' + 1$

$\because$ Last digit will be 01.

Exercise J-M

1. The coefficient of x^{256} in the expansion of $(1-x)^{101}(x^2+x+1)^{100}$ is : **[JEE(Main)July2021]**

$(1-x)^{101}(x^2+x+1)^{100}$ के प्रसार में x^{256} का गुणांक है : **[JEE(Main)July2021]**

(A) $-{}^{100}C_{15}$ (B) $-{}^{100}C_{16}$ (C) ${}^{100}C_{16}$ (D) ${}^{100}C_{15}$

Ans. D

Sol. $(1-x)((1-x)(1+x+x^2))^{100}$

$(1-x)(1-x^3)^{100}$

$(1-x^3)^{100} - x(1-x^3)^{100}$

Coefficient of $x^{256} = 0 - (-{}^{100}C_{85})$

$= {}^{100}C_{85}$

$= {}^{100}C_{15}$

2. For the natural numbers m, n, if $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + ...+a_{m+n}y^{m+n}$ and $a_1 = a_2 = 10$, then the value of $(m+n)$ is equal to : **[JEE(Main)July2021]**

यदि धन पूर्णांकों m तथा n के लिए $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + ...+a_{m+n}y^{m+n}$ तथा $a_1 = a_2 = 10$ हैं, तो $(m+n)$ बराबर है : **[JEE(Main)July2021]**

(A) 64 (B) 100 (C) 88 (D) 80

Ans. D

Sol. coeff. of $y = a_1 = {}^mC_0\,{}^nC_1 - {}^mC_1\,{}^nC_0 = 10$

$n - m = 10$ .....(1)

coeff. of $y = a_2 = {}^mC_0\,{}^nC_2 + {}^mC_2\,{}^nC_0 - {}^mC_1\,{}^nC_1 = 10$

$$\frac{n(n-1)}{2} + \frac{m(m-1)}{2} - mn = 10$$

$n^2 + m^2 - 2mn - n - m = 20$

$(m-n)^2 - n - m = 20$

$n + m = 80$

3. If b is very small as compared to the value of a, so that the cube and other higher powers of $\frac{b}{a}$ can be neglected in the identity $\frac{1}{a-b} + \frac{1}{a-2b} + \frac{1}{a-3b} + ... + \frac{1}{a-nb} = \alpha n + \beta n^2 + \gamma n^3$, then the value of γ is : **[JEE(Main)July2021]**

यदि b, a से बहुत छोटा है, जिनके लिए निम्न सर्वसमिका $\frac{1}{a-b} + \frac{1}{a-2b} + \frac{1}{a-3b} + ... + \frac{1}{a-nb} = \alpha n + \beta n^2 + \gamma n^3$ में, $\frac{b}{a}$ की D;wvkS Åph?kkr kad hmi sjkkd ht kl d r hgs$r ksy बराबर है : **[JEE(Main)July2021]**

(A) $\frac{a^2+b}{3a^3}$ (V) $\frac{a+b^2}{3a^3}$ (C) $\frac{a+b}{3a^3}$ (D) $\frac{b^2}{3a^3}$

Ans. D

Sol. $\frac{1}{a}\left(\frac{1}{1-\frac{b}{a}}+\frac{1}{1-\frac{2b}{a}}+\frac{1}{1-\frac{3b}{a}}.....+\frac{1}{1-\frac{nb}{a}}\right)$

$=\frac{1}{a}\left[\left(1-\frac{b}{a}\right)^{-1}+\left(1-\frac{2b}{a}\right)^{-1}+\left(1-\frac{3b}{a}\right)^{-1}.....+\left(1-\frac{nb}{a}\right)\right]$

$=\frac{1}{a}\left[\left(1-\frac{b}{a}+\left(\frac{b}{a}\right)^2\right)+\left(1+\left(\frac{2b}{a}\right)+\left(\frac{2b}{a}\right)^2.....\left(1+\left(\frac{nb}{a}\right)+\left(\frac{nb}{a}\right)^2\right)\right)\right]$

$\because\left(\frac{b}{a}\right)^n=0 \qquad n\geq 3$

$=\frac{1}{a}\left[n+\frac{b}{a}(1+2+3.....n)+\left(\frac{b}{a}\right)^2[1^2+2^2.....n^2]\right]$

$=\frac{1}{a}\left[n+\frac{b}{a}\frac{n(n+1)}{2}+\frac{b^2}{a^2}\frac{n(n+1)(2n+1)}{6}\right]$

$=\frac{n}{a}+\frac{b}{2a^2}[n^2+n]+\frac{b^2}{6a^3}(2n^3+3n^2+n)$

$=\left(\frac{1}{a}+\frac{b}{2a^2}+\frac{b^2}{6a^3}\right)n+\left(\frac{b}{2a^2}+\frac{b^2}{2a^3}\right)n^2+\left(\frac{b^2}{3a^3}\right)n^3$

4. The lowest integer which is greater than $\left(1+\frac{1}{10^{100}}\right)^{10^{100}}$ is : **[JEE(Main)July2021]**

व्यंजक $\left(1+\frac{1}{10^{100}}\right)^{10^{100}}$ से अधिक न्यूनतम पूर्णांक होगा : **[JEE(Main)July2021]**

(A) 2 (B) 1 (C) 3 (D) 4

Ans. C

Sol. $\left(1+\frac{1}{10^{100}}\right)^{10^{100}}$

$x = 10^{100}$

$P=\left(1+\frac{1}{x}\right)^x$

$$f(x)=\begin{cases} n+3 & n \text{ विषम है} \\ \dfrac{n}{2} & n \text{ सम है} \end{cases}$$

$$\left(1+\frac{1}{x}\right)^{x}=2+\left(\frac{1}{\lfloor 2}-\frac{1}{\lfloor 2x^{2}}\right)+\left(\frac{1}{\lfloor 3}-\right)+\ldots..$$

$$P=2+\left(\text{Positive values less than } \frac{1}{\lfloor 2}+\frac{1}{\lfloor 3}+\frac{1}{\lfloor 4}+\infty\right)$$

$$e=1+\frac{1}{\lfloor 1}+\frac{1}{\lfloor 2}+\frac{1}{\lfloor 3}+\frac{1}{\lfloor 4}+$$

$$e-2=\frac{1}{\lfloor 2}+\frac{1}{\lfloor 3}+\frac{1}{\lfloor 4}+$$

$P=2+e-2$

$P=e\ (e\in(2,3))$

Lowest integer greater than $P=3$

5. If the greatest value of the term independent of 'x' in the expansion of $\left(x\sin\alpha+a\frac{\cos\alpha}{x}\right)^{10}$ is $\frac{10!}{(5!)^2}$, then the value of 'a' is equal to : **[JEE(Main)July2021]**

यदि $\left(x\sin\alpha+a\frac{\cos\alpha}{x}\right)^{10}$ के प्रसार में 'x' से स्वतंत्र पद का अधिकतम मान $\frac{10!}{(5!)^2}$ है, तो 'a' बराबर है :

[JEE(Main)July2021]

(A) 1 (B) -2 (C) 2 (D) -1

Ans. C

Sol. $T_{r+1}={}^{10}C_r\left(x\sin\alpha\right)^{10-r}\left(\frac{a\cos\alpha}{x}\right)^{r}$

$T_{r+1}={}^{10}C_r(\sin\alpha)^{10-r}\left(a\cos\alpha\right)^{r}x^{10-r-r}$

$10-2r=0=r=5$

$T_{5+1}={}^{10}C_5\left(\sin\alpha\right)^{5}\quad\left(a\cos\alpha\right)^{5}$

$={}^{10}C_5(\sin\alpha)^5\ a^5\ (\cos\alpha)^5$

$={}^{10}C_5\frac{1}{2^5}\times\left(\sin 2\alpha\right)^5\times a^5$

$\left(\frac{a}{2}\right)^5\times{}^{10}C_5=\frac{\lfloor 10}{(\lfloor 5)^2}$

$\Rightarrow a=2$

6. The sum of all those terms which are rational numbers in the expansion of $\left(2^{1/3}+3^{1/4}\right)^{12}$ is :

[JEE(Main)July2021]

$\left(2^{1/3}+3^{1/4}\right)^{12}$ के प्रसार में, उन सभी पदों, जो परिमेय संख्याएँ हैं, का योगफल है : **[JEE(Main)July2021]**

(A) 43 (B) 89 (C) 35 (D) 27

Ans. A

Sol. $\left(2^{1/3}+3^{1/4}\right)^{12}$

$T_{r+1} = {}^{12}C_r \left(2^{1/3}\right)^{12-r} \left(3^{1/4}\right)^r$

$= {}^{12}C_r \times 2^{\frac{12-r}{3}} \times 3^{r/4}$

$\frac{12-r}{3} = I \qquad \frac{r}{4} = I$

r = 0, 3, 6, 9, 12 r = 0, 4, 8, 12

r = 0, 12

$T_1 = {}^{12}C_0\, 2^4 \times 1 = 16$

$T_{13} = {}^{12}C_{12} \times 1 \times 3^3 = 27$

Sum = 16 + 27 = 43

7. If the coefficients of x^7 in $\left(x^2+\frac{1}{bx}\right)^{11}$ and x^{-7} in $\left(x-\frac{1}{bx^2}\right)^{11}$, $b \neq 0$, are equal, then the value of b is equal to :

[JEE(Main)July2021]

यदि $\left(x^2+\frac{1}{bx}\right)^{11}$, $b \neq 0$, में x^7 का गुणांक तथा $\left(x-\frac{1}{bx^2}\right)^{11}$, में x^{-7} का गुणांक बराबर हैं, तो b का मान बराबर है :

[JEE(Main)July2021]

(A) -2 (B) -1 (C) 2 (D) 1

Ans. D

Sol. $\left(x^2+\frac{1}{bx}\right)^{11} \qquad \Rightarrow T_{r+1} = {}^{11}C_r (x^2)^{11-r} \left(\frac{1}{bx}\right)^r$

$T_{r+1} = {}^{11}C_r \frac{1}{b^r} x^{22-3r}$

Coefficient of $x^7 = {}^{11}C_5 \cdot \frac{1}{b^5}$

$\left(x-\frac{1}{bx^2}\right)^{11} \qquad \Rightarrow T_{r+1} = {}^{11}C_r\, x^{11-r} \left(\frac{-1}{bx^2}\right)^r$

$$T_{r+1} = {}^{11}C_r \left(\frac{-1}{b}\right)^r x^{11-3r}$$

Coefficient of $x^{-7} = {}^{11}C_6 \cdot \frac{1}{b^6}$

${}^{11}C_5 \frac{1}{b^5} = {}^{11}C_6 \frac{1}{b^6} \Rightarrow b = 1$

8. A possible value of 'x', for which the ninth term in the expansion of

$\left\{3^{\log_3\sqrt{25^{x-1}+7}} + 3^{\left(-\frac{1}{8}\right)\log_3(5^{x-1}+1)}\right\}^{10}$ in the increasing powers of $3^{\left(-\frac{1}{8}\right)\log_3(5^{x-1}+1)}$

is equal to 180, is : **[JEE(Main)July2021]**

'x' का एक संभव मान, जिसके लिए व्यंजक $\left\{3^{\log_3\sqrt{25^{x-1}+7}} + 3^{\left(-\frac{1}{8}\right)\log_3(5^{x-1}+1)}\right\}^{10}$ के $3^{\left(-\frac{1}{8}\right)\log_3(5^{x-1}+1)}$ की बढ़ती घातों में प्रसार नौवां पद

180 के बराबर है, है : **[JEE(Main)July2021]**

(A) 1 (B) 0 (C) – 1 (D) 2

Ans. A

Sol. Required within term in the given expansion is

${}^{10}C_8 \cdot \beta \neq 0 = 180$

$\Rightarrow \frac{10\times 9}{2}\cdot\left(\sqrt{25^{x-1}+7}\right)^2\left(3^{-\log_3\left(5^{x-1}+1\right)}\right) = 180$

$\Rightarrow 45\,(25^{x-1}+7)\cdot\left(3^{\log_3\left(\frac{1}{5^{x-1}+1}\right)}\right) = 180$

$\Rightarrow 45\left(\left(5^{x-1}\right)^2+7\right)\cdot\left(\frac{1}{5^{x-1}+1}\right) = 180$

$\Rightarrow \frac{\left(\left(5^{x-1}\right)^7+7\right)}{5^{x-1}+1} = 4$

Put $5^{x-1} = t$

$\Rightarrow \frac{t^2+7}{t+1} = 4$

$\Rightarrow t^2 - 4t + 3 = 0$

$(t-1)(t-3) = 0$

$t = 1$ or $t = 3$

$5^{x-1} = 0$ or $5^{x-1} = 3$

$x - 1 = 0$ $x = (\log_5 3) + 1$

$x = 1$

9. The number of rational terms in the binomial expansion of $(4^{\frac{1}{4}} + 5^{\frac{1}{6}})^{120}$ is _____

[JEE(Main)July2021]

$(4^{\frac{1}{4}} + 5^{\frac{1}{6}})^{120}$ के द्विपर प्रसार में परिमेय पदों की संख्या है _____ **[JEE(Main)July2021]**

Ans. 21

Sol. $T_{r+1} = {}^{120}C_r (4^{1/4})^{120-r} (5^{1/6})^r$

Powers should be integral powers

$\frac{120-r}{2} = \text{integer}$ & $\frac{r}{6} = \text{Integer}$

$\Rightarrow r \in \{0, 6, 12, 18 .....120\}$

Total 21 rational terms

10. The number of elements in the set $\{n \in \{1,2,3,.....100\} | (11)^n > (10)^n + (9)^n\}$ is ____.

[JEE(Main)July2021]

समुच्चय $\{n \in \{1,2,3,.....100\} | (11)^n > (10)^n + (9)^n\}$ में अवयवों की संख्या ____ **[JEE(Main)July2021]**

Ans. 96

Sol. $11^n > 10^n + 9^n$

$\Rightarrow 11^n - 9^n > 10^n$

$\Rightarrow (10+1)^n - (10-1)^n > 10^n$

$\Rightarrow 2\left(n_{C_1} 10^{n-1} + n_{C_3} 10^{n-3} + n_{C_5} 10^{n-5} + ...\right) > 10^n$

$\Rightarrow \frac{1}{5}\left(n_{C_1} 10^n + n_{C_3} 10^{n-2} + n_{C_5} 10^{n-4} + ...\right) > 10^n$

$\Rightarrow \frac{1}{5}\left(n_{C_1} + n_{C_3} 10^{-2} + n_{C_5} 10^{-4} + ...\right) > 1$

$\Rightarrow$ Clearly this is true for $n \geq 5$

hence such numbers in given interval are 96.

11. If the constant term, in binomial expansion of $\left(2x^r + \frac{1}{x^2}\right)^{10}$ is 180, then r is equal to.

[JEE(Main)July2021]

यदि $\left(2x^r + \frac{1}{x^2}\right)^{10}$ के द्विपद प्रसार में अचर पद 180 है, तो r बराबर है _____ **[JEE(Main)July2021]**

Ans. 8

Sol. Let, T_{n+1} be the constant term

$T_{n+1} = 180$

$${}^{10}C_n \left(2x^r\right)^n \left(\frac{1}{x^2}\right)^{10-n} = 180$$

$${}^{10}C_n \left(2^n\right)\left(x^{rn-20+2n}\right) = 180$$

Now by comparision

${}^{10}C_n\, 2^n = 180$ .....(1)

$2n + rn - 20 = 0$ .....(2)

$n = 2$

$r = 8$

12. The term independent of 'x' in the expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right)^{10}$, where $x \neq 0, 1$ is equal to _____.

[JEE(Main)July2021]

$\left(\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right)^{10}$, $x \neq 0, 1$ के प्रसार में 'x' से स्वतंत्र पद बराबर है _____ **[JEE(Main)July2021]**

Ans. 210

Sol. $$\left(\left(x^{1/3}+1\right) - \frac{(\sqrt{x}+1)}{\sqrt{x}}\right)^{10}$$

$$\left(x^{\frac{1}{3}} - \frac{1}{x^{\frac{1}{2}}}\right)^{10}$$

$$T_{r+1} = {}^{10}C_r\,(x)^{\frac{10-r}{3}}\,(x)^{-\frac{r}{2}}$$

$$= {}^{10}C_r (x)^{\frac{10-r}{3}-\frac{r}{2}}$$

$$\frac{10-r}{3} - \frac{r}{2} = 0$$

$20 - 2r - 3r = 0$

$r = 4$

T_5 is independent of 'x' ${}^{10}C_4 = 210$

$20 - 2r - 3r = 0$

$r = 4$

13. The ratio of the coefficient of the middle term in the expansion of $(1 + x)^{20}$ and the sum of the coefficients of two middle terms in expansion of $(1 + x)^{19}$ is _____. **[JEE(Main)July2021]**

$(1 + x)^{20}$ के प्रसार में मध्य पद का गुणांक तथा $(1 + x)^{19}$ के प्रसार में दो मध्य पदों के गुणांकों के योग का अनुपात है _____

[JEE(Main)July2021]

Ans. 1

Sol. $\frac{^{20}C_{10}}{^{19}C_9 + ^{19}C_{10}} = \frac{^{20}C_{10}}{^{20}C_{10}} = 1$

14. If the co-efficients of x^7 and x^8 in the expansion of $\left(2+\frac{x}{3}\right)^n$ are equal, then the value of n is equal to _____.

[JEE(Main)July2021]

यदि $\left(2+\frac{x}{3}\right)^n$ के प्रसार में x^7 तथा x^8 के गुणांक बराबर हैं, तो n बराबर है _____ **[JEE(Main)July2021]**

Ans. 55

Sol. $\left(2+\frac{x}{3}\right)^n$

$T_{r+1} = {}^nC_r (2)^{n-r}\left(\frac{x}{3}\right)^r$

$r = 7 \quad r = 8$

$T_8 = {}^nC_7 (2)^{n-7} (3)^{-7} (x)^7$

$T_9 = {}_nC^8 (2)^{n-8} (x)^8$

${}^nC_7 \times (2)^{n-7} (3)^{-7} = {}^nC_8 (2)^{n-8} (3)^{-8}$

$$\frac{^nC_7 \times 2^{n-7}}{3^7} = \frac{^nC_8 \times 2^{n-8}}{3^8}$$

$$6 \times \frac{\lfloor n}{\lfloor 7 \lfloor n-7} = \frac{\lfloor n}{\lfloor 8 \lfloor n-8}$$

$$\frac{6}{n-7} = \frac{1}{8}$$

$48 + 7 = n = n = 55$

15. Let $n \in N$ and [x] denote the greatest integer less than or equal to x. If the sum of (n + 1) terms $^nC_0, 3.^nC_1, 5.^nC_2, 7.^nC_3, ...$ is equal to $2^{100} .101$, then $2\left[\frac{n-1}{2}\right]$ is equal to _____.

[JEE(Main)July2021]

माना $n \in N$ तथा [x] महत्तम पूर्णांक $\le x$ हैं। यदि (n + 1) पदों $^nC_0, 3.^nC_1, 5.^nC_2, 7.^nC_3, ...$ का योग $2^{100} .101$ है, तो $2\left[\frac{n-1}{2}\right]$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 98

Sol. $S = {}^nC_0 + 3\,{}^nC_1 + 5\,{}^nC_2 + \ldots\ldots\ldots + (2n+1)\,{}^nC_n$

$S = (2n+1)\,{}^nC_n + (2n-1)\,{}^nC_{n-1} + \ldots\ldots\ldots + {}^nC_0$

$S = (2n+1)\,{}^nC_0 + (2n-1)\,{}^nC_1 + \ldots\ldots\ldots + {}^nC_n$

$2S = (2n+2)({}^nC_0 + {}^nC_1 + \ldots\ldots\ldots + {}^nC_n)$

$2S = 2(n+1) \times 2^n$

$S = (n+1)2^n = 2^{100} \times 101$

$n = 100$

$$2\left[\frac{n-1}{2}\right] = 2\left[\frac{100-1}{2}\right] = 2\left[\frac{99}{2}\right] = 98$$

16. If ${}^{20}C_r$ is the co–efficient of x^r in the expansion of $(1+x)^{20}$, then the value of $\sum_{r=0}^{20} r^2\,{}^{20}C_r$ is equla to : **[JEE(Main)Aug.2021]**

यदि $(1+x)^{20}$ के प्रसार में x^r का गुणांक ${}^{20}C_r$ है, तो $\sum_{r=0}^{20} r^2\,{}^{20}C_r$ का मान बराबर है : **[JEE(Main)Aug.2021]**

(A) 380×2^{18} (B) 420×2^{19} (C) 380×2^{19} (D) 420×2^{18}

Ans. D

Sol. $$\sum_{r=0}^{20} r^2\,{}^{20}C_r = \sum_{r=0}^{20} r^2\left(\frac{20}{r}\right)({}^{19}C_{r-1})$$

$$= \Sigma\, 20(r-1)\,{}^{19}C_{r-1} + \Sigma\, 20\,{}^{19}C_{r-1}$$

$$= \Sigma\,(r-1)\left(\frac{19}{r-1}\right){}^{18}C_{r-2} + (20)\Sigma\,{}^{19}C_{r-1}$$

$$= (20)(19)\Sigma\,{}^{19}C_{r-2} + 20\Sigma\,{}^{19}C_{r-1}$$

$$= (20)(19)2^{18} + (20)(2^{19})$$

$$= 2^{19}(190+20) = 210(2^{19}) = 420(2^{18})$$

17. $\sum_{k=0}^{20} ({}^{20}C_k)^2$ is equal to : **[JEE(Main)Aug.2021]**

$\sum_{k=0}^{20} ({}^{20}C_k)^2$ के बराबर है : **[JEE(Main)Aug.2021]**

(A) ${}^{40}C_{20}$ (B) ${}^{41}C_{20}$ (C) ${}^{40}C_{21}$ (D) ${}^{40}C_{19}$

Ans. A

Sol. $$\sum_{k=0}^{20} {}^{20}C_k \,.\, {}^{20}C_k = \sum_{k=0}^{20} {}^{20}C_k \,.\, {}^{20}C_{20-k}$$

$$\Rightarrow {}^{20}C_0\,{}^{20}C_{20} + {}^{20}C_1\,{}^{20}C_{19} + \ldots\ldots\ldots + {}^{20}C_{20}\,{}^{20}C_0$$

$$\Rightarrow {}^{40}C_{20}$$

18. $3\times7^{22}+2\times10^{22}-44$ when divided by 18 leaves the remainder _____. **[JEE(Main)Aug.2021]**

$3\times7^{22}+2\times10^{22}-44$ को 18 से भाग देने पर शेषफल _____है। **[JEE(Main)Aug.2021]**

Ans. 15

Sol. $= (6+1)^{22} + 2(9+1)^{22} - 44$

$= 18I_1 + 3 + 18I_2 + 2 - 44$

$= 18(I_1 + I_2) + 5 - 44$

$= 18I_3 - 39$

$= 18I_3 - 18 \times 3 - 39 + 54$

$= 18(I_3 - 3) + 15$

hence remainder is 15

19. If $\left(\frac{3^6}{4^4}\right)$k is term, independent of x, in the binomial expansion of $\left(\frac{x}{4}-\frac{12}{x^2}\right)^{12}$, then k is equal to ____.

[JEE(Main)Aug.2021]

यदि $\left(\frac{x}{4}-\frac{12}{x^2}\right)^{12}$ के द्विपद प्रसार में x से स्वतंत्र पद $\left(\frac{3^6}{4^4}\right)$k है, तो k बराबर है____। **[JEE(Main)Aug.2021]**

Ans. 55

Sol. $T_{r+1} = {}^{12}C_r \left(\frac{x}{4}\right)^{12-r} \left(\frac{-12}{x^2}\right)^r$

$T_{r+1} = {}^{12}C_r \frac{(-12)^r}{4^{12-r}}.x^{12-3r}$

$12-3r=0 \Rightarrow r=4$

$T_5 = {}^{12}C_4 \frac{12^4}{4^8} = \frac{3^6}{4^4}.k \quad \Rightarrow k = 55$

20. Let $\binom{n}{k}$ denote ${}^nC_k$ and $\left[\begin{matrix} n \\ k \end{matrix}\right] = \begin{cases} \binom{n}{k}, & \text{if } 0 \le k \le n \\ 0, & \text{otherwise} \end{cases}$.

If $A_k = \sum_{i=0}^{9} \binom{9}{i} \left[\begin{matrix} 12 \\ 12-k+i \end{matrix}\right] + \sum_{i=0}^{8} \binom{8}{i} \left[\begin{matrix} 13 \\ 13-k+i \end{matrix}\right]$ and $A_4 - A_3 = 190$ p, then p is equal to ____.

[JEE(Main)Aug.2021]

माना $\binom{n}{k}$, ${}^nC_k$ को दर्शाता है तथा $\left[\begin{matrix} n \\ k \end{matrix}\right] = \begin{cases} \binom{n}{k}, & \text{यदि } 0 \le k \le n \\ 0, & \text{अन्यथा} \end{cases}$ है।

यदि $A_k = \sum_{i=0}^{9}\binom{9}{i}\begin{bmatrix}12\\12-k+i\end{bmatrix}+\sum_{i=0}^{8}\binom{8}{i}\begin{bmatrix}13\\13-k+i\end{bmatrix}$ तथा $A_4 - A_3 = 190\,p$ है, तो p बराबर है____।

[JEE(Main)Aug.2021]

Ans. 49

Sol. $A_k = \sum_{i=0}^{9} {}^{9}C_i {}^{12}C_{k-i} + \sum_{i=0}^{8} {}^{8}C_i {}^{13}C_{k-i}$

$A_k = {}^{21}C_k + {}^{21}C_1 = 2\cdot{}^{21}C_k$

$A_4 - A_3 = 2\left({}^{21}C_4 - {}^{21}C_3\right) = 2(5985-1330)$

$190\,p = 2\,(5985 - 1330)$

$\Rightarrow p = 49$

21. If the coefficient of a^7b^8 in the expansion of $(a + 2b + 4ab)^{10}$ is $K\cdot 2^{16}$, then K is equal to_____.

[JEE(Main)Aug.2021]

यदि $(a + 2b + 4ab)^{10}$ के प्रसार में a^7b^8 का गुणांक $K\cdot 2^{16}$ है, तो K बराबर है _____। **[JEE(Main)Aug.2021]**

Ans. 315

Sol. $\frac{\lfloor 10}{\lfloor P_1 \lfloor P_2 \lfloor P_3}(a)^{P_1}(2b)^{P_2}(4ab)^{P_3}$

$P_1 + P_3 = 7 \qquad P_2 + P_3 = 8$

$P_1 + P_2 + P_3 = 10$

$7 + P_2 = 10$

$P_2 = 3 \qquad P_3 = 5$

$P_1 = 2$

$\frac{\lfloor 10}{\lfloor 2 \lfloor 3 \lfloor 5} \times 2^{13}$

$\frac{10\times 9\times 8\times 7\times 6}{6\times 2} \times 2^{13}$

$5 \times 9 \times 7 \times 2^{16} = K \times 2^{16}$

$K = 5 \times 9 \times 7$

$K = 45 \times 7 = 315$

22. If the sum of the coefficients in the expansion of $(x + y)^n$ is 4096, then the greatest coefficient in the expansion is_____.

[JEE(Main)Aug.2021]

यदि $(x + y)^n$ के प्रसार में गुणांकों का योगफल 4096 है, तब प्रसार में महत्तम गुणांक है_____

Ans. 924

Sol. for sum of coefficients

put $x = y = 1$

$(1+)^n = 4096$

$\Rightarrow 2^n = 4096$

$\Rightarrow 2^n = 2^{12}$

$n = 12$

Greatest coefficient = coefficient of middle term

$= {}^{12}C_6$

$= 924.$

23. The remainder when 3^{2022} is divided by 5 is : **[JEE(Main)June2022]**

3^{2022} को 5 से विभाजित करने पर शेषफल है : **[JEE(Main)June2022]**

(1) 1 (2) 2 (3) 3 (4) 4

Ans. Official Answer NTA (4)

Sol. $3^{2022} = 3^2 (3^{2020}) = 9(81)^{505}$

$= 9(80 + 1)^{505}$

$= 9(80k_1 + 1)$

$= 9(5k_2 + 1)$

$= 5k_3 + 9 = 5k_3 + 5 + 4 = 5k_4 + 4$

remainder = 4

24. The coefficient of x^{101} in the expression $(5 + x)^{500} + x(5 + x)^{499} + x^2(5 + x)^{498} + ... + x^{500}$, $x > 0$, is : **[JEE(Main)June2022]**

व्यंजक $(5 + x)^{500} + x(5 + x)^{499} + x^2(5 + x)^{498} + ... + x^{500}$, $x > 0$ में x^{101} का गुणांक है : **[JEE(Main)June2022]**

(1) ${}^{501}C_{101}\ (5)^{399}$ (2) ${}^{501}C_{101}\ (5)^{400}$ (3) ${}^{501}C_{100}\ (5)^{400}$ (4) ${}^{500}C_{101}\ (5)^{399}$

Ans. Official Answer NTA (1)

Sol. $(5 + x)^{500} + x\,(5 + x)^{499} + x^2(5 + x)^{498} + ... + x^{500}$

$$= (5+x)^{500}\left\{\frac{1-\left(\frac{x}{5+x}\right)^{501}}{1-\frac{x}{5+x}}\right\}$$

$$= \frac{1}{5}\left\{(5+x)^{501} - x^{501}\right\}$$

coff. at $x^{101} = \frac{1}{5}\ {}^{501}C_{101} \cdot 5^{400}$

$= {}^{501}C_{101}\ 5^{399}$

25. The remainder when $(2021)^{2023}$ is divided by 7 is : **[JEE(Main)June2022]**

$(2021)^{2023}$ को 7 से विभाजित करने पर शेषफल है : **[JEE(Main)June2022]**

(1) 1 (2) 2 (3) 5 (4) 6

Ans. Official Answer NTA (3)

Sol. $(2021)^{2023} = (7\lambda - 2)^{2023}$

$= {}^{2023}C_0(7\lambda)^{2023} - \ldots\ldots\ {}^{2023}C_{2023}2^{2023}$

$= 7t - 2^{2023}$

$\therefore -2^{2023} = -2 \times 2^{2022}$

$= -2 \times \left(2^3\right)^{674}$

$= -2(1 + 7\mu)^{674}$

$= -(7\alpha + 2)$

$\Rightarrow$ remainder $= -2$ or $+5$

26. If $\sum_{k=1}^{31}\left({}^{31}C_k\right)\left({}^{31}C_{k-1}\right) - \sum_{k=1}^{30}\left({}^{30}C_k\right)\left({}^{30}C_{k-1}\right) = \frac{\alpha(60!)}{(30!)(31!)}$, where $a \in R$, then the value of 16α is equal to :

[JEE(Main)June2022]

यदि $\sum_{k=1}^{31}\left({}^{31}C_k\right)\left({}^{31}C_{k-1}\right) - \sum_{k=1}^{30}\left({}^{30}C_k\right)\left({}^{30}C_{k-1}\right) = \frac{\alpha(60!)}{(30!)(31!)}$ है, जहाँ $a \in R$ है, तो 16α का मान बराबर है :

[JEE(Main)June2022]

(1) 1411 (2) 1320 (3) 1615 (4) 1855

Ans. Official Answer NTA (1)

Sol. $\sum_{k=1}^{31} {}^{31}C_k \cdot {}^{31}C_{k-1} = \sum_{k=1}^{31} {}^{31}C_k \cdot {}^{31}C_{32-k}$

$=$ coefficient of x^{32} in $(1 + x)^{62}$

$= {}^{62}C_{32}$

Similary $\sum_{k=1}^{30} {}^{30}C_k \cdot {}^{30}C_{31-k} = {}^{60}C_{31}$

So ${}^{62}C_{32} - {}^{60}C_{31} = \frac{62}{32} \cdot \frac{61!}{30!31!}\left[\frac{62.61}{32} - 30\right]$

$16\alpha = 31 \times 61 - 16 \times 30 = 1411$

27. The term independent of x in the expansion of $\left(1-x^2+3x^3\right)\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}$, $x \neq 0$ is :

[JEE(Main)June2022]

$\left(1-x^2+3x^3\right)\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}$, $x \neq 0$ के प्रसार में से स्वतंत्र पद है : **[JEE(Main)June2022]**

(1) $\frac{7}{40}$ (2) $\frac{33}{200}$ (3) $\frac{39}{200}$ (4) $\frac{11}{50}$

Ans. Official Answer NTA (2)

Sol. $\left(1-x^2+3x^2\right)\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}$, $x \neq 0$

general term in $\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)$ is

$$T_{r+1} = {}^{11}C_r\left(\frac{5}{2}x^3\right)^{11-r}\left(\frac{-1}{5x^2}\right)^r$$

$$= {}^{11}C_r\left(\frac{5}{2}\right)^{11-r}\left(-\frac{1}{5}\right)^r x^{33-5r}$$

So term independent of x in given expression

$$= {}^{-11}C_7\left(\frac{5}{2}\right)^4\left(-\frac{1}{5}\right)^7 = \frac{11\times10\times9\times8}{24}\times\frac{1}{16\times125}$$

$$= \frac{33}{200}$$

28. If the constant term in the expansion of $\left(3x^3-2x^2+\frac{5}{x^5}\right)^{10}$ is $2^k \cdot l$, where l is an odd integer, then the value of k is equal to : **[JEE(Main)June2022]**

यदि $\left(3x^3-2x^2+\frac{5}{x^5}\right)^{10}$ के प्रसार में अचर पद $2^k \cdot l$ है, जहाँ l एक विषम पूर्णांक है, तो k का मान बराबर है :

[JEE(Main)June2022]

(1) 6 (2) 7 (3) 8 (4) 9

Ans. Official Answer NTA (4)

Sol. $\left(3x^2-2x^2+\frac{5}{x^5}\right)^{10}$ its constant term is $2^k.l$

its general term $= \frac{10!}{r_1!r_2!r_3!}\left(3x^3\right)^{r_1}\left(-2x^2\right)^{r_2}\left(\frac{5}{x^2}\right)^{r_3}$

where $r_1 + r_2 + r_3 = 10$

$= \frac{10!}{r_1!r_2!r_3!} 3^{r_1} (-2)^{r_2} 5^{r_2} x^{3r_1+2r_2-5r_3}$

for constant term $= 3r_1 + 2r_2 - 5r_3 = 0$

$\ell.2^k = \frac{10!}{r_1!r_2!r_3!} 3^{r_1} (-2)^{r_2} .5^{r_3}$ $\quad r_1 = 1, r_2 = 8$ $r_3 = 1$ $r_2 =$ even

$= 10.9.3.5.2^8$

l. $2^k = 5^2.3^3.2^9$ $\quad r_1 = 5r_3 = 0$ $r_2 = 10$

l. $2^k = 5^2 . 3^3 . 2^9$ $\quad r_1 = r_3 = 0$ $r_2 = 10$

$\quad\quad\quad

Sol. $1+3+3^2+.....+3^{2021} = \frac{1}{2}\left[3^{2022}-1\right]$

$= \frac{1}{2}\left[9^{1011}-1\right]$

$= \frac{1}{2}\left[(10-1)^{1011}-1\right]$

$= \frac{1}{2}\left[100k+10110-1-1\right]$

$= 50k + 5055 - 1$

$= 50k + 5054$

$= 50k + 5050 + 4$

$= 50k_1 + 4$

hence remainder is =4

31. Let C_r denote the binomial coefficient of x^r in the expansion of $(1 + x)^{10}$. If for $\alpha, \beta \in R$, $C_1 + 3.2\ C_2 + 5.3\ C_3 + ...$up 10 terms $= \frac{\alpha \times 2^{11}}{2^\beta - 1}\left(C_0 + \frac{C_1}{2} + \frac{C_2}{3} + ...\text{upto 10 terms}\right)$ then the value of $\alpha + \beta$ is equal to _____. **[JEE(Main)June2022]**

माना C_r, $(1 + x)^{10}$ के प्रसार में x^r के द्विपद गुणांक को दर्शाता है। यदि $\alpha, \beta \in R$ के लिए,

$C_1 + 3.2\ C_2 + 5.3\ C_3 + ...(10 \text{ पदों तक}) = \frac{\alpha \times 2^{11}}{2^\beta - 1}\left(C_0 + \frac{C_1}{2} + \frac{C_2}{3} + ...10 \text{ पदों तक}\right)$ है, तो $\alpha + \beta$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (286)

Sol. $(1+x)^{10} = C_0 + C_1x + C_2x^2 + ... + C_{10}x^{10}$

Differentiating

$10(1+x)^9 = C_1 + 2C_2x + 3C_3x^2 + ... + 10C_{10}x^9$

replace $x \to x^2$

$10\left(1+x^2\right)^9 = C_1 + 2C_2x^2 + 3C_3x^4 + ... + 10C_{10}x^{18}$

$10 \cdot x\left(1+x^2\right)^9 = C_1x + 2C_2x^3 + 3C_3x^5 + ... + 10C_{10}x^{19}$

Differentiating

$10\left(\left(1+x^2\right)^9 \cdot 1 + x \cdot 9\left(1+x^2\right)^8 2x\right)$

$= C_1x + 2C_2 \cdot 3x^3 + 3 \cdot 5 \cdot C_3x^4 + ... + 10 \cdot 19C_{10}x^{18}$

putting $x = 1$

$10\left(2^9 + 18 \cdot 2^8\right)$

$= C_1 + 3 \cdot 2 \cdot C_2 + 5 \cdot 3 \cdot C_3 + ... + 19 \cdot 10 \cdot C_{10}$

$C_1 + 3\cdot 2\cdot C_2 + \ldots\ldots + 19\cdot 10\cdot C_{10}$

$= 10\cdot 2^9\cdot 10 = 100\cdot 2^9$

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots.. + \frac{C_9}{11} + \frac{C_{10}}{11} = \frac{2^{11}-1}{11}$$

$\uparrow$ $\uparrow$

10th term 11th term

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots.. + \frac{C_9}{11} = \frac{2^{11}-2}{11}$$

Now, $100\cdot 2^9 = \frac{\alpha\cdot 2^{11}}{2^{\beta}-1}\left(\frac{2^{11}-2}{11}\right)$

Eqn. of form $y = k(2^x - 1)$.

It has infinite solutions even if we take $x, y \in N$.

32. If the sum of the co-efficients of all the positive even power of x in the binomial expansion of $\left(2x^3 + \frac{3}{x}\right)^{10}$ is $5^{10} - \beta\cdot 3^9$, then β is equal to _____. **[JEE(Main)June2022]**

यदि $\left(2x^3 + \frac{3}{x}\right)^{10}$ के द्विपद प्रसार में x की सभी धनात्मक सम घातों के गुणांकों का योगफल $5^{10} - \beta\cdot 3^9$ है, तो β बराबर है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (83)

Sol. $T_{r+1} = {}^{10}C_r (2x^3)^{10-r}\left(\frac{3}{x}\right)^r$

$T_{r+1} = {}^{10}C_r\, 2^{10-r}\cdot 3^r x^{30-4r}$

for positive even powers

$r \neq 8, 9, 10$

Sum of coeff. $= (2+3)^{10} = 5^{10}$

Sum of coeff. of positive even powers

$= \text{Total} - (\text{sum for } r = 8, 9, 10)$

$= 5^{10} - ({}^{10}C_8\, 2^2\cdot 3^8 + {}^{10}C_9\, 2.3^9 + {}^{10}C_{10}\, 3^{10})$

$= 5^{10} - 83.3^9$

So, $\beta = 83$

33. If $\left({}^{40}C_0\right) + \left({}^{41}C_1\right) + \left({}^{42}C_2\right) + \ldots.. + \left({}^{60}C_{20}\right) = \frac{m}{n}\,{}^{60}C_{20}$ m and n are coprime, then m + n is equal to _____. **[JEE(Main)June2022]**

यदि $\left({}^{40}C_0\right) + \left({}^{41}C_1\right) + \left({}^{42}C_2\right) + \ldots.. + \left({}^{60}C_{20}\right) = \frac{m}{n}\,{}^{60}C_{20}$ है m, n असहभाज्य हैं, तो m + n बराबर है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (102)

Sol. $^{40}C_0 + ^{41}C_1 + ... + ^{60}C_{20} = \left(\frac{m}{n}\right)^{60}C_{20}$

$^{41}C_0 + ^{41}C_1 + ^{42}C_2 + ... + ^{60}C_{20}$

$^{42}C_1 + ^{42}C_2 + ... + ^{60}C_{20} = ^{60}C_{20}$

$\frac{m}{n} \times \frac{\lfloor 60}{\lfloor 20 \lfloor 40} = \frac{\lfloor 61}{\lfloor 20 \lfloor 41}$

$\frac{m}{n} \times \frac{61}{41}$

$m + n = 102$

34. If the coefficient of x^{10} in the binomial expansion of $\left(\frac{\sqrt{x}}{5^{\frac{1}{4}}} + \frac{\sqrt{5}}{x^{\frac{1}{3}}}\right)^{60}$ is $5^k.l$, where $l, k \in N$ and l is co-prime to 5, then k is equal to _____. **[JEE(Main)June2022]**

यदि $\left(\frac{\sqrt{x}}{5^{\frac{1}{4}}} + \frac{\sqrt{5}}{x^{\frac{1}{3}}}\right)^{60}$ के द्विपद प्रसार में x^{10} का गुणांक $5^k.l$ है, जहाँ $l, k \in N$ तथा l तथा l, 5 सहअभाज्य (co-prime) हैं, तो k बराबर है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (5)

Sol. $\left(\frac{\sqrt{x}}{5^{\frac{1}{4}}} + \frac{\sqrt{5}}{x^{\frac{1}{3}}}\right)^{60}$

$^{60}C_r \left(\frac{x^{\frac{1}{2}}}{5^{\frac{1}{4}}}\right)^{60-r} \left(\frac{5^{\frac{1}{2}}}{x^{\frac{1}{3}}}\right)^r$

$^{60}C_r 5^{-\left(\frac{60-r}{4}\right)}.5^{\frac{r}{2}}.x^{\frac{60-r}{2}}.x^{-\frac{r}{3}}$

$^{60}C_r 5^{\frac{3r-60}{4}}.x^{\frac{180-5r}{6}}$

$5r = 120$

$r = 24$

$^{60}C_{24} 5^3 = \frac{60!}{24!36!}5^3$

Now exponent of 5 in 60! $= \left[\frac{60}{5}\right] + \left[\frac{60}{25}\right] + \left[\frac{60}{125}\right] = 12 + 2 = 14$

$$\text{exponent of 5 in 24!} = \left[\frac{24}{5}\right] + \left[\frac{24}{25}\right] = 4 + 0 = 4$$

$$\text{exponent of 5 in 36!} = \left[\frac{36}{5}\right] + \left[\frac{36}{25}\right] = 7 + 1 = 8$$

coeff of x^{10} will be = exponent of 5

$= 3 + 14 - 4 - 8$

$\Rightarrow 5$

35. If the sum of coefficients of all positive powers of x, in the Binomial expansion of $\left(x^n + \frac{2}{x^5}\right)^7$ is 939, then the sum of all the possible integral value of n is _____. **[JEE(Main)June2022]**

यदि $\left(x^n + \frac{2}{x^5}\right)^7$ के द्विपद प्रसार में x की सभी धनात्मक घातों के गुणांकों का योग 939 है, तो n के सभी संभव पूर्णांक मानों का योग है _____. **[JEE(Main)June2022]**

Ans. Official Answer NTA (57)

Sol. $\left(x^n + \frac{2}{x^5}\right)^7 = {}^7C_0 x^{7n} + {}^7C_1 2^1 x^{6n-5} + {}^7C_2 .2^2 x^{5n-10} + {}^7C_3 2^3 x^{4n-15} + {}^7C_4 2^4 x^{3n-20}$

$+ {}^7C_5 2^5 x^{2n-25} + {}^7C_6 2^6 x^{n-30} + {}^7C_7 2^7 x^{-35}$

${}^7C_0 = 1$ ${}^7C_2(2^2) = 84$ ${}^7C_4(2^4) = 560$

${}^7C_1(2) = 14$ ${}^7C_3(2^3) = 280$

Sum = 939

$3n - 20 \geq 0$ & $2n - 25 < 0$

$n \geq \frac{20}{3}$ $n < 12.5$

$\Rightarrow n = 7, 8, 9, 10, 11, 12$

Sum = 57

36. The number of positive integers k such that the constant term in the binomial expansion of $\left(2x^3 + \frac{3}{x^k}\right)^{12}, x \neq 0$ is $2^8 . \ell$, where ℓ is an odd integer, is _____. **[JEE(Main)June2022]**

धनात्मक पूर्णांकों k जिनके लिए $\left(2x^3 + \frac{3}{x^k}\right)^{12}, x \neq 0$ के द्विपद प्रसार में अचर पद $2^8 . \ell$ है, जहाँ ℓ एक विषम संख्या है, की संख्या है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (2)

Sol. $\left(2x^3 + \frac{3}{x^k}\right)^{12}$

$t_{r+1} = {}^{12}C_r \left(2x^3\right)^r \left(\frac{3}{x^k}\right)^{12-r}$

$x^{3r-(12-r)k} \rightarrow$ constant

$\therefore$ 3r -12k + rk = 0

$\Rightarrow k = \frac{3r}{12-r}$

$\therefore$ possible values of r are 3,6,8,9,10 and corresponding values of k are 1,3,6,9,15

Now ${}^{12}C_r$ = 220, 924, 495, 220, 66

$\therefore$ possible values of k for which we will get 2^8 are 3, 6

37. Let the coefficients of x^{-1} and x^{-3} in the expansion of $\left(2x^{\frac{1}{5}} - \frac{1}{\frac{1}{x^5}}\right)^{15}$, $x > 0$, be m and n respectively. If r is a positive integer such that $mn^2 = {}^{15}C_r \cdot 2^r$, then the value of r is equal to _____. **[JEE(Main)June2022]**

माना $\left(2x^{\frac{1}{5}} - \frac{1}{\frac{1}{x^5}}\right)^{15}$, $x > 0$ के प्रसार में x^{-1} तथा x^{-3} के गुणांक क्रमशः m तथा n है। यदि $\left(2x^{\frac{1}{5}} - \frac{1}{\frac{1}{x^5}}\right)^{15}$, $x > 0$, धनपूर्णांक r के लिए $mn^2 = {}^{15}C_r \cdot 2^r$ है, तो r का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (5)

Sol. $\left(2x^{\frac{1}{5}} - \frac{1}{\frac{1}{x^5}}\right)^{15}$, $x > 0$

$T_{r+1} = {}^{15}C_r \left(2x^{\frac{1}{5}}\right)^{15-r} \left(\frac{-1}{x^{\frac{1}{5}}}\right)^r$

for coefficient of x^{-1}

$\frac{15-r}{5} - \frac{r}{5} = -1 \Rightarrow 15 - 2r = -5 \Rightarrow r = 10$

$\therefore m = {}^{15}C_{10} \cdot 2^5$

& coefficient of x^{-3}, $\frac{15-2r}{5} = -3$

r = 15

$n = -{}^{15}C_{15}$

Given, $mn^2 = {}^{15}C_r \cdot 2^r$

$^{15}C_{10}.2^5 = {}^{15}C_r.2^r$

$\Rightarrow r = 5$

So, $(a-b)^2 = \left(5\sqrt{5}-3\sqrt{5}\right)^2 = 3.4 = 12$

38. The remainder when $(11)^{1011} + (1011)^{11}$ is divided by 9 is : **[JEE(Main)July2022]**

$(11)^{1011} + (1011)^{11}$ को 9 से विभाजित करने पर शेषफल है : **[JEE(Main)July2022]**

(1) 1 (2) 4 (3) 6 (4) 8

Ans. Official Answer NTA (4)

Sol. $\text{Re}\left(\frac{(11)^{1011}+(1011)^{11}}{9}\right) = \text{Re}\left(\frac{2^{1011}+3^{11}}{9}\right)$

For $\text{Re}\left(\frac{2^{1011}}{9}\right)$

$2^{1011} = (9-1)^{337} = {}^{337}C_0 9^{337}(-1)^0 + {}^{337}C_1 9^{336}(-1)^1 + {}^{337}C_2 9^{335}(-1)^2 + \ldots + {}^{337}C_{337} 9^0(-1)^{337}$

so, remainder is 8

and $\text{Re}\left(\frac{3^{11}}{9}\right) = 0$

so, remainder is 8

39. $\sum_{\substack{i,j=0 \\ i\neq j}}^{n} {}^nC_i \, {}^nCj$ is equal to : **[JEE(Main)July2022]**

$\sum_{\substack{i,j=0 \\ i\neq j}}^{n} {}^nC_i \, {}^nCj$ बराबर है : **[JEE(Main)July2022]**

(1) $2^{2n} - {}^{2n}C_n$ (2) $2^{2n-1} - {}^{2n-1}C_{n-1}$

(3) $2^{2n} - \frac{1}{2}{}^{2n}C_n$ (4) $2^{n-1} + {}^{2n-1}C_n$

Ans. Official Answer NTA (2)

Sol. $\sum_{\substack{i,j=0 \\ i\neq j}}^{n} {}^nC_i \, {}^nC_j$

$= \sum_{i=0}^{n} {}^nC_i . \sum_{j=0}^{n} {}^nC_j - \sum_{i=j=0}^{n} \left({}^nC_i\right)^2$

$= (2^n)(2^n) - {}^{2n}C_n$

$= 2^{2n} - {}^{2n}C_n$

40. The remainder when $(2021)^{2022} + (2022)^{2021}$ is divided by 7 is : **[JEE(Main)July2022]**

$(2021)^{2022} + (2022)^{2021}$ को 7 से विभाजित करने पर शेषफल है : **[JEE(Main)July2022]**

(1) 0 (2) 1 (3) 2 (4) 6

Ans. Official Answer NTA (1)

Sol. $(2021)^{2022} + (2022)^{2021}$

$= (2023 - 2)^{2022} + (2023 - 1)^{2021}$

$= 7n_1 + 2^{2022} + 7n_2 - 1$

$= 7(n_1 + n_2) + 8^{674} - 1$

$= 7(n_1 + n_2) + (7 - 1)^{674} - 1$

$= 7(n_1 + n_2) + 7n_3 + 1 - 1$

$= 7(n_1 + n_2 + n_3)$

$\therefore$ Given number is divisible by 7 hence remainder is zero

41. The remainder when $7^{2022} + 3^{2022}$ is divided by 5 is : **[JEE(Main)July2022]**

$7^{2022} + 3^{2022}$ को 6 से विभाजित करने पर शेषफल है : **[JEE(Main)July2022]**

(1) 0 (2) 2 (3) 3 (4) 4

Ans. Official Answer NTA (3)

Sol. $7^{2022} + 3^{2022}$

$= (49)^{1011} + (9)^{1011}$

$= (50 - 1)^{1011} + (10 - 1)^{1011}$

$= 5\lambda - 1 + 5K - 1$

$= 5m - 2$

Remainder $= 5 - 2 = 3$

42. If the maximum value of the term independent of t in the expansion of $\left(t^2 x^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}$, $x \geq 0$, is K, then 8 K is equal to _____. **[JEE(Main)July2022]**

यदि $\left(t^2 x^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}$, $x \geq 0$ के प्रसार t से स्वतंत्र पद का अधिकतम K है, तो 8 K बराबर है _____.

[JEE(Main)July2022]

Ans. Official Answer NTA (6006)

Sol. General Term $= 15C_r \left(t^2 x^{\frac{1}{5}}\right)^{15-r} \left(\frac{(1-x)^{\frac{1}{10}}}{t}\right)^r$

for tem independent on t

$2(15-r)-r=0$

$\Rightarrow r=10$

$\therefore T_{11}={}^{15}C_{10}x(1-x)$

Maximum value of $x(1-x)$ occur at $x=\frac{1}{2}$

i.e., $(x(1-x))_{max}=\frac{1}{4}$

$\Rightarrow K={}^{15}C_{10}\times\frac{1}{4}$

$\Rightarrow 8K=2\left({}^{15}C_{10}\right)=6006$

43. If the coefficients of x and x^2 in the expansion of $(1+x)^p(1-x)^q$, $p, q \le 15$, are –3 and –5 respectively, then the coefficient of x^3 is equal to _____. **[JEE(Main)July2022]**

$(1+x)^p(1-x)^q$, $p, q \le 15$, के प्रसार में x तथा x^2 के गुणांक क्रमशः –3 तथा –5 हैं, तो x^3 का गुणांक बराबर है _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (23)

Sol. Since coefficient of x is –3

$\Rightarrow {}^pC_1-{}^qC_1=-3$

$\Rightarrow p-q=-3$ ....(1)

Comparing coefficients of x^2

$-{}^pC_1\,{}^qC_1+{}^pC_2+{}^qC_2=-5$

$-pq+\frac{p(p-1)}{2}+\frac{q(q-1)}{2}=-5$ ....(2)

Solving (1) and (2)

$p=8, q=11$

Coefficient of x^3 is

$-{}^qC_3+{}^pC_3+{}^pC_1\,{}^qC_2-{}^pC_2\,{}^qC_1$

$=-{}^{11}C_3+{}^8C_3+{}^8C_1\,{}^{11}C_2-{}^8C_2\,{}^{11}C_1$

$=23$

44. Let for the 9th term in the binomial expansion of $(3+6x)^n$, in the increasing powers of 6x, to be the greatest for $x=\frac{3}{2}$, the least value of n is n_0. If k is the ratio of the coefficient of x^6 to the coefficient of x^3, then $k+n_0$ is equal to : _____. **[JEE(Main)July2022]**

माना 6x की बढ़ती घात में $(3+6x)^n$ के द्विपद प्रसार में $x=\frac{3}{2}$ पर 9वें पद का मान अधिकतम होने के लिए, n का fuEur e eku n_0 है। यदि x^6 के गुणांक का x^3 के गुणांक से अनुपात k हैं, तो $k+n_0$ बराबर है _____.

[JEE(Main)July2022]

Ans. Official Answer NTA (24)

Sol. $(3+6x)^n = {}^nC_0 3^n + {}^nC_1 3^{n-1}(6x)^1 + \ldots$

$T_{r+1}\ {}^nC_r 3^{n-r}.(6x)r = {}^nC_r 3^{n-r}.6_r.\ x_r$

$= {}^nC_r 3^{n-r}.3^r.2^r.\left(\frac{3}{2}\right)^r = {}^nC_r 3^n.3^r \qquad [\text{for } x=\frac{3}{2}]$

T_9 is greatest of $x=\frac{3}{2}$

So, $T_9 > T_{10}$ and $T_9 > T_8$

(concept of numerically greatest term)

Here, $\frac{T_9}{T_{10}}>1$ and $\frac{T_9}{T_8}>1$

$\Rightarrow \frac{{}^nC_8 3^n.3^8}{{}^nC_9 3^n.3^9}>1$ and $\frac{{}^nC_8 3^n.3^8}{{}^nC_7 3^n.3^7}>1$

and $\frac{{}^nC_8}{{}^nC_7}>\frac{1}{3}$

and $\frac{n-7}{8}>\frac{1}{3}$

$\Rightarrow \frac{29}{3}<n<11 \Rightarrow n=10=n_0$

So, in $(3+6x)^n$ for $n=n_0=10$

i.e., in $(3+6x)^{10}$, here $T_{r+1} = {}^{10}C_r 3^{10-r}6^r x^r$

$T_7 = {}^{10}C_6 3^4.6^6.x^6 = 210.3^{10}.2^6x^6$

$T_4 = {}^{10}C_3 3^7 6^3 x^3 = 120.3^{10}.2^3x^3$

Ratio of coefficient of x6 and coefficient of $x^3 = k$

$\therefore k=\frac{210.3.{}^{10}2^6}{120.3^{10}.2^3}=\frac{7}{4}\times 2^3=14$

So, $k+n_0 = 14+1=24$

45. If $1+(2+{}^{49}C_1+{}^{49}C_2+\ldots\ldots+{}^{49}C_{49})({}^{50}C_2+{}^{50}C_4+\ldots\ldots+{}^{50}C_{50})$ is equal to $2^n.$ m, where m is odd, then n + m is equal to _____.

[JEE(Main)July2022]

यदि $1+(2+{}^{49}C_1+{}^{49}C_2+\ldots\ldots+{}^{49}C_{49})({}^{50}C_2+{}^{50}C_4+\ldots\ldots+{}^{50}C_{50}) = 2^n.$ m हैं, जहाँ m एक विषम संख्या हैं, तो n + m बराबर है _____

[JEE(Main)July2022]

Ans. Official Answer NTA (99)

Sol. $1+(1+2^{49})(2^{49}-1)=2^{98}$

$m=1, n=98$

$m+n=99$

46. Let the ratio of the fifth term from the beginning to the fifth from the end in the binomial expansion of $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^n$, in the increasing powers of $\frac{1}{\sqrt[4]{3}}$ be $\sqrt[4]{6}$: 1. If the sixth term from the beginning is $\frac{\alpha}{\sqrt[4]{3}}$, then α is equal to _____. **[JEE(Main)July2022]**

माना $\frac{1}{\sqrt[4]{3}}$ की बढ़ती घातों में $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^n$ के द्विपद प्रसार में आरंभ पाँचवें पद का अंत से पाँचवें पद से अनुपात $\sqrt[4]{6}$: 1 है। यदि आरंभ से छठा पद $\frac{\alpha}{\sqrt[4]{3}}$ है, तो α बराबर है _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (84)

Sol. $\frac{T_5}{T_{n-3}}=\frac{{}^nC_4\left(2^{1/4}\right)^{n-4}\left(3^{-1/4}\right)^4}{{}_nC_{n-4}\left(2^{1/4}\right)^4\left(3^{-1/4}\right)^{n-4}}=\frac{\sqrt[4]{6}}{1}$

$\Rightarrow 2^{\frac{n-8}{4}}3^{\frac{n-8}{4}}=6^{1/4}$

$\Rightarrow 6^{n-8}=6$

$\Rightarrow n-8=1\Rightarrow n=9$

$T_6={}^9C_5\left(2^{1/4}\right)^4\left(3^{-1/4}\right)^5=\frac{84}{\sqrt[4]{3}}$

$\therefore \alpha=84$

47. If $\sum_{k=1}^{10}K^2\left(10_{C_k}\right)^2=22000L$, then L is equal to _____. **[JEE(Main)July2022]**

यदि $\sum_{k=1}^{10}K^2\left(10_{C_k}\right)^2=22000L$ है, तो L बराबर _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (221)

Sol. $\sum_{K=1}^{10}K^2\left({}^{10}C_K\right)^2$

$\sum_{K=1}^{10}\left(K\cdot{}^{10}C_K\right)^2=\sum_{K=1}^{10}\left(10.{}^9C_{K-1}\right)^2$

$$= 100\sum_{K=1}^{10} {}^{9}C_{K-1} \cdot {}^{9}C_{10-K}$$

$$= 100\left({}^{18}C_{9}\right) = 100\left(\frac{18!}{9!9!}\right)$$

$$\Rightarrow 4862000 = 22000\,\text{L ence L} = 221$$

Exercise J - A

1. The coefficients of three consecutive terms of $(1 + x)^{n+5}$ are in the ratio 5 : 10 : 14. Then n =

[JEE(Adv.)2013]

$(1 + x)^{n+5}$ के तीन क्रमागत पदों के गुणांक 5 : 10 : 14 के अनुपात में हैं। तब, n = **[JEE(Adv.)2013]**

Ans. 6

Sol. Let the three consecutive terms be

$${}^{n+5}C_{r-1},\ {}^{n+5}C_r,\ {}^{n+5}C_{r+1}$$

$$\therefore \quad \frac{{}^{n+5}C_{r-1}}{{}^{n+5}C_r} = \frac{1}{2} \quad \Rightarrow \quad \frac{r}{n-r+6} = \frac{1}{2}$$

$$\Rightarrow \quad n = 3r - 6 \qquad .....(1)$$

Alao, $$\frac{{}^{n+5}C_r}{{}^{n+5}C_{r+1}} = \frac{5}{7} \quad \Rightarrow \quad \frac{r+1}{n-r+5} = \frac{5}{7}$$

$$\Rightarrow \quad 12r = 5n + 18 \qquad .....(2)$$

Solving (1) and (2), we get n = 6

2. Coefficient of x^{11} in the expansion of $(1+x^2)^4 (1+x^3)^7(1+x^4)^{12}$ is : **[JEE(Adv.)2014]**

$(1+x^2)^4 (1+x^3)^7(1+x^4)^{12}$ के प्रसार में x^{11} का गुणांक होगा **[JEE(Adv.)2014]**

(A) 1051 (B) 1106 (C) 1113 (D) 1120

Ans. C

Sol. Coefficent of $x^{11} \equiv \frac{(1+x^2)^4(1+x^3)^7(1+x^4)^{12}(1-x^2)^4}{(1-x^2)^4}$

Coefficent of $x^{11} \equiv (1 - x^8)^4 (1 + x^4)^8 (1 + x^3)^7 (1 - x^2)^{-4}$

$= (1 - 4x^8) (1 + x^4)^8 (7x^3 + 35x^9) (1 - x^2)^{-4}$

$= (7x^3 + 35x^9 - 28x^{11}) (1 + x^4)^8 (1 - x^2)^{-4}$

Coefficent of $x^8 = (7x + 35x^6 - 28x^8) (1 + 8x^4 + 28x^8) (1 - x^2)^{-4}$

$= (7 + 35x^6 - 28x^8 + 56x^4 + 196x^8) (1 - x^2)^{-4}$

Coefficent of $t^4 \equiv (7 + 56t^2 + 35t^3 + 168t^4) (1 - t)^{-4}$

$= 7 \cdot {}^7C_3 + 56 \cdot {}^5C_3 + 35 \cdot {}^4C_3 + 168$

$= 245 + 700 + 168 = 1113.$

Alterantive :

$2x + 3y + 4z = 11$

$(x, y, z) = (0, 1, 2)\ {}^4C_0 \times {}^7C_1 \times {}^{12}C_2$

$(1, 3, 0)\quad {}^4C_1 \times {}^7C_3$

$(2, 1, 1)\ {}^4C_2 \times {}^7C_1 \times {}^{12}C_1$

$(4, 1, 0)\quad {}^7C_1$

coefficient of $x^{11} = 66 \times 7 + 35 \times 4 + 42 \times 12 + 7$

$= 1113$

3. The coefficient of x^9 in the expansion of $(1 + x)\ (1 + x^2)\ (1 + x^3)\ldots.\ (1 + x^{100})$ is : **[JEE(Adv.)2015]**

$(1+x)\ (1+x^2)\ (1+x^3)\ldots.\ (1+x^{100})$ के विस्तार में x^9 के गुणांक का मान है। **[JEE(Adv.)2015]**

Ans. 8

Sol. $9 = (0, 9)\ (1, 8), (2, 7), (3, 6), (4, 5)$ # 5 cases

$9 = (1,2,6), (1,3,5), (2, 3, 4)$ # 3 cases

total = 8

4. Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1 + x)^2 + (1 + x)^3 + \ldots.. + (1 + x)^{49} + (1 + mx)^{50}$ is $(3n + 1)\ {}^{51}C_3$ for some positive integer n. Then the value of n is **[JEE(Adv.)2016]**

माना कि m ऐसा न्यूनतम धनात्मक पूर्णांक है कि $(1 + x)^2 + (1 + x)^3 + \ldots.. + (1 + x)^{49} + (1 + mx)^{50}$ के विस्तार में x^2 का गुणांक $(3n + 1)\ {}^{51}C_3$ किसी धनात्मक पूर्णांक n के लिए है। तब n का मान है। **[JEE(Adv.)2016]**

Ans. 5

Sol. ${}^2C_2 + {}^3C_2 + {}^4C_2 + \ldots\ldots\ldots + {}^{49}C_2 + {}^{50}C_2\ m^2 = (3n + 1)\ {}^{51}C_3$

$\Rightarrow {}^{50}C_3 + {}^{50}C_2\ m^2 = (3n + 1)\ {}^{51}C_3$

$\Rightarrow {}^{50}C_3 + {}^{50}C_2 + (m^2 - 1)\ {}^{50}C_2 = 3n\ {}^{51}C_3 + {}^{51}C_3$

$\Rightarrow n = \dfrac{m^2 - 1}{51}$

$\Rightarrow m^2 = 51n + 1$ must be a perfect squared

$\Rightarrow n = 5$ and $m = 16$

5. Let

$$X = \left({}^{10}C_1\right)^2 + 2\left({}^{10}C_2\right)^2 + 3\left({}^{10}C_3\right)^2 + \ldots. + 10\left({}^{10}C_{10}\right)^2,$$

where ${}^{10}C_r, r \in \{1, 2,\ldots., 10\}$ denote binomial coefficients. Then, the value of $\dfrac{1}{1430}X$ is _____.

[JEE(Adv.)2018]

माना कि

$$X = \left({}^{10}C_1\right)^2 + 2\left({}^{10}C_2\right)^2 + 3\left({}^{10}C_3\right)^2 + ... + 10\left({}^{10}C_{10}\right)^2,$$

जहाँ ${}^{10}C_r$, $r \in \{1, 2,...., 10\}$, द्विपद गुणांकों को दर्शाते हैं। तब $\frac{1}{1430}$ X का मान है। **[JEE(Adv.)2018]**

Ans. 646

Sol. $$X = \sum_{r=1}^{10} r.\,{}^{10}C_r = 10.\sum_{r=1}^{10} {}^{9}C_{r-1}.\,{}^{10}C_{10-r} = 10.\ {}^{19}C_9$$

$$\text{Now } \frac{X}{1430} = \frac{10.\,{}^{19}C_9}{1430} = \frac{{}^{19}C_9}{143} = \frac{{}^{19}C_9}{11\times13} = \frac{19.17.16}{8} = 19\times34 = 646$$

6. For nonnegative integers s and r, let $\binom{s}{r} = \begin{cases} \frac{s!}{r!(s-r)!} & \text{if } r \le s, \\ 0 & \text{if } r > s. \end{cases}$

For positive integers m and n, let $g(m,n) = \sum_{p=0}^{m+n} \frac{f(m,n,p)}{\binom{n+p}{p}}$

where for any nonnegative integers p, $f(m, n, p) = \sum_{i=0}^{p} \binom{m}{i}\binom{n+i}{p}\binom{p+n}{p-i}$

Then which of the following statements is / are TRUE ? **[JEE(Adv.)2020]**

(A) g (m, n) = g (n, m) for all positive integers m, n

(B) g(m, n + 1) = g(m+1, n) for all positive integers m, n

(C) g (2m, 2n)=2g (m, n) for all positive integers m, n

(D) g (2m, 2n) = (g (m, n))^2 for all positive integers m, n

माना कि ऋणेतर पूर्णांकों s एवं r के लिए $\binom{s}{r}=\begin{cases}\dfrac{s!}{r!(s-r)!} & \text{if } r\le s,\\ 0 & \text{if } r>s.\end{cases}$

है। माना कि धनात्मक पूर्णांको m एवं n के लिए $g(m,n)=\sum_{p=0}^{m+n}\dfrac{f(m,n,p)}{\binom{n+p}{p}}$

है, जहाँ किसी ऋणेतर पूर्णांक p के लिए $f(m,n,p)=\sum_{i=0}^{p}\binom{m}{i}\binom{n+i}{p}\binom{p+n}{p-i}$

है। तब निम्न में से कौन सा (से) कथन सही है (हैं) ? **[JEE(Adv.)2020]**

(A) सभी धनात्मक पूर्णांकों m, n के लिए g (m, n) = g (n, m) है

(B) सभी धनात्मक पूर्णांकों m, n के लिए g (m, n + 1) = g (m + 1, n) है

(C) सभी धनात्मक पूर्णांकों m, n के लिए g (2m, 2n) = 2g (m, n) है

(D) सभी धनात्मक पूर्णांकों m, n के लिए g (2m, 2n) = $(g(m,n))^2$ है

Ans. ABD

Sol. $f(m,n,p)=\sum_{i=0}^{p}\left({}^{m}C_{i}\dfrac{(n+i)!}{(n+i-p)!p!}\times\dfrac{(p+n)!}{(n+i)!(p-i)!}\right)$

$=\sum_{i=0}^{p}{}^{m}C_{i}\left(\dfrac{n!}{(n+i-p)!(p-i)!}\right)\left(\dfrac{p+n!}{(n)!(p)!}\right)$

$={}^{n+p}C_{p}\sum_{i=0}^{p}{}^{m}C_{i}\,{}^{n}C_{p-i}$

$f(m,n,p)={}^{n+p}C_{p}\left({}^{m}C_{0}\,{}^{n}C_{p}+{}^{m}C_{1}\,{}^{n}C_{p-1}+.....+{}^{m}C_{p}\,{}^{n}C_{0}\right)$

$={}^{n+p}C_{p}.{}^{m+n}C_{p}$

$g(m,n)=\sum_{p=0}^{m+n}\dfrac{{}^{n+p}C_{p}\;{}^{m+n}C_{p}}{{}^{n+p}C_{p}}=2^{m+n}$

(A) g(m, n) =

$(2)^{m+n}=(2)^{n+m}=g(n,m)$

(B) $g(m, n+1) = (2)^{m+n+1} = (2)^{m+1+n}$

(C) $g(2m, 2n) = (2)^{2m+2n} = (2)^{2(m+n)}$

(D) $g(2m, 2n) = (2)^{2m+2n} = (2)^{2(m+n)}$

$= (2^{m+n})^2 = (g(m, n))^2$

Hence, option (A), (B), (D) are correct answer.

7. Let a and b be two nonzero real numbers. If the coefficient of x^5 in the expansion of $\left(ax^2 + \frac{70}{27bx}\right)^4$ is equal to the coefficient of x^{-5} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^7$, then the value of 2b is _____. **[JEE(Adv.)2023]**

माना कि a एवं b दो शून्येतर (nonzero) वास्तविक संख्याएं है। यदि $\left(ax^2 + \frac{70}{27bx}\right)^4$ के प्रसार (expansion) में x^5 का गुणांक (coefficient), $\left(ax - \frac{1}{bx^2}\right)^7$ के प्रसार में x^{-5} के गुणांक के बराबर है, तब 2b का मान है _____ **[JEE(Adv.)2023]**

Ans. 3

Sol. $T_{r+1} = {}^4C_r \left(a \cdot x^2\right)^{4-r} \cdot \left(\frac{70}{27bx}\right)^r$

$= {}^4C_r \cdot a^{4-r} \cdot \frac{70^r}{(27b)^r} \cdot x^{8-3r}$

here $8 - 3r = 5$

$8 - 5 = 3r \Rightarrow r = 1$

$\therefore$ coeff. $= 4 \cdot a^3 \cdot \frac{70}{27b}$

$T_{r+1} = {}^7C_r (ax)^{7-r} \left(\frac{-1}{bx^2}\right)^r$

$= {}^7C_r \cdot a^{7-r} \left(\frac{-1}{b}\right)^r \cdot x^{7-3r}$

$7 - 3r = -5 \Rightarrow 12 = 3r \Rightarrow r = 4$

Coeff. : ${}^7C_4 \cdot a^3 \cdot \left(\frac{-1}{b}\right)^4 = \frac{35a^3}{b^4}$

Now $\frac{35a^3}{b^4} = \frac{280a^3}{27b}$

$b^3 = \frac{35 \times 27}{280} = b = \frac{3}{2} \Rightarrow 2b = 3$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. The coefficient of x^{53} in the expansion $\sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$ is :

$\sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$ के प्रसार में x^{53} का गुणांक होगा :

(A) ${}^{100}C_{47}$ (B) ${}^{100}C_{53}$ (C) $-{}^{100}C_{53}$ (D) $-{}^{100}C_{100}$

Ans. C

Sol. $S = \sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$

$S = {}^{100}C_0 (x-3)^{100} + {}^{100}C_1 (x-3)^{99} \cdot 2 + ... + {}^{100}C_{100} \cdot 2^{100}$

$S = (2 + (x-3))^{100} = (x-1)^{100}$

Co-efficient of $x^{53} = -{}^{100}C_{53} = -{}^{100}C_{47}$

2. If the sum of the coefficients in the expansion of $\left(x^2 + \frac{2}{x^3}\right)^n$ is 243, then the term independent of x, is equal to :

$\left(x^2 + \frac{2}{x^3}\right)^n$ के प्रसार में गुणांको का योगफल 243 है तब x से स्वतन्त्र पद होगा :

(A) 11 (B) $\frac{7}{2}$ (C) 15 (D) 40

Ans. D

Sol. $\left(x^2 + \frac{2}{x^3}\right)^n = \sum_{r=0}^{n} {}^nC_r x^{2(n-r)} \left(\frac{2}{x^3}\right)^r$

$= \sum_{r=0}^{n} {}^nC_r x^{2n-5r} \cdot 2^r$

For independent term

$2n = 5r$

$\Rightarrow r = 2$

Given

$\left(1 + \frac{2}{1}\right)^n = \sum_{r=0}^{n} {}^nC_r 2^r$

$= 243$

$\Rightarrow 3^n = 243$

$\Rightarrow n = 5$

$\Rightarrow$ Coefficient

$= {}^5C_2 \cdot 2^2 = 10 \cdot 4 = 40$

3. The value of $4\{{}^nC_1 + 4 \cdot {}^nC_2 + 4^2 \cdot {}^nC_3 + ... + 4^{n-1}\}$ is :

$4\{{}^nC_1 + 4 \cdot {}^nC_2 + 4^2 \cdot {}^nC_3 + ... + 4^{n-1}\}$ का मान होगा :

(A) 0 (B) $5^n + 1$ (C) 5^n (D) $5^n - 1$

Ans. D

Sol. $4\left\{{}^nC_1 + 4 \cdot {}^nC_2 + ... + 4^{n-1} \cdot {}^nC_n\right\}$

$= \sum_{r=1}^{n} {}^nC_r \cdot 4^r$

$= -{}^nC_0 + \sum_{r=0}^{n} {}^nC_r 4^r$

$= -1 + (1+4)^n$

$= 5^n - 1$

4. The value of $\sum_{r=0}^{n} \frac{1}{(2r)!(2n-2r)!}$, is equal to :

$\sum_{r=0}^{n} \frac{1}{(2r)!(2n-2r)!}$, का मान होगा :

(A) $\frac{2^{2n}}{(2n)!}$ (B) $\frac{2^{2n-1}}{(2n)!}$ (C) $\frac{2^{n-1}}{n!}$ (D) $\frac{2^n}{n!}$

Ans. B

Sol. $\sum_{r=0}^{n} \frac{1}{\lfloor 2r \lfloor 2n-2r} = \frac{1}{\lfloor 2n} \sum_{r=0}^{n} \frac{\lfloor 2n}{\lfloor 2r \lfloor 2n-2r}$

$= \frac{1}{\lfloor 2n} \sum_{r=0}^{n} {}^{2n}C_{2r}$

$= \frac{1}{\lfloor 2n} \left[{}^{2n}C_0 + {}^{2n}C_2 + {}^{2n}C_4 + ... + {}^{2n}C_{2n}\right]$

$$= \frac{1}{\lfloor 2n} \cdot \frac{2^{2n}}{2} = \frac{2^{2n-1}}{\lfloor 2n}$$

5. In the expansion of $(1+x)^n (1+y)^n (1+z)^n$, the sum of the co-efficients of the terms of degree 'r' is :

; 1. गुणांक $(1+y)^n (1+z)^n$ के प्रसार में पदों की घात 'r' के रूप में गुणांको का योगफल होगा :

(A) ${}^{n^3}C_r$ (B) ${}^{n}C_{r^3}$ (C) ${}^{3n}C_r$ (D) $3 . {}^{2n}C_r$

Ans. C

Sol. Put $x = y = z$, we get $(1+x)^{3n}$

$\therefore$ coeff. of terms of degree $r = {}^{3n}C_r$

6. The sum of the last n coefficients in the expansion of $(1+x)^{2n-1}$ when expanding in ascending power of x, is equal to :

$(1+x)^{2n-1}$ के प्रसार में अन्त से n पदों के गुणांको का योगफल क्या होगा जबकि प्रसार x की बढ़ती हुई घात के रूप में किया जाए:

(A) 2^{2n-1} (B) 2^{2n-2} (C) 2^{2n} (D) $2^{2n} - 2^n$

Ans. B

Sol. $(1+x)^{2n-1} = {}^{2n-1}C_0 + {}^{2n-1}C_1 x + ... + \underbrace{{}^{2n-1}C_n x^n + ... + {}^{2n-1}C_{2n-1} x^{2n-1}}_{\text{last n terms}}$

$\Rightarrow {}^{2n-1}C_n + + {}^{2n-1}C_{n+1} + ... + {}^{2n-1}C_{2n-1} = \frac{2^{2n-1}}{2} = 2^{2n-2}$ $\left\{\because {}^{n}C_r = {}^{n}C_{n-r}\right\}$

7. The fractional part of $\frac{3^{2n}}{8}$, is equal to :

(A) $\frac{3}{8}$ (B) $\frac{7}{8}$ (C) $\frac{1}{8}$ (D) None of these

$\frac{3^{2n}}{8}$ का भिन्नात्मक भाग होगा :

(A) $\frac{3}{8}$ (B) $\frac{7}{8}$ (C) $\frac{1}{8}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\frac{3^{2n}}{8} = \frac{9^n}{8} = \frac{(8+1)^n}{8}$

$$= \frac{\sum_{r=0}^{n} {}^{n}C_r 8^r}{8} = \frac{{}^{n}C_0 + 8\lambda}{8}; \lambda \in I$$

$$= \frac{1}{8} + \lambda; \lambda \in I$$

Fractional part of $\frac{3^{2n}}{8}$ is $\frac{1}{8}$.

8. Given positive integers r > 1, n > 2, and the coefficients of the $(3r)^{th}$ and $(r + 2)^{th}$ terms in the binomial expansion of $(1 + x)^{2n}$ are equal, which, if any, of the following statements is true ?

दिया गया है धनात्मक पूर्णांक r > 1 के लिए $(r + x)^{2n}$ के प्रसार में $(3r)^{th}$ व $(r + 2)$ पदों के गुणांक बराबर है जहां n > 2 तब ?

(A) n = 2r (B) n = 3r (C) n = 2r + 1 (D) None of these

Ans. A

Sol. r > 1, n > 2

$$(1+x)^{2n} = \sum_{r=0}^{2n} {}^{2n}C_r x^r$$

Given

Coefficient of $3r^{th}$ term = Coefficient of $(r + 2)^{th}$ term

$\Rightarrow {}^{2n}C_{3r-1} = {}^{2n}C_{r+1}$

$\Rightarrow 3r - 1 = r + 1$ or $3r - 1 + r + 1 = 2n$

$\Rightarrow 2r = 2$ or $4r = 2n$

$\Rightarrow r = 1$ or $n = 2r$

as r > 1

$\Rightarrow n = 2r$

9. If in the expansion of $(1 + x)^m (1 - x)^n$, coefficients of x and x^2 are 3 and - 6 respectively, then m is :

; 1. गद्धm $(1 - x)^n$ के प्रसार में x तथा x^2 के गुणांक 3 तथा - 6 है तब m बराबर होगा :

(A) 6 (B) 9 (C) 12 (D) 24

Ans. C

Sol. Consider $(1+x)^m(1-x)^n$

$\Rightarrow [{}^mC_0 + {}^mC_1x + {}^mC_2x^2 + ... + {}^mC_mx^m][{}^nC_0 + {}^nC_1(-x) + {}^nC_2(-x)^2 + ... {}^nC_n(-x)^n]$

$[\because (1+x)^n = {}^nC_0 + {}^nC_1x + ... {}^nC_nx^n]$

On multiplying and comparing

Now coefficient of $x = {}^mC_0 . {}^nC_1(-1) + {}^mC_1 . {}^nC_0 = m - n = 3$ ...(i)

Coeff. of $x^2 = {}^mC_0 . {}^nC_2(-1)^2 + {}^mC_2 . {}^nC_0 + {}^mC_1 . {}^nC_1(-1)$

$= \frac{n(n-1)}{2} + \frac{m(m-1)}{2} - mn = -6$

$= \frac{n(n-1)}{2} + \frac{(3+n)(3+n-1)}{2} - (3+n)n = -6$ From (i)

$= \frac{n^2-n}{2} + \frac{n^2+5n+6}{2} - 3n - n^2 = -6$

$= \frac{n^2-n+n^2+5n+6-6n-2n^2}{2} = -6$

$= -2n + 6 = -12$

$n! = 9$ Now $m = 3 + n = 12$

10. ${}^{n-1}C_r = (K^2-3).{}^nC_{r+1}$, if $K \in$

(A) $[-\sqrt{3}, \sqrt{3}]$ (B) $(-\infty, -2)$ (C) $(2, \infty)$ (D) $(\sqrt{3}, 2]$

Ans. D

Sol. ${}^{n-1}C_r = (k^2-3) \cdot {}^nC_{r+1}$

$\frac{(n-1)!}{r!(n-r-1)!} = (k^2-3) \cdot \frac{n!}{(r+1)!(n-r-1)!}$

$\Rightarrow \frac{r+1}{n} + 3 = k^2$

$\Rightarrow \because 0 \le r \le n-1$

$1 \le r+1 \le n$

$\Rightarrow \frac{1}{n} \le \frac{r+1}{n} \le 1 \Rightarrow \frac{1}{n} \le k^2 - 3 \le 1$

$\Rightarrow 3 + \frac{1}{n} \le k^2 \le 4 \Rightarrow \sqrt{3+\frac{1}{n}} \le k \le 2$

as $n \to \infty \Rightarrow \sqrt{3} < k \le 2$

$\Rightarrow k \in \left(\sqrt{3}, 2\right]$

11. The value of $\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} ..... + \binom{30}{20}\binom{30}{30}$ is, where $\binom{n}{r} = {}^nC_r$:

$\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} ..... + \binom{30}{20}\binom{30}{30}$ is, where $\binom{n}{r} = {}^nC_r$. का मान है:

(A) $\binom{30}{10}$ (B) $\binom{30}{15}$ (C) $\binom{60}{30}$ (D) $\binom{31}{10}$

Ans. A

Sol. ${}^{30}C_0\ {}^{30}C_{10} - {}^{30}C_1\ {}^{30}C_{11} + {}^{30}C_2\ {}^{30}C_{12} + ... + {}^{30}C_{20}\ {}^{30}C_{30}$

= Coefficient of x^{20} in $(1 - x^2)^{30}$

as

$(1 + x)^{30} = {}^{30}C_0 + {}^{30}C_1\ x + {}^{30}C_2\ x^2 + ... + {}^{30}C_{20}\ x^{20} + ... + {}^{30}C_{30}\ x^{30}$

$(1 - x)^{30} = {}^{30}C_0 - {}^{30}C_1\ x + ... + {}^{30}C_{30}\ x^{30}$

$= {}^{30}C_{30} - {}^{30}C_{29}\ x + {}^{30}C_{28}\ x^2 + ... + {}^{30}C_0\ x^{30}$

$= {}^{30}C_{30} - {}^{30}C_{29}\ x + {}^{30}C_{28}\ x^2 + ... - {}^{30}C_{11}\ x^{19} + ... + {}^{30}C_{30}\ x^{30}$

$= {}^{30}C_{10}$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

12. The polynomial $1 - x + x^2 - x^3 + ... + x^{16} - x^{17}$ may be written in the form $\sum_{i=0}^{17} a_i y^i$ where $y = x + 1$ and a_i's are constants. Then a_2 equals –

(A) The coefficient of t^3 in the expansion of $1 - (1 - t)^{18}$

(B) $\sum_{n=2}^{17} \binom{n}{2}$

(C) 816

(D) The coefficient of t^3 in the expansion of $1 - (1 - t)^{17}$

यदि बहुपद $1 - x + x^2 - x^3 + ... + x^{16} - x^{17}$ को $\sum_{i=0}^{17} a_i y^i$ के रूप में लिखा जा सकता हो, जहाँ $y = x + 1$ तथा

a_i अचर है, तो a_2 बराबर होगा –

(A) $1 - (1 - t)^{18}$ के प्रसार में t^3 का गुणांक

(B) $\sum_{n=2}^{17} \binom{n}{2}$

(C) 816

(D) $1 - (1 - t)^{17}$ के प्रसार में t^3 का गुणांक

Ans. ABC

Sol. $1 - x + x^2 - x^3 + ...+ x^{16} - x^{17}$

$= (1 - x)\,[1 + x^2 + ... + x^{16}]$

$y = x + 1$

$\Rightarrow x = y - 1$

$$\frac{(1-x)(1-x^{18})}{(1-x^2)}$$

$$= \frac{1-x^{18}}{(1+x)}$$

$$= \frac{1-(y-1)^{18}}{y} = \frac{1-\sum_{r=0}^{18} {}^{18}C_r y^r (-1)^r}{y}$$

$$= \frac{1-\left[{}^{18}C_0 + {}^{18}C_1 y(-1) + {}^{18}C_2 y^2 (-1)^2 + ... + {}^{18}C_{18} y^{18}\right]}{y}$$

$$= -\frac{1}{y}\left[-{}^{18}C_1 y + {}^{18}C_2 y^2 + ... + {}^{18}C_{18} y^{18}\right]$$

$$= \left[{}^{18}C_1 - {}^{18}C_2 y + {}^{18}C_3 y^2 + ... - {}^{18}C_{18} y^{17}\right]$$

$$a_2 = {}^{18}C_3 = \frac{18 \times 17 \times 16}{6} = 816$$

= Coefficient of t^3 in the expansion of $1 - (1 - t)^{18}$

$$= \sum_{n=2}^{17} {}^{n}C_2$$

13. In the expansion of $\left(x^3 + 3.2^{-\log_{\sqrt{2}} \sqrt{x^3}}\right)^{11}$

(A) there appears a term with the power x^2

(B) there does not appear a term with the power x^2

(C) there appears a term with the power x^{-3}

(D) the ratio of the co-efficient of x^3 to that of x^{-3} is 1/3

$\left(x^3 + 3.2^{-\log_{\sqrt{2}} \sqrt{x^3}}\right)^{11}$ के प्रसार में

(A) x^2 का पद आयेगा

(B) x^2 का पद नहीं आयेगा

(C) x^{-3} का पद आयेगा

(D) x^3 एवं x^{-3} के गुणांको का अनुपात 1/3 होगा।

Ans. BCD

Sol. $\left(x^3 + 3\cdot 2^{-\log_{\sqrt{2}} \sqrt{x^3}}\right)^{11}$

$$\left\{\log_{\sqrt{2}}^{\sqrt{x^3}} = \frac{1}{2}\log_{\sqrt{2}} x^3 = \frac{3}{2}\log_{\sqrt{2}} x = 3\log_2 x\right\}$$

$$= \left(x^3 + 3\cdot 2^{\log_2 x^{-3}}\right)^{11}$$

$$= \left(x^3 + \frac{3}{x^3}\right)^{11}$$

$$= \sum_{r=0}^{11} {}^{11}C_r x^{33-3r} \cdot \frac{3^r}{x^{3r}}$$

$$= \sum_{r=0}^{11} {}^{11}C_r \cdot 3^r \cdot x^{33-6r}$$

For x^2 $\quad 33 - 6r = -3$

$\Rightarrow 36 = 6r$

For x^3

$33 - 6r = 3$

$\Rightarrow r = 5$

$$\frac{\text{Coeff. of } x^3}{\text{Coeff. of } x^{-3}} = \frac{{}^{11}C_5 \cdot 3^5}{{}^{11}C_6 \cdot 3^6} = \frac{6}{11-6+1} \cdot \frac{}{3} = \frac{2}{6} = \frac{1}{3}$$

14. If it is known that the third term of the binomial expansion $\left(x + x^{\log_{10} x}\right)^5$ is 10^6 then x is equal to :

यदि $\left(x + x^{\log_{10} x}\right)^5$ के प्रसार में तीसरा पद 10^6 हो, तो x का मान होगा :

(A) 10 (B) $10^{-5/2}$ (C) 100 (D) 5

Ans. AB

Sol. $\left(x + x^{\log_{10} x}\right)^5 = \sum_{r=0}^{5} {}^5C_r x^{5-r} \left(x^{\log_{10} x}\right)^r$

Given

$${}^5C_2 x^3 \left(x^{\log_{10} x}\right)^2 = 10^6$$

$$\Rightarrow x^3 \left(x^{\log_{10} x}\right)^2 = 10^5$$

Take log both sides

$$3\log_{10} x + 2\log_{10}\left(x^{\log_{10} x}\right) = 5$$

$$\Rightarrow 3\log_{10} x + 2\left(\log_{10} x\right)^2 = 5$$

Let $\log_{10} x = t$

$2t^2 + 3t - 5 = 0$

$$\Rightarrow t = \frac{-3 \pm \sqrt{9+40}}{4}$$

$$t = \frac{-3 \pm 7}{4}$$

$\Rightarrow t = 1$ or $t = -\frac{10}{4}$

$\Rightarrow x = 10$ or $\log_{10} x = -\frac{5}{2}$

$\Rightarrow x = 10$ or $x = 10^{-\frac{5}{2}}$

15. The expansion of $\left(\sqrt{x}+\frac{1}{2\sqrt[4]{x}}\right)^n$ is arranged in decreasing power of x. If coefficient of first three terms form an A.P. then in expansion, the integral powers of x are :

$\left(\sqrt{x}+\frac{1}{2\sqrt[4]{x}}\right)^n$ के प्रसार को x की अवरोही घातों के क्रम में लिखा जाता है। यदि प्रथम पदों के गुणांक एक A.P. बनाते हो, तो प्रसार में, x की पूर्णांक घातें है :

(A) 0 (B) 2 (C) 4 (D) 8

Ans. B

Sol. $\left(\sqrt{x}+\frac{1}{2\sqrt[4]{x}}\right)^n=\sum_{r=0}^{n}{}^nC_r\left(\sqrt{x}\right)^{n-r}\cdot\frac{1}{2^r}\cdot\left(\frac{1}{x}\right)^{\frac{r}{4}}$

$=\sum_{r=0}^{n}{}^nC_r\left(x\right)^{\frac{n-r}{2}}\cdot\frac{1}{2^r}\cdot x^{-\frac{r}{4}}$

$=\sum_{r=0}^{n}{}^nC_r x^{\frac{2n-2r-r}{4}}\cdot\frac{1}{2^r}$

${}^nC_0\cdot\frac{1}{2^0},{}^nC_1\cdot\frac{1}{2},{}^nC_2\cdot\frac{1}{2^2}\rightarrow$ A.P.

$1+\frac{n(n-2)}{8}=\frac{1}{2}\cdot n\cdot 2$

$\Rightarrow 8+n^2-n=8n$

$\Rightarrow n^2-9n+8=0$

$\Rightarrow n^2-8n-n+8=0$

$\Rightarrow (n-1)(n-8)=0$

$\Rightarrow n=1$ (wrong) or $n=8$

[For A.P. we need three terms in this question]

For integral power of x

$\frac{2n-3r}{4}\in I$

$\Rightarrow 2n-3r\in 4I$

$n = 8$; $r = 0, ..., 7, 8$

$16 - 3r = 4k$; $k \in Z$

$$\Rightarrow r = \frac{16-4k}{3}$$

$$\Rightarrow r = \underset{\downarrow}{0}, \underset{\downarrow}{8}, \underset{\downarrow}{4}$$
$$\quad\; 4 \;\; -2 \;\; 1$$

16. $\sum_{i=0}^{n}\sum_{j=0}^{n}\sum_{k=0}^{n}\binom{n}{i}\binom{n}{j}\binom{n}{k}$, where $\binom{n}{r} = {}^nC_r$:

(A) is less than 5000 if $n = 3$ (B) is greater than 600 if $n = 3$

(C) is less than 5000 if $n = 4$ (D) is greater than 4000 if $n = 4$

$\sum_{i=0}^{n}\sum_{j=0}^{n}\sum_{k=0}^{n}\binom{n}{i}\binom{n}{j}\binom{n}{k}$ जहाँ $\binom{n}{r} = {}^nC_r$:

(A) 5000 से छोटा है यदि $n = 3$ (B) 600 से बड़ा है यदि $n = 3$

(C) 5000 से छोटा है यदि $n = 4$ (D) 4000 से बड़ा है यदि $n = 4$

Ans. ACD

Sol. $$S = \sum_{i=0}^{n}\sum_{j=0}^{n}\sum_{k=0}^{n}\binom{n}{i}\binom{n}{j}\binom{n}{k}$$

$$= \sum_{i=0}^{n}\sum_{j=0}^{n}\left[{}^nC_i\,{}^nC_j\sum_{K=0}^{n}{}^nC_k\right]$$

$$= 2^n\sum_{i=0}^{n}{}^nC_i\left(\sum_{i=0}^{n}{}^nC_j\right)$$

$= 2^n \cdot 2^n \cdot 2^n$

$= (2^n)^3$

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Let $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17} = a_0 + a_1 (1 + x) + a_2 (1 + x)^2 + ... + a_{17} (1 + x)^{17}$,

find the value of a_2.

माना $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17} = a_0 + a_1 (1 + x) + a_2 (1 + x)^2 + ... + a_{17} (1 + x)^{17}$ तो a_2 का मान ज्ञात कीजिए।

Ans. 816

Sol. $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17}$

$$= \frac{1\left(1-(-x)^{18}\right)}{1-(-x)}$$

$$= \frac{1-x^{18}}{1+x}$$

$$\frac{1-x^{18}}{1+x} = a_0 + a_1(1+x) + a_2(1+x)^2 + ... + a_{17}(1+x)^{17}$$

$1 - x^{18} = a_0 (1 + x) + a_1 (1 + x)^2 + a_2 (1 + x)^3 + ... + a_{17} (1 + x)^{18}$

Let $1 + x = t$

$x = t - 1$

$1 - (t - 1)^{18} = a_0 t + a_1 t^2 + a_2 t^3 + ... + a_{17} t^{18}$

So, we need to find coefficient of t^3

$1 - (1 - t)^{18}$

$1 - \{{}^{18}C_0 - {}^{18}C_1 t_1 + {}^{18}C_2 t^2 - {}^{18}C_3 t^3 + ...\}$

So ${}^{18}C_3 \Rightarrow \frac{18}{3} \times \frac{17}{2} \times \frac{16}{1} = 816$

2. Let $a = \left(4^{1/401} - 1\right)$ and let $b_n = {}^nC_1 + {}^nC_2 \cdot a + {}^nC_3 \cdot a^2 + ... + {}^nC_n \cdot a^{n-1}$.

Find the value of $(b_{2006} - b_{2005})$.

माना $a = \left(4^{1/401} - 1\right)$ और माना $b_n = {}^nC_1 + {}^nC_2 \cdot a + {}^nC_3 \cdot a^2 + ... + {}^nC_n \cdot a^{n-1}$ है।

तो $(b_{2006} - b_{2005})$ का मान ज्ञात कीजिए।

Ans. 2^{10}

Sol. $a = 4^{\frac{1}{401}} - 1$

$b_n = {}^nC_1 + {}^nC_2\, a + {}^nC_3\, a^2 + ... + {}^nC_n\, a^{n-1}$

$ab_n = {}^nC_1\, a^1 + {}^nC_2\, a^2 + {}^nC_3\, a^3 + ... + {}^nC_n\, a_n$

$+ \qquad +$

${}^nC_0 \qquad {}^nC_0$

$ab_n + {}^nC_0 = [1 + a]^n$

$ab_n = (1 + a)^n - 1$

$n = 2006$

$$ab_{2006} = \frac{(1+a)^{2006}}{a} - \frac{1}{a}$$

$$b_{2005} = \frac{(1+a)^{2005}}{a} - \frac{1}{a}$$

$$b_{2006} - b_{2005} = \frac{(1+a)^{2006}}{a} - \frac{1}{a} - \frac{(1+a)^{2005}}{a} + \frac{1}{a}$$

$$= \frac{(1+a)^{2006} - (1+a)^{2005}}{a}$$

$$= \frac{[1+a]^{2005}\{1+a-1\}}{a}$$

$\Rightarrow (1 + a)^{2005}$

$$1 + a = 4^{\frac{1}{401}}$$

$$(1+a)^{2005} = 4^{\frac{2005}{401}} = 4^5 = 2^{10}$$

3. Let $a = 3^{\frac{1}{223}} + 1$ and let $f(n) = {}^nC_0 \cdot a^{n-1} - {}^nC_1 \cdot a^{n-2} + {}^nC_2 \cdot a^{n-3} - ..... + (-1)^{n-1} \cdot {}^nC_{n-1} \cdot a^0$.

If the value of $f(2007) + f(2008) = 9k$ where $k \in N$, then find k.

माना $a = 3^{\frac{1}{223}} + 1$ और माना $f(n) = {}^nC_0 \cdot a^{n-1} - {}^nC_1 \cdot a^{n-2} + {}^nC_2 \cdot a^{n-3} - \ldots\ldots + (-1)^{n-1} \cdot {}^nC_{n-1} \cdot a^0$.

यदि $f(2007) + f(2008) = 9k$ जहां $k \in N$ तो k का मान ज्ञात कीजिए।

Ans. 2187

Sol. $a = 3^{\frac{1}{223}} + 1$

$f(n) = {}^nC_0\, a^{n-1} - {}^nC_1\, a^{n-2} + {}^nC_2\, a^{n-3} - \ldots\ldots + (-1)^{n-1}\, {}^nC_{n-1}\, a_0$

$af(n) = {}^nC_0\, a^n - {}^nC_1\, a^{n-1} + {}^nC_2\, a^{n-2} - {}^nC_n\, a^0 \ldots\ldots + (-1)^{n-1}\, {}^nC_{n-1}\, a + (-1)^n\, {}^nC_n\, a^0$

+

$(-1)^n \cdot {}^nC_n\, a^0$

$= [a-1]^n$

$af(n) + (-1)^n = (a-1)^n$

$af(2007) - 1 = (a-1)^{2007}$

$af(2008) + 1 = (a-1)^{2008}$

$a\{f(2007) + f(2008)\} = (a-1)^{2007}\ \{1 + (a-1)\}$

$f(2007) + f(2008) = (a-1)^{2007}$

$\Rightarrow 3^{\frac{2007}{223}} \Rightarrow 3^9 = 9k$, $k = 2187$

4. Let $P = \left(2+\sqrt{3}\right)^5$ and $f = P - [P]$, where $[P]$ denotes the greatest integer function.

Find the value of $\left(\frac{f^2}{1-f}\right)$.

माना $P = \left(2+\sqrt{3}\right)^5$ तथा $f = P - [P]$, जहां $[P]$ महत्तम पूर्णांक फलन है। तो $\left(\frac{f^2}{1-f}\right)$ का मान ज्ञात कीजिए।

Ans. 722

Sol. $P = \left(2+\sqrt{3}\right)^5$

$f = P - [P]$

$\frac{f^2}{1-f}$

$I + f = \left(2+\sqrt{3}\right)^5 = {}^5C_0 2^5 + {}^5C_1 2^4\sqrt{3}^1 + {}^5C_2 2^3\sqrt{3}^2 + ... + {}^5C_5\left(\sqrt{3}\right)^5$

$F = \left(2-\sqrt{3}\right)^5 = {}^5C_0 2^5 - {}^5C_1 2^4\sqrt{3} + ... - {}^5C_5\left(\sqrt{3}\right)^5$

$I + f + F = 2[{}^5C_0\, 2^5 + {}^5C_2\, 2^3\, 3 + {}^5C_4\, 2\cdot 3^2]$

$= 2\,[32 + 10\cdot 24 + 5\cdot 18]$

$= 724$

$0 < f < 1$

$0 < F < 1$

$0 < f + F < 2$

$\Rightarrow f + F = 1$

$I = 723$

$F = 1 - f$

$([P] + f)\, F = 1$

$$[P] = \frac{1}{F} - f$$

$$= \frac{1}{1-f} - f$$

$$= \frac{1-f+f^2}{1-f}$$

$$723 = 1 + \frac{f^2}{1-f}$$

$$\frac{f^2}{1-f} = 722$$

5. Prove that :
सिद्ध कीजिये

(a) $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + ... + \frac{n.C_n}{C_{n-1}} = \frac{n(n+1)}{2}$

(b) $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + ... + \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$

(c) $2 \cdot C_o + \frac{2^2 \cdot C_1}{2} + \frac{2^3 \cdot C_2}{3} + \frac{2^4 \cdot C_3}{4} + ... \frac{2^{n+1} \cdot C_n}{n+1} = \frac{3^{n+1}-1}{n+1}$

(d) $C_o - \frac{C_1}{2} + \frac{C_2}{3} - ..... + (-1)^n \frac{C_n}{n+1} = \frac{1}{n+1}$

Ans.

Sol. (a) $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + ... + \frac{nC_n}{C_{n-1}}$

$$\sum_{r=1}^{n} \frac{r\ {}^nC_r}{{}^nC_{r-1}}$$

$$\sum_{r=1}^{n} n \frac{{}^{n-1}C_{r-1}}{{}^nC_{r-1}}$$

$$n\sum_{r=1}^{n} \frac{\frac{(n-1)!}{(n-1-r+1)!(r-1)!}}{\frac{n!}{(n-r+1)!(r-1)!}}$$

$$n\sum \frac{(n-1)!}{n!} \times \frac{(n-r+1)!}{(n-r)!}$$

$$n\sum \frac{1}{n} \times (n-r+1)$$

$$\sum_{r=1}^{n} (n-r+1)$$

$$n^2 - \frac{n(n+1)}{2} + n$$

$$\frac{2n^2 - n^2 - n + 2n}{2}$$

$$\frac{n^2+n}{2} \Rightarrow \frac{n(n+1)}{2}$$

(b) ${}^nC_0 + \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} + ... + \frac{{}^nC_n}{n+1}$

$$\sum_{r=0}^{n} \frac{{}^{n}C_r}{r+1}$$

$$\frac{1}{n+1}\sum_{r=0}^{n} \frac{n+1}{r+1}\cdot {}^{n}C_r$$

$$\frac{1}{n+1}\sum_{r=0}^{n} {}^{n+1}C_{r+1}$$

$$\frac{1}{n+1}\left\{{}^{n+1}C_1 + {}^{n+1}C_2 + ... + {}^{n+1}C_{n+1}\right\}$$

$$+ \qquad\qquad - {}^{n+1}C_0$$

$${}^{n+1}C_0$$

$$\frac{1}{n+1}\cdot\left\{2^{n+1}-1\right\}$$

(c) $$2\cdot {}^{n}C_0 + 2^2\cdot\frac{{}^{n}C_1}{2} + 2^3\cdot\frac{{}^{n}C_2}{3} + ... + \frac{2^{n+1}\ {}^{n}C_n}{n+1}$$

$$\sum_{r=0}^{n} \frac{2^{r+1}\ {}^{n}C_r}{r+1}$$

$$\frac{1}{n+1}\sum 2^{r+1}\cdot\frac{n+1}{r+1}\cdot\ {}^{n}C_r$$

$$\frac{1}{n+1}\sum_{r=0}^{n} 2^{r+1}\ {}^{n+1}C_{r+1}$$

$$\frac{1}{n+1}\left\{2\cdot {}^{n+1}C_1 + 2^2\ {}^{n+1}C_2 + ... + 2^{n+1}\ {}^{n+1}C_{n+1}\right\}$$

$$+\ {}^{n+1}C_0 \qquad\qquad - {}^{n+1}C_0$$

$$\frac{1}{n+1}\left\{(1+2)^{n+1}-1\right\}$$

$$\frac{1}{n+1}\left\{3^{n+1}-1\right\}$$

(d) $^{n}C_0 - \frac{^{n}C_1}{2} + \frac{^{n}C_2}{3} - .... + (-1)^n \frac{^{n}C_n}{n+1}$

$\sum_{r=0}^{n} (-1)^r \frac{^{n}C_r}{r+1}$

$\frac{1}{n+1} \sum_{r=0}^{n} (-1)^r \ {}^{n+1}C_{r+1}$

$\frac{1}{n+1} \left\{ {}^{n+1}C_1 - {}^{n+1}C_2 + {}^{n+1}C_3 - ..... (-1)^n \ {}^{n+1}C_{n+1} \right\}$

$\frac{1}{n+1} \{1\}$

6. Let $\sum_{r=0}^{100} \sum_{s=0}^{100} (C_r^2 + C_s^2 + C_r C_s) = m \left({}^{2n}C_n \right) + 2^p$ where m, n and p are even natural numbers and C_r represents the coefficient of x^r in the expansion of $(1 + x)^{100}$. Find the value of $(m + n + p)$.

माना $\sum_{r=0}^{100} \sum_{s=0}^{100} (C_r^2 + C_s^2 + C_r C_s) = m \left({}^{2n}C_n \right) + 2^p$ जहां m, n और p सम प्राकृत संख्याऐं हैं तथा $C_r (1+x)^{100}$ के प्रसार में x^r के गुणांक है। तब $(m + n + p)$ का मान ज्ञात कीजिए।

Ans. 502

Sol. $\sum_{r=0}^{100} \sum_{s=0}^{100} (C_r^2 + C_s^2 + C_r C_s) = \sum_{r=0}^{100} \left(\sum_{s=0}^{100} \left[\left({}^{n}C_r \right)^2 + \left({}^{n}C_s \right)^2 + {}^{n}C_r . {}^{n}C_s \right] \right)$

$$= \sum_{r=0}^{100} \left(101. \left({}^{n}C_r \right)^2 + \left[\underbrace{\left({}^{n}C_0 \right)^2 + \left({}^{n}C_1 \right)^2 + ... + \left({}^{n}C_{100} \right)^2}_{p} \right] + {}^{n}C_r \left[\underbrace{\left({}^{n}C_0 \right) + \left({}^{n}C_1 \right) + ... + \left({}^{n}C_{100} \right)}_{q} \right] \right) .....(A)$$

Now, for P : -

$\because (1+n)^n = {}^{n}C_0 + {}^{n}C_1 . x + ... {}^{n}C_n . x^n$ .....(1)

$(x+1)^n = {}^{n}C_0 . x^n + {}^{n}C_1 . x^{n-1} + ... + {}^{n}C_n$ .....(2)

Multiply equation (1) & (2) and compare coefficient of x^n both side, we get .....

$${}^{2n}C_n = \left({}^{n}C_0 \right)^2 + \left({}^{n}C_1 \right)^2 + ... C_0 = p$$

$p = {}^{200}C_{100}$

Put x = 1 in equation (1) we get

$2^n = 2^{100} = {}^nC_0 + {}^nC_1 + ...+ {}^nC_n = q$

Now, from (A)

$$\sum_{r=0}^{100}\left[101\times\left({}^nC_r\right)^2 + {}^{200}C_{100} + 2^{100}.\ {}^nC_r\right]$$

$$= 101\sum_{r=0}^{100}\left({}^nC_r\right)^2 + {}^{200}C_{100}\sum_{r=0}^{100}1 + 2^{100}\sum_{r=0}^{100}{}^nC_r$$

$= 101 \times {}^{200}C_{100} + {}^{200}C_{100} \times 101 + 2^{100} \times 2^{100}$

$= 202 \times {}^{200}C_{100} + 2^{200} = m({}^{2n}C_n) + 2^p$ (Given)

by comapre both side we get

m = 202, n = 100, p = 200

$\therefore$ m + n + p = 502

7. If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_n x^n$, then show that the sum of the products of the C_i's taken two at a time, represented by $\underset{0 \le i < j \le n}{\Sigma\,\Sigma\, C_i\, C_j}$ is equal to $2^{2n-1} - \frac{2n!}{2(n!)^2}$.

यदि $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_n x^n$, तब सिद्ध कीजिए कि दो C_i's को एक साथ लेकर गुणन फलों का योग यदि $\underset{0 \le i < j \le n}{\Sigma\,\Sigma\, C_i\, C_j}$ से परिभाषित है, का मान $2^{2n-1} - \frac{2n!}{2(n!)^2}$ होगा।

Ans.

Sol. $(1+x)^n = C_0 + C_1x + ... + C_n x^n$

$S = \Sigma\Sigma\, C_iC_j = \frac{1}{2}[(C_0 + C_1 + C_2 + ...+ C_n)^2 - (C_0^2 + C_1^2 + C_2^2 + ...+ C_n^2)]$

$\Rightarrow 2S = 2^{2n} - {}^{2n}C_n \Rightarrow S = 2^{2n-1} - \frac{{}^{2n}C_n}{2}$

8. Let $S_1 = \sum\sum_{0\le i<j\le 100} C_i\, C_j$, $S_2 = \sum\sum_{0\le j<i\le 100} C_i\, C_j$ and $S_3 = \sum\sum_{0\le i=j\le 100} C_i\, C_j$

where C_r represents coefficient of x^r in the binomial expansion of $(1 + x)^{100}$.

If $S_1 + S_2 + S_3 = a^b$ where $a, b \in N$ then find the least value of $(a + b)$.

माना $S_1 = \sum\sum_{0\le i<j\le 100} C_i\, C_j$, $S_2 = \sum\sum_{0\le j<i\le 100} C_i\, C_j$ तथा $S_3 = \sum\sum_{0\le i=j\le 100} C_i\, C_j$

जहां C_r, $(1 + x)^{100}$ में x^r के गुणांक हैं, यदि $S_1 + S_2 + S_3 = a^b$ जहां $a, b \in N$ तब $(a + b)$ का न्यूनतम मान ज्ञात कीजिए।

Ans. 66

Sol. Given that $S_1 + S_2 + S_3 = \sum_{i=0}^{100}\sum_{j=0}^{100} C_i C_j$

$=\left({}^{100}C_0 + {}^{100}C_1 + {}^{100}C_2 + ... + {}^{100}C_{100}\right)^2$

$=\left(2^{100}\right)^2 = 2^{200} = a^b$

$\Rightarrow a^b = 2^{200} = 4^{100} = 16^{50} = 32^{40} = ...$

So, least value of a + b = 16 + 50 = 66

9. The expressions $1 + x$, $1+x + x^2$, $1 + x + x^2 + x^3$,..... $1 + x + x^2 + ... + x^n$ are multiplied together and the terms of the product thus obtained are arranged in increasing powers of x in the form of $a_0 + a_1x + a_2x^2 + ...$, then,

(a) how many terms are there in the product.

(b) show that the coefficients of the terms in the product, equidistant from the beginning and end are equal.

(c) show that the sum of the odd coefficients = the sum of the even coefficients = $\frac{(n+1)!}{2}$

यदि व्यंजक $1 + x$, $1+x + x^2$, $1 + x + x^2 + x^3$,..... $1 + x + x^2 + ... + x^n$ को गुणा किया जाता है तथा सभी पदों को x की बढ़ती घातों में निम्न $a_0 + a_1x + a_2x^2 + ...$, प्रकार से लिखा जाता है तो –

(a) गुणनफल में पदों की संख्या कितनी होगी।

(b) सिद्ध कीजिए कि गुणनफल में प्रारम्भ तथा अन्तः से समान दूरी पर स्थित पद समान है।

(c) सिद्ध कीजिए कि विषम गुणांकों का योग = सम गुणांकों का योग = $\frac{(n+1)!}{2}$

Ans. (a) $\frac{n^2+n+2}{2}$

Sol. $(1 + x)(1 + x + x^2)(1 + x + x^2 + x^3) ..... (1 + x + x^2 + ... + x^n)$

$= a_0 + a_1x + a_2x^2 + ... +$

(a) As the largest power of 'x' present in the expansion will be

$1 + 2 + 3 + ... + n$

$\Rightarrow \frac{n(n+1)}{2}$

So number of terms will be $1 + \frac{n(n+1)}{2}$

$\Rightarrow \frac{n^2+n+2}{2}$

(b) $(1 + x)(1 + x + x^2)(1 + x + x^2 + x^3) \ldots\ldots (1 + x + x^2 + \ldots\ldots + x^n)$

$= a_0 + a_1x + a_2x^2 + \ldots. + a_n x^{\frac{n(n+1)}{2}}$ ...(A)

Replace $x \to \frac{1}{x}$

$$\left(1+\frac{1}{x}\right)\left(1+\frac{1}{x}+\frac{1}{x^2}\right)\left(1+\frac{1}{x}+\frac{1}{x^2}+\frac{1}{x^3}\right).....\left(1+\frac{1}{x}+\frac{1}{x^2}+....+\frac{1}{x^n}\right)$$

$$= a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + ...\frac{a_n}{x^{\frac{n(n+1)}{2}}}$$

$$\frac{(x+1)}{x}\left(\frac{x^2+x+1}{x^2}\right)\left(\frac{x^3+x^2+x+1}{x^3}\right).....\left(\frac{x^n+x^{n-1}+....+1}{x^n}\right)$$

$$= \frac{a_0x^{\frac{n(n+1)}{2}} + a_1x^{\frac{n(n+1)}{2}-1} + ... + a_n}{x^{\frac{n(n+1)}{2}}}$$

$(1 + x)(1 + x + x^2)(1 + x + x^2 + x^3) \ldots\ldots (1 + x + x^2 + \ldots. + x^n) = a_0x^{\frac{n(n+1)}{2}} + a_1x^{\frac{n^2+n-2}{2}} + ... + a_n$ ...(B)

from (A) & (B)

$a_0 + a_1x + a_2x^2 + \ldots. + a_n x^{\frac{n(n+1)}{2}} = a_0x^{\frac{n(n+1)}{2}} + a_1x^{\frac{n^2+n-2}{2}} + \ldots + a_n$

Comparing coefficients

$$\left.\begin{array}{l} a_0 = a_n \\ a_1 = a_{n-1} \\ \vdots \\ a_n = a_0 \end{array}\right\} \Rightarrow a_r = a_{n-r}$$

(c) Put x = 1 in (A)

$2 \cdot 3 \cdot 4 \cdot \ldots\ldots\ldots (n + 1) = a_0 + a_1 + a_2 + \ldots\ldots\ldots + a_n$

Put $x = -1$ in (A)

$0 = a_0 - a_1 + a_2 - \ldots\ldots\ldots + a_n$

$$a_0 + a_2 + \ldots + a_n = \frac{(n+1)!}{2}$$

10. Let $x = \left(3\sqrt{6} + 7\right)^{89}$. If $\{x\}$ denotes the fractional part of 'x' then find the remainder when $x\{x\} + (x\{x\})^2 + (x\{x\})^3$ is divided by 31.

माना $x = \left(3\sqrt{6} + 7\right)^{89}$ है। यदि $\{x\}$, 'x' के भिन्नात्मक भाग को निरूपित करता है तो शेषफल ज्ञात कीजिए जब $x\{x\} + (x\{x\})^2 + (x\{x\})^3$, 31 से विभाजित किया जाये।

Ans. 0

Sol. $x = \left(3\sqrt{6} + 7\right)^{89}$

Find $x\{x\} + \left(x\{x\}\right)^2 + \left(x\{x\}\right)^3$

$1 + y + y^2 + y^3 - 1$

$(1 + y) + y^2 (1 + y) - 1$

$(1 + y) (1 + y^2) - 1$

$I + f = \left(7 + 3\sqrt{6}\right)^{89} = {}^{89}C_0 7^{89} + {}^{89}C_1 7^{88}\left(3\sqrt{6}\right) + \ldots\ldots$

$F = \left(7 - 3\sqrt{6}\right)^{89} = \ldots\ldots\ldots\ldots$

$$2\left[{}^{89}C_0 7^{89} + {}^{89}C_2 7^{87}\left(3\sqrt{6}\right)^1 + \ldots\ldots {}^{89}C_{88}(7)^1\left(3\sqrt{6}\right)^{88}\right]$$

$I + f + F =$ Even Integer

$0 < f < 1$

$-1 < F < 0$

$-1 < f + F < 1$

$I =$ Even Integer

$\Rightarrow f + F = 0$

$\{x\} = -F$

$= -\left(7 - 3\sqrt{6}\right)^{89}$

$\{x\} = \left(3\sqrt{6} - 7\right)^{89}$

$x = \left(3\sqrt{6} + 7\right)^{89}$

$x\{x\} = \left\{\left(3\sqrt{6} - 7\right)\left(3\sqrt{6} + 7\right)\right\}^{89}$

$= \{54 - 49\}^{89}$

$= 5^{89}$

Now 5^{89} {5 + 25 + 125}

$5^{89} \times 5 \times 31$

Divide by 31

So remainder = 0

11. Find the sum of possible real values of x for which the sixth term of $\left(3^{\log_3 \sqrt{9^{|x-2|}}} + 7^{\left(\frac{1}{5}\right)\log_7 (3^{|x-2|}-9}\right)^7$ equals 567.

x के संभव वास्तविक मानों का योग ज्ञात कीजिए, जिनके लिए $\left(3^{\log_3 \sqrt{9^{|x-2|}}} + 7^{\left(\frac{1}{5}\right)\log_7 (3^{|x-2|}-9}\right)^7$ का छठा पद 567 के बराबर है।

Ans. 4

Sol. $\left\{3^{\log_3\left(9^{|x-2|}\right)^{\frac{1}{2}}} + 7^{\frac{1}{5}\log_7 3^{|x-2|}-9}\right\}^7$

$\left\{\left(9^{|x-2|}\right)^{\frac{1}{2}} + \left\{3^{|x-2|}-9\right\}^{\frac{1}{5}}\right\}^7$

$\left\{3^{|x-2|} + 3^{\frac{|x-2|-9}{5}}\right\}^7$

$${}^{7}C_{5}\left(3^{|x-2|}\right)^{2}\left(3^{\frac{|x-2|-9}{5}}\right)^{5}=567$$

$$3^{2|x-2|}\cdot 3^{|x-2|-9}=\frac{567}{21}$$

$3^{|x-2|}=t$

$t^{2}\cdot\frac{t}{3^{9}}=27$

$t^{3}=3^{12}$

$t=3^{9}$

$3^{|x-2|}=3^{9}$

$|x-2|=9$

$x-2=\pm 9$

$x=2\pm 9$

$x=11,-7$

$11+(-7)$

$\Rightarrow 4$

12. Let $1+\sum_{r=1}^{10}\left(3^{r}\cdot{}^{10}C_{r}+r.{}^{10}C_{r}\right)=2^{10}(\alpha\,.\,4^{5}+\beta)$ where $\alpha,\beta\in N$ and $f(x)=x^{2}-2x-k^{2}+1$. If α,β lies between the roots of $f(x)=0$. Then find the smallest positive integral value of k.

माना $1+\sum_{r=1}^{10}\left(3^{r}\cdot{}^{10}C_{r}+r.{}^{10}C_{r}\right)=2^{10}(\alpha\,.\,4^{5}+\beta)$ है, जहाँ $\alpha,\beta\in N$ तथा $f(x)=x^{2}-2x-k^{2}+1$ है। यदि α,β, $f(x)=0$ के मूलों के बीच स्थित है, तो k का न्यूनतम संभव पूर्णांक मान ज्ञात कीजिए।

Ans. 5

Sol. $1+\sum_{r=1}^{10}3^{r}\ {}^{10}C_{r}+\sum_{r=1}^{10}r\ {}^{10}C_{r}$

$$=1+3^{0}\ {}^{10}C_{0}+3^{1}\ {}_{10}C_{1}+3^{2}\ {}^{10}C_{2}+....+3^{10}\ {}^{10}C_{10}-3^{0}.{}^{10}C_{0}+\sum_{r=1}^{10}r.{}^{10}C_{r}$$

$$=(1+3)^{10}+\sum_{r=1}^{10}r\cdot\frac{10}{r}\cdot{}^{9}C_{r-1}$$

$$=4^{10}+10\left[{}^{9}C_{0}+{}^{9}C_{1}+{}^{9}C_{2}+.....+{}^{9}C_{9}\right]$$

$$=4^{10}+10\left(2^{9}\right)$$

$= 4^{10} + 10 \cdot 2^9$

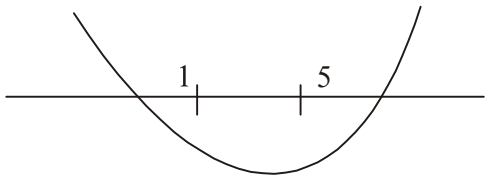
$= 4^{10} + 5 \cdot 2^{10}$

$\Rightarrow 2^{10}\left[4^5 + 5\right] = 2^{10}\left[\alpha \cdot 4^5 + \beta\right]$

$\alpha = 1, \beta = 5$

Now,

$f(x) = x^2 - 2x - k^2 + 1$



$f(1) < 0$ $\qquad$ $f(5) < 0$

$1 - 2 - k^2 + 1 < 0$ $\qquad$ $25 - 10 - k^2 + 1 < 0$

$-k^2 < 0$ $\qquad$ $16 - k^2 < 0$

$k^2 > 0$ $\qquad$ $k \in (-\infty, -4) \cup (4, \infty)$

$k \neq 0$

Smallest +ve integral value = 5

13. Let $n \in N$; $S_n = \sum_{r=0}^{3n}\left({}^{3n}C_r\right)$ and $T_n = \sum_{r=0}^{n}\left({}^{3n}C_{3r}\right)$. Find $|S_n - 3T_n|$.

माना $n \in N$; $S_n = \sum_{r=0}^{3n}\left({}^{3n}C_r\right)$ तथा $T_n = \sum_{r=0}^{n}\left({}^{3n}C_{3r}\right)$ है, तो $|S_n - 3T_n|$ का मान ज्ञात कीजिए।

Ans. 2

Sol. $S_n = \sum_{r=0}^{3n} {}^{3n}C_r$

$= {}^{3n}C_0 + {}^{3n}C_1 + {}^{3n}C_2 + ... + {}^{3n}C_{3n}$ ...(A)

$(1 + x)^{3n} = {}^{3n}C_0\, x^0 + {}^{3n}C_1\, x^1 + ... + {}^{3n}C_{3n}\, x^{3n}$

Put $x = 1$

$S_n = 2^{3n}$

$T_n = \sum_{r=0}^{n} {}^{3n}C_{3r}$

$= {}^{3n}C_0 + {}^{3n}C_3 + {}^{3n}C_6 + ... + {}^{3n}C_{3n}$

$(1 + x)^{3n} = {}^{3n}C_0\, x^0 + {}^{3n}C_1\, x + {}^{3n}C_2\, x^2 + {}^{3n}C_3\, x^3 + {}^{3n}C_4\, x^4 + ... + {}^{3n}C_{3n}\, x^{3n}$

$x = w$

$(1 + w)^{3n} = {}^{3n}C_0 + {}^{3n}C_1\, w + {}^{3n}C_2\, w^2 + {}^{3n}C_3 + ... + {}^{3n}C_{3n}$

$x = w^2$

$(1 + w^2)^{3n} = {}^{3n}C_0 + {}^{3n}C_1\, w^2 + {}^{3n}C_2\, w + {}^{3n}C_3 + ... + {}^{3n}C_{3n}$

$(1 + w)^{3n} + (1 + w^2)^{3n} = 2\ {}^{3n}C_0 + {}^{3n}C_1\,(-1) + {}^{3n}C_2\,(-1) + 2\ {}^{3n}C_3 + ...$

Now Add (A) in it

$2^{3n} + (1 + w)^{3n} + (1 + w^2)^{3n} = 3[T_n]$

$2^{3n} + (-w^2)^{3n} + (-w)^{3n} = 3T_n$

$2^{3n} + (-1)^{3n} + (-1)^{3n} = 3T_n$

$|S_n - 3T_n|$

$= |(-1)^{3n} + (-1)^{3n}|$

$\Rightarrow 2$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 14 to 16

Consider the binomial expansion $R = (1 + 2x)^n = I + f$ where I is the integral part of R and f is the fractional part of R $n \in N$. Also the sum of the coefficients of R is 6561.

माना द्विपद प्रसार $R = (1 + 2x)^n = I + f$ है, जहाँ I तथा f, क्रमशः R के पूर्णांक भाग तथा भिन्नात्मक भाग है $n \in N$. R के गुणांकों का योग 6561 है।

14. The value of $(n + R - Rf)$ for $x = \frac{1}{\sqrt{2}}$ equals :

$x = \frac{1}{\sqrt{2}}$ के लिए $(n + R - Rf)$ का मान होगा:

(A) 7 (B) 8 (C) 9 (D) 10

Ans. C

Sol. $I + f = (1 + 2x)^n = R$

Sum of coefficient 6561

Put $x = 1$

$3^n = 6561 = 3^8$

$n = 8$

$R = I + f = \left(1+\sqrt{2}\right)^8$

$F = \left(1-\sqrt{2}\right)^8$

$I + f = {}^8C_0 + {}^8C_1\sqrt{2} + ... + {}^8C_8\left(\sqrt{2}\right)^8$

$F = {}^8C_0 - {}^8C_1\sqrt{2} ..... + {}^8C_8\left(\sqrt{2}\right)^8$

$I + f + F = 2[{}^8C_0 + {}^8C_2 \cdot 2 + {}^8C_4\, 2^2 + {}^8C_6\, 2^3 + {}^8C_8\, 2^4]$

$I + f + F = 2 \times 577 = 1154$

$0 < f < 1$

$0 < F < 1$

$0 < f + F < 2$

$f + F = 1$

$I = 1153$

Now $n + R(1 - f)$

$8 + \left(1+\sqrt{2}\right)^8\left(1-\sqrt{2}\right)^8$

$8 + \{1 - 2\}^8$

9

15. If i^{th} terms is the greatest term for $x = 1/2$, then 'i' equals :

यदि $x = 1/2$ के लिए iवाँ पद महतम पद है, तो 'i' बराबर होगा :

(A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. $\frac{n+1}{1+\left|\frac{a}{b}\right|} - 1 \le i \le \frac{n+1}{1+\left|\frac{a}{b}\right|}$

Now $\frac{9}{1+\left|\frac{1}{2x}\right|} \Rightarrow \frac{9}{1+1} \Rightarrow \frac{9}{2} \Rightarrow 4.5$

$3.5 \le i \le 4.5$

$\Rightarrow i = 4$

So term will be 5th term.

16. If k^{th} terms is having greatest coefficient then sum of all possible value(s) of k is :

यदि kवें पद का गुणांक महतम है, तो k के सभी संभव मानों का योग होगा :

(A) 6 (B) 7 (C) 11 (D) 13

Ans. D

Sol. $n = 8$

So, greatest coefficient will be middle term.

Paragraph for Question Nos. 17 to 19

Let us consider the expansion $(3 + 4x)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + ... + a_nx^n$, then

माना प्रसार $(3 + 4x)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + ... + a_nx^n$ है, तो

17. $a_1 + a_3 + a_5 + a_7 + ...$ is equal to :

$a_1 + a_3 + a_5 + a_7 + ...$ बराबर है :

(A) $\frac{7^n + (-1)^n}{2}$ (B) $\frac{7^n - (-1)^n}{2}$ (C) $\frac{7^{n-1} - (-1)^{n-1}}{2}$ (D) $\frac{7^{n-1} + (-1)^{n-1}}{2}$

Ans. B

Sol. $(3 + 4x)^n = a_0 + a_1x + a_2x^2 + ... + a_nx^n$

$x = 1$

$7^n = a_0 + a_1 + a_2 + a_3 + ... + a_n$

$x = -1$

$(-1)^n = a_0 - a_1 + a_2 - a_3 - .....$

$$\frac{7^n + (-1)^n}{2} = a_0 + a_2 + a_4 + ...$$

$$\frac{7^n - (-1)^n}{2} = a_1 + a_3 + a_5 + ...$$

18. $a_0 - a_2 + a_4 - a_6 + ...$ is equal to :

$a_0 - a_2 + a_4 - a_6 + \ldots\ldots\ldots$ बराबर है :

(A) $5^n \cos\left(n \tan^{-1}\frac{4}{3}\right)$ (B) $7^n \cos\left(n \tan^{-1}\frac{3}{4}\right)$ (C) $7^n \cos\left(n \tan^{-1}\frac{4}{3}\right)$ (D) $5^n \cos\left(n \tan^{-1}\frac{3}{4}\right)$

Ans. A

Sol. $(3 + 4i)^n = a_0 + a_1 i - a_2 - a_3 i + a_4 + a_5 i - a_6 \ldots\ldots\ldots\ldots$

$(3 + 4i)^n = (a_0 - a_2 + a_4 - a_6 + \ldots\ldots) + i(a_1 - a_3 + a_5 - a_7 + \ldots\ldots)$

$$\left\{5e^{i\left(\tan^{-1}\frac{4}{3}\right)}\right\}^n$$

$$5^n e^{in \tan^{-1}\frac{4}{3}}$$

$$5^n \left\{\cos\left(n \tan^{-1}\frac{4}{3}\right) + i \sin\left(n \tan^{-1}\frac{4}{3}\right)\right\}$$

$= (a_0 - a_2 + a_4 - a_6 + \ldots\ldots) + i(a_1 - a_3 + a_5 - a_7 + \ldots\ldots)$

Now $a_0 - a_2 + a_4 - a_6 + \ldots\ldots = 5^n \cos\left(n \tan^{-1}\frac{4}{3}\right)$

19. $(a_0 - a_2 + a_4 - a_6 + \ldots\ldots\ldots)^2 + (a_1 - a_3 + a_5 - \ldots\ldots\ldots)^2$ equals :

$(a_0 - a_2 + a_4 - a_6 + \ldots\ldots\ldots)^2 + (a_1 - a_3 + a_5 - \ldots\ldots\ldots)^2$ बराबर है :

(A) 7^{2n} (B) 5^n (C) 5^{2n} (D) 7^n

Ans. C

Sol. $(a_0 - a_2 + a_4 - a_6 + \ldots\ldots)^2 + (a_1 - a_3 + a_5 - a_7 + \ldots\ldots)$

$= 5^{2n}$ [from above $\sin^2\theta + \cos^2\theta = 1$]

MATCH THE COLUMN TYPE QUESTION(S)

20. If $\left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + \ldots\ldots\ldots + a_{2n} x^{2n}$, then :

यदि $\left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + \ldots\ldots\ldots + a_{2n} x^{2n}$, तो:

Column -I	**Column-II**
(A) $a_0 + a_1 + \ldots\ldots a_{2n}$	(P) n+1
(B) $a_0 - a_1 + a_2 - a_3 + \ldots\ldots + a_{2n}$	(Q) 1

(C) $a_0 - 2a_1 + 3a_2 + \ldots\ldots + (2n+1)a_{2n}$ (R) $3n$

(D) $a_0 - 2a_1 + 4a_2 - 8a_3 + \ldots\ldots + 2^{2n}.a_{2n}$ (S) 3^n

Ans. A – S, B – Q, C – P, D – S

Sol. $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$

$[A] \rightarrow S$

$x = 1$

$3^n = a_0 + a_1 + a_2 + \ldots\ldots + a_{2n}$

$[B] \rightarrow Q$

$x = -1$

$(1 - 1 + 1)^n = a_0 - a_1 + a_2 - a_3 + \ldots\ldots + a_{2n}$

$= 1$

$[C] \rightarrow Q$

$(1 + x + x^2)^n \cdot x = a_0x + a_1x^2 + a_2x^3 + \ldots\ldots + a_2nx^{2n+1} = (x^n + x^{n+1} + x^{n+2})^n$

Now differentiating

$n(x^n + x^{n+1} + x^{n+2})^{n-1} \{nx^{n-1} + (n+1)x^n + (n+2)x^{n+1}\} = a_0 + 2a_1x + 3a_2x^2 + \ldots + (2n+1)a_{2n}x^{2n}$

Put $x = -1$

$n((-1)^n + (-1)^{n+1} + (-1)^n)^{n-1} \{n(-1)^{n-1} + (n+1)(-1)^n + (n+2)(-1)^{n+1}\}$

$= a_0 - 2a_1 + 3a_2 - \ldots\ldots\ldots\ldots + (2n+1)a_{2n}$

$n(-1 + 1 - 1)^{even} \{+n - n - 1 + n + 2\}$

$n(n+1)$